The Optimization Space of Training-Free Reinforcement Learning Is Governed by Reward Spread, Not Search Effort or Agent Wrapping:
Theory, a Speech-Agent Testbed, and a Cautionary Paralinguistic Study

Exploring-L4-Intelligence Project

Abstract

This is a *proposal / position paper*: its central falsifiable question—whether *agentic* decomposition adds anything beyond a single frozen model—is left *open* as the paper's motivation, not settled as a result, and its empirical validation (Phases 2–3) is future work. Modern omni speech LLMs absorb broad cross-modal, multi-granularity knowledge during pretraining, yet the gain that training-free reinforcement learning can extract at inference time is *not* a property of how powerful, or how "agentic," the system is: it is governed by the *reward range*—the spread spread $= \max R - \min R$ over $\mathrm{supp}(q_0)$—of the (action-space, reward) pair the practitioner actually optimizes over, and to leading order by the reward variance $\mathrm{Var}_{q_0}(R)/(2\beta)$. This spread is *model-induced*: different model classes realize different spreads for the *same* reward—which is exactly why our paralinguistic fix later switches encoder—so the slogan means the lever is the reward structure rather than search effort or agent wrapping, *not* that the spread is model-independent. We make this precise. Casting training-free RL as Gibbs tilting of a frozen base policy, $q^\star \propto q_0 \exp(R/\beta)$, we develop an *optimization-space adequacy* (OSA) account whose qualitative core—and a new strict-positivity lemma—are machine-checked sorry-free in Lean 4, while the quantitative ceilings are Lean theorems that *consume* their concentration inputs as named hypotheses (Hoeffding, which we discharge on paper; Pinsker and a best-of-N order-statistics bound, which we do not). **OSA-1**: the KL-regularized gain obeys $\mathcal{G} = \beta\mathrm{KL}(q_0 \,\|\, q^\star) \le \mathrm{spread}^2/(8\beta)$, a bound on the *reward structure*, not the model class (a remark refines the certified range-based ceiling to the variance $\mathrm{Var}_{q_0}(R)/(2\beta)$ to leading order; it is not a separate theorem). The same frozen model yields vanishing gain under a flat reward—e.g. instruction-conditioning of a frozen embedding read-out—and large gain under a high-spread reward—e.g. task-level verifiable selection (our own single-model Operator-B: SLURP intent +0.330). **OSA-2**: under context isolation the optimum over isolated blocks *equals* the monolithic single-model optimum on the product reward $R = \sum_i R_i$ (`qstar_product`); the gain is additive (`gain_product`) and each *non-degenerate* block contributes *strictly positive* gain (new theorem `gain_pos_of_nonconstant`), so the only route to more gain is introducing additional non-degenerate verifiable rewards. These are the two load-bearing results; alongside them a *static*, nonnegative rollout-deficit corollary (`rollout_deficit` $= \beta\mathrm{KL}(q_0 \,\|\, q^\star) \ge 0$) records the cost of a frozen rollout, with finite-time convergence of the loop left open. Because the isolated optimum equals the monolithic one, whether *agentic* decomposition adds anything beyond a single model is untested: by `qstar_product` any such gain can come only from genuinely new non-degenerate rewards. We build a self-evolving omni *speech* agent as a *testbed* for this falsifiable question—a generative-omni policy under a verifiable-reward acceptance gate, paired with a memory that uses the omni embedding for *content* and classic speech encoders (ECAPA/SER) for paralinguistics. The empirical picture is deliberately humbling. For *content/intent* the high-spread regime is real and the single-model gains are large. For *paralinguistics* we report an honest negative, precisely scoped to what we measured: on *one* content bi-encoder (omni-embed), a linear/kNN speaker probe over its frozen embeddings sits near chance (≈ 0.04) at all 37 layers, and a multi-seed re-run of the emotion-pooling gain (dev-selected layer, paired-delta bootstrap, 5 seeds) is a *null* at the across-seed level—mean $\Delta = +0.037$, 95% t-CI $[-0.043, +0.116]$ (spans zero)—retiring an earlier single-seed +0.097 as a winner's-curse artifact of oracle test-layer selection. This negative is established on a content bi-encoder probe and does *not* bound the proposed generative Operator-B policy, which we never test for paralinguistics; we do not generalize it to frozen omni models as a class. The agent mechanism is not novel; the contributions are the spread-governs-gain theory, the domain transfer, verifiable speech rewards, and the honest, precisely-scoped paralinguistic negative that bounds where the program works.

1 Introduction

A modern omni or speech-capable multimodal LLM is, by the time it is released, an extraordinary repository of latent capability. From large-scale unsupervised audio, parallel speech-text corpora, and instruction mixtures, models such as Qwen2-Audio, Qwen2.5-Omni, Qwen3-Omni, and SALMONN [qwe, 2024, 2025a,b, sal, 2023] acquire transcription, translation, speaker identity, paralinguistic affect, and dialogue competence simultaneously. The animating question of this program is deliberately conservative: *how far can one go without touching a single weight?* We treat the pretrained model as frozen and ask what reward-guided optimization, applied purely at inference time, can elicit. This is the regime of *training-free* reinforcement learning—best-of-N selection, reward-guided decoding, reranking, task-conditioning, and representation read-out—that changes no parameters and no architecture [bei, 2024, mud, 2023, zuo, 2025].

This is a proposal, and its interesting claim is open. We state the paper's status plainly up front, because it governs how every later claim should be read. This is a *proposal / position paper*. Its *proven* content is deliberately modest and, granted the Gibbs framing we adopt, close to definitional: a variational identity for the KL-regularized gain with a range-based ceiling (OSA-1), and an accounting identity showing that under a *fixed* separable reward the context-isolated optimum coincides with the monolithic single-model optimum (OSA-2). The paper's *interesting* claim—the surprising reading of our slogan, that "agent wrapping adds nothing beyond a single frozen model"—is *open*. It is the paper's central *motivation*, not a result: by `qstar_product` any gain from agentic decomposition can come only from genuinely new, non-degenerate verifiable rewards, and whether a frozen omni model *affords* such rewards per component is exactly the falsifiable question we set up but do not yet answer. The empirical validation that would answer it (Phases 2–3) is future work; what we report here is theory, single-model feasibility measurements that locate the regime, and one honest paralinguistic negative. Read the proven identities as the floor and the open question as the point.

The lever is the reward structure, not the model class. The premise that organizes this paper is that the gain available to inference-time optimization is a property of *which reward, over which action space* one optimizes—not of whether the system is a single model or a multi-component agent. Some reward structures are nearly *flat*: the degrees of freedom they expose—the prompt, the demonstrations, the sampling temperature applied to a fixed read-out—move the model through a narrow envelope of near-identical rewards. In-context demonstrations frequently fix *format* rather than *accuracy*: their labels can be permuted or corrupted with little effect on the answer distribution [min, 2022, ali, 2026], and for speech in particular, models tend to "read" a transcribable surface rather than "listen" to the acoustic substrate, so paralinguistic conditioning barely perturbs the output [vox, 2026, hsu, 2023]. Our own measurements corroborate this for one specific reward—instruction-conditioning of a frozen omni embedding read-out—where the conditional output distribution is close to flat in the reward dimension. But the very same frozen model, optimized against a *different* reward—task-level verifiable selection over generative candidates—exposes a wide reward spread and a large gain. We name the flat case *flat conditioning*; the central object of the theory is the reward spread (its range, and to leading order its variance) that separates it from the high-spread case, on *one and the same frozen model*. Crucially this spread is *model-induced*: a fixed reward induces different spreads on different model classes—which is exactly why our paralinguistic fix later switches encoder—so the slogan "the lever is the reward structure" asserts that gain tracks the $(\text{action}, \text{reward})$ pair rather than search effort or agent wrapping, *not* that the spread is independent of the model.

Training-free RL as Gibbs tilting. To reason about this quantitatively we adopt the now-standard variational view of inference-time alignment. Optimizing the KL-regularized objective

$$F(q) \;=\; \mathbb{E}_{z\sim q}[R(z)] \;-\; \beta\,\mathrm{KL}(q\,\|\,q_0)$$

over search distributions q on the inference-time action space $\mathcal{Z}$ has the closed-form Gibbs optimum $q^\star(z) \propto q_0(z)\exp(R(z)/\beta)$, with partition function Z [kor, 2022, bei, 2024]. Here β multiplies the KL penalty and the tilt is $\exp(R/\beta)$. Every practical training-free method—best-of-N and its soft and asymptotic refinements [ver, 2025, yan, 2024, gui, 2024], self-consistency [wan, 2022], MBR decoding [ich, 2025, mbr, 2025], and controlled decoding [mud, 2023]—is an estimator of this tilt over some realization of $\mathcal{Z}$. The attainable benefit of *any* such estimator is therefore bounded by how far $q^\star$ can depart from q_0, which in turn is governed by the spread of R that the base distribution q_0 actually places mass on. Flat conditioning is precisely the statement that this spread is small *for that reward*—and the same machinery makes "high-spread reward, large gain" equally precise.

Central thesis: optimization-space adequacy. Our contribution is to turn this observation into a sharp, verifiable account that we call *optimization-space adequacy* (OSA). **OSA-1 (spread bound on the (action,reward) pair).** Granted the Gibbs framing, this result is a *variational identity plus a concentration bound*, and we present it as such rather than as a deep discovery. The gain is exactly $\mathcal{G} = F(q^\star) - F(q_0) = \beta \mathrm{KL}(q_0 \,\|\, q^\star)$ (the identity), and a Hoeffding argument caps it: $\mathcal{G} \le \frac{1}{8\beta}\mathrm{spread}^2$, where spread is the range of R on the support of q_0. To leading order the same argument sharpens the discriminator to the reward variance, $\mathcal{G} \approx \mathrm{Var}_{q_0}(R)/(2\beta)$; this is a remark, not a separate theorem—the *certified* ceiling is the range-based one. The bound is a statement about the *(action-space, reward) pair*, not the model class. When the spread is small—the flat-conditioning regime—the Gibbs tilt is a vanishing perturbation and more samples, cleverer reranking, and more elaborate search cannot manufacture headroom the conditional distribution does not contain. When the spread is large, the gain can be large *on the very same frozen model*: our own single-model Operator-B task-level selection lifts SLURP intent accuracy by +0.330, where flat instruction-conditioning of the same model's embedding read-out moves the factor accuracies by under 0.01. This mirrors, from the optimization side, the reward-model over-optimization scaling laws observed empirically [gao, 2022]. **OSA-2 (isolation is additive but adds nothing by itself).** This result is an *accounting identity*, not a claim about what agents can do. Decomposing a task across a *context-isolated* system—each isolated context exposing an independent operator on $\mathcal{Z}$, whether the embedding/Operator-A view $\mathcal{Z}_\mathrm{A}$ or the generative/Operator-B view $\mathcal{Z}_\mathrm{B}$—makes the joint action space a product. On a *fixed* separable product reward $R(z_1, z_2) = R_1(z_1) + R_2(z_2)$ the isolated optimum *equals* the monolithic single-model optimum: $q^\star$ factorizes (`qstar_product`) and the gain is exactly additive, $\mathcal{G} = \sum_i \mathcal{G}_i$ (`gain_product`). We are therefore explicit that *isolation per se adds nothing*—it neither creates nor destroys gain on a fixed reward. The *only* new gain comes from introducing additional *non-degenerate* verifiable rewards, and a new sorry-free lemma makes the contribution of each one strict: if a block's reward is non-constant then its gain is strictly positive (`gain_pos_of_nonconstant`). Whether a given frozen model actually *affords* such non-degenerate per-block rewards—rather than each block collapsing to its own flat band—is an *empirical* question, which we state as Conjecture 1 and test in this paper, not a theorem we prove. Because the isolated optimum equals the single-model one, the theory does *not* credit *agentic* decomposition with any gain over a monolithic model; by `qstar_product` the only way agentic structure can help is by carrying genuinely new non-degenerate rewards, which we do not test here. **A static rollout-deficit corollary (not a third pillar).** OSA-1 and OSA-2 are the two load-bearing results. Alongside them we record a static accounting identity: rolling out under the frozen base rather than the tilted optimum leaves a nonnegative deficit, `rollout_deficit` $= \beta\mathrm{KL}(q_0 \,\|\, q^\star) \ge 0$. This is a *static* nonnegative deficit, not a bounded "tax" and not a convergence guarantee—finite-time convergence of the rollout loop is left open. It matters because optimizing a product action space by rollout invites variance and feedback-loop pathologies—context collapse, error propagation in append-only memories, and "memory that hurts" [ace, 2025, exp, 2025, rem, 2026]—so the additive gain of OSA-2 is realizable in practice only under algorithm-level credit assignment [jit, 2026, ahm, 2024] and an explicit β-KL trust region that keeps each context's search distribution close to its own base. The resulting design stance: *do not search harder on one model and do not isolate for its own sake; introduce additional non-degenerate verifiable rewards—then measure, per factor, whether the frozen model affords them.*

A self-evolving omni speech agent as a testbed. We instantiate the theory in the most demanding modality, as a *testbed* that makes Conjecture 1 falsifiable rather than as a system we claim is independently worth building now. The system pairs (i) a *generative-omni policy* that acts under retrieved context; (ii) a *memory* that is content-addressable over past episodes; and (iii) a *verifiable-reward acceptance gate* that admits a candidate skill or memory only when an automatically checkable speech reward (WER for ASR, verifiable matches for translation and intent, and learned speech reward models for paralinguistics) certifies improvement [tul, 2024, rlv, 2025, gsr, 2026, par, 2025]. We are candid about the memory design in light of our own evidence below: on the one content bi-encoder we probe, the frozen omni embedding [omn, 2025, 2026] does *not* afford load-bearing paralinguistic retrieval—it is at chance on speaker identity at every layer—so we reserve the omni embedding for *content* and route paralinguistic memory through classic speech encoders (ECAPA-style speaker embeddings, dedicated SER read-outs). The two operators—memory (Operator-A) and policy (Operator-B)—are the context-isolation substrate OSA-2 describes, and the acceptance gate is the trust region the rollout-deficit corollary motivates. We build it not because the system is independently "worth building now," but because, to our knowledge, no speech-native, no-gradient, self-evolving agent of this form yet exists; the omni-policy-plus-verifiable-speech-reward pairing is, at present, an unfilled slot that makes the falsifiable question concrete.

What is and is not novel. We are explicit and honest about positioning. The *agent mechanism*—reflective self-improvement, skill libraries, memory-augmented policies—is not novel; the general-agent frontier is crowded with strong, closely related designs [ref, 2023, voy, 2023, jit, 2026, mem, 2025a, 2023a]. Our novelty lies elsewhere: in the *optimization-space theory* that shows the gain is governed by the reward spread (its range, and to leading order its variance) rather than model class and pins down exactly what an isolated decomposition can and cannot add; in the *domain transfer* of training-free RL to frozen omni speech models; in *verifiable speech rewards* as the acceptance criterion; and—as a first-class contribution—in an honest, precisely-scoped *empirical* demarcation of where the program works (content/intent) and where our measurement found no gain (a paralinguistic probe on one content bi-encoder). The theory is the load-bearing contribution; the system, and the cautionary study around it, are its honest test.

Contributions. We make five. **(C1) Theory (stated at its true strength).** We formalize optimization-space adequacy, and we are precise about what is proven versus what the slogan suggests. OSA-1 is a *spread ceiling that is, granted the Gibbs framing, a variational identity* ($\mathcal{G} = \beta\mathrm{KL}(q_0 \,\|\, q^\star)$) *plus a Hoeffding bound* ($\mathcal{G} \le \mathrm{spread}^2/(8\beta)$); it constrains the (action, reward) pair, not the model class. OSA-2 (`qstar_product`) is an *accounting identity*: under a *fixed* separable reward the context-isolated optimum coincides with the monolithic single-model optimum, the gain is additive (`gain_product`), and each non-degenerate block contributes strictly positive gain (`gain_pos_of_nonconstant`). Consequently only the *accounting-identity* reading of our slogan is proven; the *surprising* reading—that agent wrapping adds nothing beyond a single frozen model—is left *open*, since by `qstar_product` any such gain must come from genuinely new non-degenerate rewards we do not yet test. Alongside these sits a static nonnegative rollout-deficit corollary. The qualitative core (the gain identity $\mathcal{G} = \beta\mathrm{KL}(q_0 \,\|\, q^\star)$, its non-negativity, the flat-reward no-gain case, the product factorizations `qstar_product`/`gain_product`, and the rollout-deficit decomposition) *and* the new strict-positivity lemma `gain_pos_of_nonconstant` (reward non-constant $\Rightarrow \mathcal{G} > 0$) are sorry-free in Lean 4. The *quantitative* results consume their concentration inputs as named hypotheses: the $\mathrm{spread}^2/(8\beta)$ ceiling rests on a Hoeffding bound, which we discharge on paper, and on Pinsker's inequality, which we do *not* discharge but carry as a named hypothesis; the $O(\sqrt{\log N})$ best-of-N regret rests on a Beirami order-statistics bound, which is the single documented Lean `sorry` (`klBoN_le_klBoundBoN_TODO`). We label this scope plainly rather than overstate what the machine has checked. **(C2) Convergence analysis.** We connect the rollout-deficit corollary to a survey of agentic instability (context collapse, append-only error propagation, memory-induced regression) and give conditions, under β-KL trust regions and credit assignment, for the agent loop to converge rather than plateau or diverge [ace, 2025, exp, 2025, rem, 2026]; finite-time convergence rates remain open. **(C3) An honest empirical demarcation, including a cautionary negative.** We report single-model measurements that locate the operating regime. For *content/intent* the high-spread regime is real: single-model Operator-B selection yields large, well-separated gains (+0.330 on SLURP intent, +0.335 on URO spoken-QA). These are *single-model* high-spread gains, not evidence of agentic recovery—OSA-2 shows the isolated optimum equals the monolithic one, so whether agentic decomposition adds anything can come only from genuinely new non-degenerate rewards, which we do not test. For *paralinguistics* we report an honest negative, precisely scoped to what we measured. On *one* content bi-encoder (omni-embed), a linear/kNN speaker probe over its frozen embeddings is near chance (≈ 0.04, chance 0.011) at all 37 layers; and a rigorous multi-seed re-run of the emotion-pooling gain—dev-selected layer (no test-set oracle), paired-delta bootstrap, seeds $\{42, 7, 123, 2024, 31337\}$—is a *null at the across-seed level*: mean $\Delta = +0.037$ with a 95% t-CI of $[-0.043, +0.116]$ that spans zero.[1] The previously reported single-seed +0.097 was the best seed under *oracle test-layer* selection: a winner's-curse / selection-leak artifact, not a stable effect. This negative is established on a content bi-encoder probe; it does *not* bound the proposed generative Operator-B policy, which we never test for paralinguistics, and we do not generalize it to frozen omni models as a class. We treat it as a contribution: it is the reason the agent's paralinguistic memory uses classic speech encoders rather than the omni embedding. The re-run is committed at `projects/speech-mllm-omni-embedding-rl/_repro/emotion_pool_paired_v2.json`. **(C4) System design.** We specify the self-evolving omni speech agent—generative-omni policy, content-addressable memory (omni embedding for content, classic speech encoders for paralinguistics), verifiable-reward gate—as a concrete, reproducible *testbed* for Conjecture 1, not as a system we claim is independently worth building now. **(C5) Research plan and benchmark.** We lay out a staged research plan and propose a *cross-session paralinguistic memory* benchmark for speech agents, addressing a gap in current memory and multi-turn audio evaluations [lon, 2024, loc, 2024, aud, 2025].

Organization. Section 1 has framed the spread-governs-gain thesis and the OSA account. The remainder of the paper develops the formal setting and the Gibbs-tilt objective; states OSA-1 (the spread bound on the action–reward pair) and OSA-2 (additivity under isolation, the equality of the isolated and monolithic optima, the strict-positivity lemma, and the empirical Conjecture 1 it rests on), together with the static rollout-deficit corollary, with an explicit account of which Lean inequalities are sorry-free and which consume named external inputs; analyzes convergence of the agent loop under trust-region credit assignment; presents the self-evolving omni speech agent and its verifiable-reward gate together with feasibility, the single-model results, and the paralinguistic cautionary study; introduces the proposed cross-session paralinguistic memory benchmark; and closes with related work, limitations, and the staged research plan.

2 Preliminaries and Problem Formulation

We study *training-free* reinforcement learning (RL): a regime in which a frozen multimodal speech/omni model is steered at inference time by a reward signal, with no gradient ever propagated into its parameters and no architectural surgery. This section fixes notation, states the invariant that distinguishes training-free RL from conventional alignment, isolates the two model classes and their associated inference-time operators, lifts the formulation to an agentic action space with context isolation, and specifies the reward class we admit. The objects defined here are reused verbatim in the Theory section.

2.1 Training-free RL as constrained inference-time reward maximization

Let g_θ denote a pretrained speech/omni model with *frozen* parameters θ, and let x be an input (an utterance, a prompt, or a paired context). The inference-time degrees of freedom—the only quantities we are permitted to vary—are collected into an *action* $z \in \mathcal{Z}$, where $\mathcal{Z}$ is the model- and task-dependent action space made precise in Sections 2.2 and 2.3. Executing an action produces an output $g_\theta(x, z)$ that is scored by a bounded scalar *reward* $R \colon \mathcal{Z} \to [0, 1]$ (we write $R(z)$ for $R(g_\theta(x, z))$ when x is fixed). A pretrained *base* (or prior) distribution $q_0 \in \Delta(\mathcal{Z})$ over actions is induced by the frozen model and the default inference procedure—for a generative model, the temperature-scaled autoregressive sampler; for an embedding model, the canonical encoding policy. Training-free RL searches over distributions $q \in \Delta(\mathcal{Z})$ for one that raises expected reward while remaining close to the prior.

Definition 1 (Training-free RL objective)**.** Fix x, a base distribution q_0, a reward R, and a temperature $\beta > 0$. The KL-regularized inference-time objective is

$$F(q) \;=\; \mathbb{E}_{z \sim q}\big[R(z)\big] \;-\; \beta\, \mathrm{KL}(q \,\|\, q_0)\,, \tag{1}$$

and the training-free RL problem is $q^\star \;=\; \arg\max_{q \in \Delta(\mathcal{Z})} F(q)$.

[1] The paired bootstrap CI excludes zero in only 2/5 seeds taken individually; aggregated across seeds, by the t-CI above, the effect is not significant. We therefore report the across-seed CI as the headline, not the per-seed vote count.

The objective (1) is strictly concave in q over the simplex, and its maximizer is the classical Gibbs/exponential tilt of the prior [kor, 2022].

Proposition 1 (Gibbs optimum)**.** *The unique maximizer of* (1) *is*

$$q^\star(z) \;=\; \frac{1}{Z}\, q_0(z)\, \exp\Big(\tfrac{1}{\beta}\, R(z)\Big), \qquad Z \;=\; \mathbb{E}_{z\sim q_0}\Big[\exp\big(\tfrac{1}{\beta} R(z)\big)\Big], \tag{2}$$

with optimal value $F(q^\star) = \beta \log Z$*.*

Proof sketch. Adding a Lagrange multiplier for the normalization constraint and differentiating the concave functional (1) with respect to $q(z)$ yields $R(z) - \beta(\log q(z) - \log q_0(z)) - \beta = \text{const}$, i.e. $\log q(z) = \log q_0(z) + R(z)/\beta + \text{const}$; exponentiating and normalizing gives (2). Substituting back collapses the objective to $\beta \log Z$. □

Equation (2) is the formal anchor of the paper: every concrete inference-time procedure we analyze—best-of-N, soft best-of-N, minimum-Bayes-risk decoding, reward-guided decoding, and search over embedding configurations—is read as an *estimator of, or a sampler from, the tilted target* $q^\star$, and is therefore comparable through the lens of how well it approximates (2) at a given query budget. The temperature β interpolates between the prior ($\beta \to \infty$, $q^\star \to q_0$) and greedy reward maximization ($\beta \to 0^+$, $q^\star \to$ the $\arg\max_z R$ point mass), recovering best-of-N alignment in the low-temperature limit [bei, 2024, ver, 2025, kha, 2025].

Definition 2 (No-weight-update invariant)**.** A procedure is *training-free* if, for every input and every query budget, the model parameters θ and the model's computational graph are held fixed; all adaptation is realized through the choice of q over the inference-time action space $\mathcal{Z}$. Formally, the map $\theta \mapsto \theta$ is the identity throughout, and only q is optimized.

Definition 2 is the operational contract of this work. It excludes parameter-efficient fine-tuning and any backpropagation through g_θ; it admits only *selection, reweighting, search, and prompt/context construction* over $\mathcal{Z}$. Because $q^\star$ depends on θ only through q_0 and on the task only through R, the frozen model's pretrained competence is *activated*, never overwritten—the central premise the paper develops empirically and theoretically.

2.2 Two model classes and their inference-time operators

The action space $\mathcal{Z}$ factorizes by model class. We distinguish a *vector* (embedding) class and a *generative* (thinker–talker) class, with corresponding operators $\mathcal{Z}_\mathrm{A}$ and $\mathcal{Z}_\mathrm{B}$, $\mathcal{Z} = \mathcal{Z}_\mathrm{A} \cup \mathcal{Z}_\mathrm{B}$.

Vector / embedding omni (Operator A). A contrastive bi-encoder omni model maps a (conditioned) input to a single ℓ_2-normalized vector, typically by masked-mean pooling over a chosen hidden layer followed by a projection [omn, 2025, cla, 2022, nve, 2024]. Here g_θ emits a representation rather than a token sequence, the reward is a representation-quality functional (retrieval hit, probe accuracy, alignment/uniformity surrogate), and the prior q_0 is the default encoding policy.

Definition 3 (Operator A: embedding search space $\mathcal{Z}_\mathrm{A}$)**.** For the vector class, an action $z_\mathrm{A} = (c, \ell, \pi, P, S) \in \mathcal{Z}_\mathrm{A}$ specifies (i) the *conditioning* c (instruction/prompt prepended to the audio), (ii) the *layer* ℓ whose hidden states are read, (iii) the *pooling* rule π (masked mean, attentive statistics, last-token), (iv) the *projection* P applied to the pooled vector, and (v) the *selection* S over candidate views/segments. The output embedding is $u(z_\mathrm{A}) = P\big(\pi_\ell(g_\theta(x, c))\big)/\|\cdot\|_2$. Operator A is the search over $\mathcal{Z}_\mathrm{A}$ that maximizes a verifiable representation reward.

Generative thinker–talker omni (Operator B). An autoregressive omni model defines a distribution over token sequences y conditioned on (x, c); the default sampler is the prior q_0 [qwe, 2025b]. Inference-time alignment reshapes q_0 toward $q^\star$ either by drawing N candidates and selecting/aggregating (best-of-N, minimum-Bayes-risk) or by tilting the per-step transition with a value estimate (reward-guided/controlled decoding) [mud, 2023, bei, 2024, ver, 2025].

Definition 4 (Operator B: generative decoding space $\mathcal{Z}_{\mathrm{B}}$)**.** For the generative class, an action $z_{\mathrm{B}} \in \mathcal{Z}_{\mathrm{B}}$ specifies a decoding policy over output sequences: a sampling temperature, a candidate budget N, and an aggregation rule $A \in \{\text{best-of-}N,\ \text{MBR},\ \text{reward-guided}\}$. Best-of-$N$ returns $\arg\max_{i\leq N} R(y_i)$ for $y_i \sim q_0$; MBR returns $\arg\max_i \mathbb{E}_{y'\sim q_0}[u(y_i, y')]$ for a utility u [ich, 2025, mbr, 2025]; reward-guided decoding tilts each transition by $\exp(V(\cdot)/\beta)$ to approximate (2) token-by-token [mud, 2023].

Operators A and B are not interchangeable: A reweights a *geometric* object (a point on the unit sphere) and contributes no entropy to the output, whereas B reweights a *combinatorial* object (a sequence) drawn from a high-entropy prior. This asymmetry—a single-sample deterministic readout versus an N-sample stochastic search—governs the budget–quality trade-offs analyzed later and motivates treating the two as distinct instantiations of the same tilt (2).

2.3 The agent action space and context isolation

When the frozen model is embedded in an agent, the action is no longer a single decode but a *pipeline*: the agent first *recalls* memory, then *invokes* a skill, then *constructs* context, then *decodes*. We make this explicit.

Definition 5 (Agentic action)**.** An agentic action is a tuple $z = (m, s, c, d)$ where m selects which memory item(s) to recall, s selects which skill/tool to invoke, c builds the context window from (x, m, s), and $d \in \mathcal{Z}_{\mathrm{A}} \cup \mathcal{Z}_{\mathrm{B}}$ is the terminal embedding or decoding action of Definitions 3 and 4. The base distribution q_0 factorizes as $q_0(z) = q_0(m)\, q_0(s \mid m)\, q_0(c \mid m, s)\, q_0(d \mid c)$.

Many agentic systems decompose a task across K sub-agents that operate on disjoint slices of the action. We formalize the structure that makes such a decomposition tractable.

Definition 6 (Context isolation / separable structure)**.** The action space exhibits *context isolation* across K sub-agents if it factorizes as a product $\mathcal{Z} = \prod_{k=1}^{K} \mathcal{Z}^{(k)}$ with $z = (z^{(1)}, \ldots, z^{(K)})$, the base distribution is independent across blocks, $q_0(z) = \prod_k q_0^{(k)}(z^{(k)})$, and the reward is *additively separable*, $R(z) = \sum_{k=1}^{K} R^{(k)}(z^{(k)})$.

Remark 1. Under Definition 6, the Gibbs optimum (2) factorizes blockwise, $q^\star(z) = \prod_k q^{\star(k)}(z^{(k)})$ with $q^{\star(k)} \propto q_0^{(k)} \exp(R^{(k)}/\beta)$ and $Z = \prod_k Z^{(k)}$. Consequently the search decouples: each sub-agent may sample/select within its own block without cross-block interference, and the partition function—hence the optimization gain—is the product (equivalently, sum in log-space) of per-block contributions. This separability is the formal object the second operator-level result (OSA-2) requires, and it is precisely what *context contamination* between sub-agents destroys.

2.4 Verifiable versus model-judged rewards

The validity of every guarantee in Section 2.1 rests on R being trustworthy. We admit only *verifiable* rewards.

Definition 7 (Verifiable reward)**.** A reward R is *verifiable* if it is a deterministic, label-derived function of the output and a ground-truth annotation $y^\star$, computable without invoking a learned scorer. Canonical instances are (i) negative word error rate $1 - \mathrm{WER}(y, y^\star)$ for ASR/ST, (ii) exact-match/accuracy for classification and intent, (iii) retrieval hit@k for embedding queries, and (iv) probe accuracy for a fixed linear probe. A reward is *model-judged* if it is the output of a separate generative or preference model evaluating y.

Assumption 1 (Verifiability)**.** Throughout, R is verifiable in the sense of Definition 7: bounded in $[0, 1]$, deterministic given $y^\star$, and independent of g_θ.

We adopt Assumption 1 for two reasons. First, model-judged rewards are susceptible to *reward hacking*—the optimizer exploits gaps between the proxy and the true objective, so that increasing $\mathbb{E}_{q^\star}[R]$ no longer increases task quality [ska, 2022, gao, 2022]. Because the tilt (2) sharpens precisely along the direction of R, any reward misspecification is amplified rather than averaged out as $\beta \to 0$. Second, learned judges carry systematic artifacts—position bias and self-preference—that contaminate the ranking used by Operator B [pos, 2024, sel, 2024]; verifiable rewards are invariant to these. Restricting to label-derived R is therefore not a convenience but a precondition for the no-weight-update invariant of Definition 2 to translate into genuine, contamination-free activation of pretrained competence [tul, 2024]. With Definitions 1 to 4, 6 and 7 and Assumption 1 in place, the notation required by the Theory section is complete.

3 Related Work

Our contribution sits at the confluence of three literatures that have, until recently, evolved in relative isolation: the theory of inference-time reward optimization as exponential tilting, the empirical study of in-context learning (ICL) in audio language models, and the geometric characterization of what frozen speech encoders represent. We review each in turn, emphasizing not a catalogue of methods but the structural tensions that motivate our framework: that training-free RL is a search over a fixed action space $\mathcal{Z}$whose efficacy is bounded by the support of the base distribution q_0, and that this boundedness manifests differently for the embedding operator $\mathcal{Z}_{\mathrm{A}}$than for the generative operator $\mathcal{Z}_{\mathrm{B}}$.

3.1 Inference-Time / Training-Free RL and the Tilting View

The unifying abstraction behind training-free RL is that a reward-guided search distribution $q$is, in its optimal form, an exponential tilt of the base policy: the KL-regularized objective $F(q) = \mathbb{E}_q[R] - \beta\,\mathrm{KL}(q \,\|\, q_0)$ is maximized by the Gibbs measure $q^\star(z) \propto q_0(z)\exp(R(z)/\beta)$, with partition function Z[kor, 2022]. This identity recasts a spectrum of seemingly disparate inference-time procedures as approximations to the same tilt. Best-of-N (BoN) sampling is the most studied instance: bei [2024] establish that the KL divergence between the BoN distribution and $q_0$grows as $\log N - (N-1)/N$, giving a closed-form exchange rate between compute and the distributional shift purchased. ver [2025] sharpen this with a soft, temperature-controlled BoN whose suboptimality decays as $O(1/N)$, while ami [2025] recast BoN through a smoothing lens, bounding its regret against the Gibbs optimum and exposing when additional samples cease to help. The crucial—and for our purposes load-bearing—observation across this line is that *every* tilt reweights existing samples: the support of $q$is contained in the support of q_0. No amount of N creates mass where $q_0$assigns none. Minimum Bayes risk (MBR) decoding [ich, 2025], including its specialization to ASR [mbr, 2025], and controlled decoding [mud, 2023] inherit the same ceiling: they are consensus or value-guided reweightings, powerful when the correct hypothesis is already sampled with non-negligible probability and inert otherwise.

A more recent strand attempts to extract reward signal without any labels at all. TTRL [zuo, 2025] and TPO [li2, 2025] use self-consistency or majority vote as a surrogate reward at test time, and JitRL [jit, 2026] performs gradient-free credit assignment over a frozen policy. These methods are attractive precisely because they touch no weights, but they sharpen rather than dissolve the support constraint: a self-generated reward can only amplify modes the model already prefers, which is why majority-vote rewards can entrench confident errors. The over-optimization literature quantifies the resulting hazard. gao [2022] show that reward and true objective diverge as KL grows, following a predictable scaling law; kha [2025] characterize the optimal effective N^* (the Best-of-Poisson regime) beyond which tilting harms. The over-optimization risk is governed by the reward spread spreadand the realized gain $\mathcal{G}$: when the proxy reward is mis-specified, larger $\mathcal{G}$trades calibrated base knowledge for spurious peaks.

These dynamics intersect the ongoing RLVR boundary debate, which directly concerns whether inference-time or verifiable-reward optimization can expand a model's capabilities or merely re-allocate them. rlv [2025] argue that RLVR predominantly sharpens the base distribution rather than adding skills, a position echoed by the finding that even spurious or random rewards can improve benchmark scores by exploiting format and prior structure [spu, 2025]. The countervailing evidence that RLVR can induce genuinely new skills [new, 2026] does not contradict the tilting view so much as locate its exception: new behavior emerges only when the action space $\mathcal{Z}$is rich enough—e.g., multi-step generative composition—that the relevant trajectory had non-zero base mass that single-shot sampling rarely surfaced. This distinction—reweighting versus genuine expansion—is exactly the axis along which we separate $\mathcal{Z}_{\mathrm{A}}$from $\mathcal{Z}_{\mathrm{B}}$, and the tilting formalism makes the separation precise rather than rhetorical.

3.2 In-Context Learning in Audio LLMs and the Two Model Classes

If training-free RL reweights the base distribution, the natural question is whether in-context demonstrations can *move* it. The text-domain ICL literature offers a sobering prior. min [2022] demonstrate that demonstrations are largely label-insensitive: corrupting ground-truth labels barely degrades performance, implying that demonstrations specify the task and output format rather than teaching the input-output mapping. ali [2026] reach a congruent conclusion for instruction-following demonstrations, which fix format but not accuracy. ICL is moreover scale-gated—its more sophisticated, override-the-prior behaviors emerge only in larger models [wei, 2023]—a property that carries into audio, where few-shot competence appears as an emergent phenomenon in larger audio LLMs [mim, 2025]. The function- and task-vector view [tod, 2023, tas, 2025] explains the mechanism: demonstrations compress to a latent task direction that conditions decoding without altering the underlying perceptual representation. For a generative operator $\mathcal{Z}_{\mathrm{B}}$, this means demonstrations can re-route *how* content is verbalized, but they do not re-open perceptual channels the encoder has discarded.

This perceptual ceiling is starkly visible in audio. A growing body of evidence shows that audio LLMs frequently "read rather than listen": vox [2026] document that models answer paralinguistic questions from textual priors and transcript content rather than the acoustic signal, and lis [2025] find that performance often depends on whether the model transcribes versus genuinely attends to non-lexical cues. In-context demonstrations do little to repair this when the requisite acoustic information has already been pooled away upstream—consistent with the broader finding that few-shot prompting fixes format, not the missing percept. SER is a partial and instructive exception: ser [2025] report that few-shot demonstrations *do* improve speech emotion recognition, plausibly because emotion is one of the paralinguistic factors a generative omni model retains with enough base probability for tilting and ICL to surface. The pattern—ICL helping for retained factors, failing for discarded ones—mirrors the support constraint of Section 3.1 and foreshadows our operator-dependent treatment.

The contrast with embedding models is decisive for our two-class taxonomy. Text embedders do not acquire ICL for free: bge [2024] show that exploiting in-context examples in an embedder requires dedicated training, and instruction-conditioned embedders such as INSTRUCTOR [su2, 2022] must be explicitly trained to follow task instructions at encode time. An off-the-shelf embedding operator $\mathcal{Z}_{\mathrm{A}}$therefore exposes *no* prompt-conditioned degree of freedom analogous to $\mathcal{Z}_{\mathrm{B}}$'s in-context steering: its action space is the choice of pooling, layer, and query, not a re-conditioning of the forward pass. This asymmetry—$\mathcal{Z}_{\mathrm{B}}$admits demonstration-driven tilts, $\mathcal{Z}_{\mathrm{A}}$does not—is, in our framework, not an implementation detail but a defining structural difference between the two model classes, and it dictates which training-free interventions are even available for each.

3.3 Capability Map, Embedding Geometry, and Probing

Whether a frozen model *can* be tilted toward a task ultimately depends on whether the relevant information survives in its representations—a question the probing and embedding-geometry literatures answer with unusual clarity. Contrastive and alignment objectives are now understood to actively discard nuisance variation: xia [2020] show that contrastive learning suppresses factors not aligned with the training augmentations, and wan [2020] formalize the alignment–uniformity trade-off that drives semantically similar inputs together while spreading the hypersphere uniformly. For a content-oriented embedder, speaker identity, emotion, and channel are precisely the nuisances such objectives are designed to remove. This is the geometric root of why an embedding operator $\mathcal{Z}_{\mathrm{A}}$, however cleverly searched, cannot recover a factor its training objective erased: the base distribution $q_0$over $\mathcal{Z}_{\mathrm{A}}$simply has no mass on the discarded direction.

Pooling compounds the loss. Mean- or last-token pooling imposes an information bottleneck that NV-Embed's learned latent-attention pooling was introduced to mitigate [nve, 2024], and the speaker-verification community has long known that attentive statistics pooling [att, 2018] and ECAPA-TDNN's multi-scale aggregation [eca, 2020] are necessary precisely because naive temporal averaging destroys speaker-discriminative structure. The implication is that the *same* encoder can be near-blind or highly informative for a factor depending solely on the read-out, which is exactly the degree of freedom a training-free search over $\mathcal{Z}_{\mathrm{A}}$should exploit. Layer-wise probing makes this concrete: pas [2021] chart an autoencoder-like trajectory across wav2vec 2.0 layers in which phonetic and word-level information peaks at intermediate depths, and spk [2025] show speaker attributes are distributed non-uniformly across layers. Capability is thus not a scalar property of a model but a function of (layer, pooling, query)—the very coordinates of our action space $\mathcal{Z}_{\mathrm{A}}$.

Crucially, evidence that paralinguistic content *persists* even when unused is now substantial. fro [2024] demonstrate that a frozen LLM, given speech features, perceives paralinguistic attributes it never verbalizes; emo [2026] localize emotion-selective units inside such models. These findings establish the optimistic precondition for our flagship study: the information needed to disentangle content, speaker, emotion, and intent is present in the frozen omni model's hidden states, even when the default forward pass—or a generative decode $\mathcal{Z}_{\mathrm{B}}$—suppresses it. The task is therefore one of *activation* via search over read-outs, not of teaching. The SUPERB and Dynamic-SUPERB taxonomies [sup, 2021, dyn, 2023] supply the benchmark scaffolding for evaluating such activation across content, speaker, semantic, and paralinguistic axes, and we adopt their factorization to structure our capability map. What this literature lacks—and what our tilting formalism supplies—is a principled account of *which* of these latent capabilities are reachable by reweighting a fixed $q_0$and which demand a different action space altogether.

3.4 Agent Memory

A memory module is the substrate on which a training-free agent accumulates experience without touching weights, and the literature has moved from flat buffers to increasingly structured stores. Early designs append raw interaction logs and retrieve by recency or embedding similarity: the memory stream of generative agents records timestamped observations and synthesizes higher-order reflections [gen, 2023], while MemGPT introduces an operating-system metaphor in which a fixed context window pages information to and from an external store [mem, 2023a]. Subsequent systems impose explicit organization. A-MEM grows a Zettelkasten-style network whose notes are linked and revised as new evidence arrives [ame, 2025]; AriGraph maintains an integrated semantic–episodic knowledge graph that supports structured planning [ari, 2024]; Zep adds a bi-temporal model that distinguishes event time from ingestion time so that stale facts can be invalidated rather than overwritten [zep, 2025]; MemoryOS again borrows OS hierarchy to schedule short- and long-term memory [mem, 2025c]. Mem0 makes the curation operators first-class, framing maintenance as explicit ADD/UPDATE/DELETE decisions over extracted facts [mem, 2025a].

Retrieval is the second axis. Dense passage retrieval and the broader retrieval-augmented generation paradigm established the read path that most memory systems still inherit [rag, 2020], with late-interaction scoring offering finer token-level matching at scale [col, 2020]. HippoRAG draws on hippocampal indexing to support multi-hop associative recall over a knowledge graph, improving over flat dense retrieval on questions that require chaining facts [hip, 2024].

Curation quality, not raw capacity, governs whether memory helps. Agentic Context Engineering documents *context collapse*, in which iterative rewriting erodes accumulated detail, and argues for incremental, structure-preserving updates [ace, 2025]; Mem0's selective edit operators are partly a response to the same failure mode [mem, 2025a]; MemoryBank instead imports an Ebbinghaus forgetting curve so that unreinforced traces decay [mem, 2023b]. A parallel line exposes memory as an attack surface and a liability: AgentPoison shows that a handful of poisoned entries can hijack downstream retrieval [age, 2024a], MemoryGraft demonstrates implanted false memories that persist across sessions [mem, 2025b], and, even absent an adversary, accumulated memory can *degrade* performance through distraction and error propagation [rem, 2026]. This motivates treating memory writes as decisions under reward rather than unconditional logging. Memory-R1 takes exactly this step, using reinforcement learning to assign credit over memory-management actions [mem, 2025d] — a *gradient-based* answer to a problem this work attacks training-free.

Evaluation remains immature and almost entirely text-centric. Audits of memory benchmarks reveal that many ostensibly memory-dependent tasks are solvable without genuine cross-session recall [ana, 2026], while LongMemEval and LoCoMo provide the most credible long-horizon and conversational probes [lon, 2024, loc, 2024]. On the speech side, retrieval over audio is nascent: WavRAG retrieves directly over acoustic content [wav, 2025], VoxRAG operates in the spoken domain end-to-end [vox, 2025], and asynchronous retrieval has been folded into full-duplex spoken dialogue [mos, 2026]. Yet no benchmark isolates *cross-session* memory for a speech agent: the closest, an emotion-in-dialogue suite, is text-derived and single-axis [amb, 2026]. The gap — structured, reward-curated, contamination-resistant memory evaluated for spoken interaction — is precisely where a training-free agent built on a frozen omni model can contribute.

3.5 Agent Skills and Self-Evolution

A complementary form of inference-time accumulation is the *skill*: a reusable procedure the agent acquires and later invokes. A recent systematization formalizes a skill as a tuple of context, policy, transferable trigger, and reward (C, π, T, R), separating *when* a skill applies from *what* it does and *how* its utility is scored [sok, 2026]. Voyager pioneered an ever-growing, executable skill library for open-ended control, with new skills written as code and retrieved by description [voy, 2023]; agentic workflow memory generalizes this to reusable workflows distilled from successful trajectories [awm, 2024]; and recent systems automate skill discovery for multimodal use [mus, 2026]. A central empirical finding tempers the self-generation enthusiasm: on controlled benchmarks, *curated* skill sets substantially outperform self-generated ones, because unverified skills accumulate noise [ski, 2026b]. Two responses follow. Co-evolutionary skill learning trains a surrogate verifier jointly with the skill library so that candidate skills are filtered before admission [coe, 2026], and body-first skill routing retrieves by the executable body rather than a possibly-misleading natural-language description [ski, 2026a]. Both can be read as variance-reduction and reward-shaping mechanisms over a discrete skill action space.

Because the target models are frozen, skill optimization here is necessarily gradient-free, aligning with a mature line of prompt-, workflow-, and graph-level optimizers that improve agents without backpropagation: GEPA evolves prompts along a Pareto frontier of reflective mutations [gep, 2025]; AFlow searches over workflow graphs [afl, 2024]; AgentSquare performs modular agent search [age, 2024b]; DSPy compiles declarative pipelines [dsp, 2023]; TextGrad propagates natural-language "gradients" [tex, 2024]; ADAS searches the space of agentic designs [ada, 2024]; and ExpeL extracts and reuses experiential rules [exp, 2023]. Speech-specific skills are only beginning to appear: rare-word and entity correction via generative error correction [ger, 2025], voice-driven tool use [aur, 2025], and paralinguistic style control under explicit reward [par, 2025]. None of these assembles a frozen omni model's latent competencies into a self-curated, reward-verified skill library — the construction this work pursues.

3.6 Agent-Level Convergence and Stability

Iterative self-improvement is not monotone, and a body of evidence catalogues where it stalls or reverses. Reflexion improves through verbal self-feedback but plateaus once reflections cease to surface new information [ref, 2023]. Context engineering collapses when rewriting overwrites detail [ace, 2025], and append-only experience logs propagate early errors forward because mistaken entries are never retracted [exp, 2025]. More starkly, added memory can *hurt*, degrading task accuracy through distraction and compounding of stale facts [rem, 2026]. Tree-structured search offers one principled remedy: language-agent tree search couples Monte Carlo rollouts with value backups [lat, 2023], inheriting the regret guarantees of UCT bandit exploration [koc]. Convergence under a frozen policy is the explicit object of just-in-time RL, which performs no-gradient credit assignment and characterizes when inference-time self-correction provably converges [jit, 2026]; our analysis treats the same stability question through the lens of a reward-tilted search distribution over a fixed action space, connecting plateau, collapse, and error-propagation phenomena to properties of the tilting reward and its spread spread.

3.7 Speech Agents and the Open Moat

Spoken-language agents are now richly benchmarked, yet almost exclusively in a *single-episode* regime that evaluates a fresh agent on each independent interaction. URO-Bench probes spoken dialogue across understanding, reasoning, and oral conversation [uro, 2025]; VoiceBench stress-tests LLM-based voice assistants under realistic perturbations [voi, 2024]; EVA targets expressive and emotional voice interaction [eva, 2026]; and a dedicated suite even measures multi-turn memory *within* a single conversation [aud, 2025]. What unifies these is the absence of persistence and self-evolution across sessions: the agent that finishes a dialogue is identical to the one that began it. Where the field does pursue improvement, the mechanism is weight-updating — continual or alignment training of the speech model itself, as in recent end-to-end conversational systems [cor, 2026, sou, 2026]. The consequence is an open moat: the *training-free, self-evolving speech agent* — one that, over its lifetime, accrues reward-curated memory and a verified skill library atop a *frozen* omni model, with no weight or structural change — is an unfilled slot. This work targets exactly that slot, importing the inference-time reward-tilting and verification machinery of Sections 3.4 to 3.6 into the spoken-agent setting.

4 Theory: Optimization-Space Adequacy

The thesis of this paper is that the optimization gain achievable by training-free reinforcement learning is governed not by the cleverness of the search operator, and not by whether the system is a single model or a multi-component agent, but by the *reward spread of the (action-space, reward) pair* over which the search is performed — which we abbreviate *OSA* (Optimization-Space Adequacy) throughout. The central object is a property of the *reward structure*, not of the *model class*: a flat reward (for instance, instruction-conditioning of a frozen embedding read-out; Table 2) yields vanishing gain, while a high-spread reward (for instance, task-level verifiable selection) yields large gain *even on a single frozen model* (our own single-model Operator-B results, e.g. SLURP +0.330 in Table 4). We make this precise with a sequence of theorems that we have additionally formalized in Lean 4. We are deliberately careful about what "machine-checked" means here (see Remark 2): the *qualitative core* — the gain identity (`gain_eq`), its nonnegativity (`gain_nonneg`), the flat-reward collapse (`flat_no_gain`), the strictly-positive-gain theorem for a non-degenerate reward (`gain_pos_of_nonconstant`), the partition-function and Gibbs-optimum product factorizations, and the rollout deficit (`rollout_deficit`) — is *sorry-free*, whereas the *quantitative* results (the spread$^2/(8\beta)$ ceiling via Hoeffding, the best-of-N KL bound carrying the one documented `sorry`, and the $O(\sqrt{\log N})$ regret via Pinsker) are Lean theorems stated *conditionally on named, undischarged hypotheses*. The development fixes a finite action space $\mathcal{Z}$ (the inference-time degrees of freedom of a frozen model — an embedding code in $\mathcal{Z}_\mathrm{A}$, a generative continuation in $\mathcal{Z}_\mathrm{B}$, or a product of such), a strictly positive base measure $q_0 : \mathcal{Z} \to \mathbb{R}_{>0}$ with $\sum_{w\in\mathcal{Z}} q_0(w) = 1$ induced by the model at temperature, a bounded reward $R : \mathcal{Z} \to \mathbb{R}$, and an inverse-temperature parameter $\beta > 0$. All distributions below are probability mass functions on $\mathcal{Z}$. We write $\mathbb{E}_q[\cdot]$ for expectation under q and adopt the convention $0 \log 0 = 0$.

Remark 2 (Honesty about the formalization). Every result in Section 4 except the two we explicitly flag is *sorry-free* in our Lean development (file `OptSpace.lean`, supported by the tilting core in `Tilting.lean`). The two flagged dependencies are isolated as named hypotheses and are *never* discharged silently: (i) Hoeffding's lemma, used only in Theorem 3, is taken as an assumption (`gain_le_of_hoeffding`); and (ii) the order-statistics step in the best-of-N KL bound (the Beirami–Mroueh integral, `klBoN_le_klBoundBoN`) carries the single documented `sorry` in the wider development and is consumed elsewhere as a hypothesis, not in this section. In particular the new Theorem 5 (`gain_pos_of_nonconstant`) is sorry-free, resting only on the strict Gibbs inequality (`kl_pos_of_ne`). We state both flagged results as assumptions so that no result here over-claims machine verification.

4.1 The Tilting Objective and its Optimum

The object of study is the KL-regularized reward functional, whose maximizer is the exponential tilt of the base policy. This is the inference-time specialization of the well-known equivalence between KL-regularized control and Bayesian posterior inference [kor, 2022]; it underlies controlled decoding [mud, 2023] and the best-of-N literature [bei, 2024].

Definition 8 (Partition function, Gibbs optimum, regularized objective)**.** Define the *partition function*

$$Z \;:=\; \sum_{w\in\mathcal{Z}} q_0(w)\, \exp\!\big(R(w)/\beta\big) \;=\; \mathbb{E}_{q_0}\!\big[\exp(R/\beta)\big] \;>\; 0,$$

the *Gibbs optimum* (exponential tilt)

$$q^\star(z) \;:=\; \frac{q_0(z)\, \exp\!\big(R(z)/\beta\big)}{Z},$$

which is a probability distribution because $q_0 > 0$ and $Z > 0$, and the *KL-regularized objective*

$$F(q) \;:=\; \sum_{z\in\mathcal{Z}} q(z)\, R(z) \;-\; \beta \sum_{z\in\mathcal{Z}} q(z)\, \log \frac{q(z)}{q_0(z)} \;=\; \mathbb{E}_q[R] \;-\; \beta\, \mathrm{KL}(q \,\|\, q_0)\,.$$

The next lemma is the algebraic spine of the whole section: it converts the gap to the optimum into a single nonnegative KL divergence.

Lemma 1 (Master identity)**.** *For every probability distribution q on $\mathcal{Z}$,*

$$F(q) \;=\; \beta \log Z \;-\; \beta\,\mathrm{KL}(q \,\|\, q^\star)\,, \qquad \textit{hence} \qquad F(q^\star) - F(q) \;=\; \beta\,\mathrm{KL}(q \,\|\, q^\star)\,.$$

Proof. Taking logarithms in Definition 8, $\log q^\star(z) = \log q_0(z) + R(z)/\beta - \log Z$, so that, pointwise,

$$R(z) \;=\; \beta\Big(\log \tfrac{q^\star(z)}{q_0(z)} + \log Z\Big).$$

Substituting into $F(q)$ and using $\sum_z q(z) = 1$,

$$\begin{aligned}
F(q) &= \sum_z q(z)\Big[\beta \log \tfrac{q^\star(z)}{q_0(z)} + \beta \log Z\Big] - \beta \sum_z q(z) \log \tfrac{q(z)}{q_0(z)} \\
&= \beta \log Z + \beta \sum_z q(z)\Big[\log \tfrac{q^\star(z)}{q_0(z)} - \log \tfrac{q(z)}{q_0(z)}\Big] = \beta \log Z + \beta \sum_z q(z) \log \tfrac{q^\star(z)}{q(z)} \\
&= \beta \log Z - \beta\,\mathrm{KL}(q \,\|\, q^\star)\,.
\end{aligned}$$

Evaluating at $q = q^\star$ gives $F(q^\star) = \beta \log Z$ (the KL term vanishes), and subtracting yields the stated gap. □

Theorem 1 (Tilting optimality; `tilting_optimal`, T1)**.** *For every probability distribution q on $\mathcal{Z}$, $F(q) \le F(q^\star)$, with equality iff $q = q^\star$. Thus the exponential tilt $q^\star$ is the unique maximizer of the inference-time objective $F(\cdot)$.*

Proof. By Lemma 1, $F(q^\star) - F(q) = \beta\,\mathrm{KL}(q \,\|\, q^\star)$. Nonnegativity of KL (Gibbs' inequality) follows from $\log x \le x - 1$ for $x > 0$: writing the ratio $r(z) = q^\star(z)/q(z)$ on the support of q,

$$-\mathrm{KL}(q \,\|\, q^\star) = \sum_z q(z) \log r(z) \;\le\; \sum_z q(z)\big(r(z) - 1\big) = \sum_z q^\star(z) - \sum_z q(z) = 0,$$

so $\mathrm{KL}(q \,\|\, q^\star) \ge 0$ and $F(q) \le F(q^\star)$. Since $\beta > 0$ and $\log x = x - 1$ only at $x = 1$, equality forces $q = q^\star$ pointwise. □

Theorem 1 is the entire ceiling of any training-free method: no reranker, decoder, or search distribution $q \in \Delta(\mathcal{Z})$ can exceed $F(q^\star) = \beta \log Z$. The rest of the section asks the operational question this raises — *how large is the headroom $F(q^\star) - F(q_0)$ between the model's own sampling distribution q_0 and this ceiling?* — because that headroom is exactly the gain a training-free optimizer can ever realize.

4.2 OSA-1: A Flat or Degenerate Reward Yields Vanishing Gain

Definition 9 (Optimization gain)**.** The *gain* of the action space under reward R at temperature β is the headroom $\mathcal{G} := F(q^\star) - F(q_0)$.

Proposition 2 (Closed form for the gain; `gain_eq`, OSA-1a)**.**

$$\mathcal{G} \;=\; \beta \log Z - \mathbb{E}_{q_0}[R] \;=\; \beta \log \mathbb{E}_{q_0}\big[\exp(R/\beta)\big] - \mathbb{E}_{q_0}[R].$$

Proof. Apply Lemma 1 at $q = q_0$. Since $\mathrm{KL}(q_0 \,\|\, q_0) = 0$, the definition gives $F(q_0) = \mathbb{E}_{q_0}[R] - \beta\mathrm{KL}(q_0 \,\|\, q_0) = \mathbb{E}_{q_0}[R]$, while $F(q^\star) = \beta \log Z$. Subtract, and recall $Z = \mathbb{E}_{q_0}[\exp(R/\beta)]$ from Definition 8. □

Proposition 3 (Nonnegativity of the gain; `gain_nonneg`, OSA-1b)**.** *$\mathcal{G} \ge 0$, with equality iff $q^\star = q_0$.*

Proof. Two equivalent routes. (i) By Lemma 1, $\mathcal{G} = F(q^\star) - F(q_0) = \beta\,\mathrm{KL}(q_0 \,\|\, q^\star) \ge 0$, strict unless $q_0 = q^\star$. (ii) Directly from Proposition 2, Jensen's inequality applied to the strictly convex $t \mapsto e^{t/\beta}$ gives $\beta \log \mathbb{E}_{q_0}[e^{R/\beta}] \ge \beta\,\mathbb{E}_{q_0}[\log e^{R/\beta}] = \mathbb{E}_{q_0}[R]$. □

The gain is therefore a divergence between what the model already does, q_0, and what the reward would have it do, $q^\star$. When the reward is *degenerate* — so that it cannot distinguish the elements of $\mathcal{Z}$ — this divergence collapses.

Theorem 2 (Flat reward, no gain; `flat_no_gain`, OSA-1c $\Rightarrow$ T3)**.** *If $R \equiv c$ is constant on $\mathcal{Z}$, then $q^\star = q_0$ and $\mathcal{G} = 0$.*

Proof. With $R(z) = c$, $Z = \sum_w q_0(w)e^{c/\beta} = e^{c/\beta}$, so $q^\star(z) = q_0(z)e^{c/\beta}/e^{c/\beta} = q_0(z)$ (this is exactly T3, `qstar_eq_q0_of_const`). By Proposition 2, $\mathcal{G} = \beta \log e^{c/\beta} - c = c - c = 0$. □

Theorem 2 is the formal content of the flat-reward regime, and it is a statement about the *reward*, not about the model class. Its empirical signature in our own data is *instruction-conditioning of a frozen embedding read-out*: in Table 2 the conditioning instruction does not measurably move the factor accuracies (overlapping CIs, not diagonally dominant), so the effective R is approximately flat on $\mathrm{supp}(q_0)$, the tilt does essentially nothing, and there is no leverage for the optimizer to exploit. We stress that this is *not* an artifact of using a single model: the very same single frozen model affords a high-spread reward under task-level Operator-B selection (Table 4). The flat case is one regime of the reward structure, not a verdict on single-model systems. This is consistent with reports that purely self-referential test-time RL plateaus quickly [zuo, 2025, ref, 2023] and with spurious-reward phenomenology [spu, 2025, ska, 2022]. The next theorem makes the dependence on reward dispersion quantitative.

Assumption 2 (Hoeffding's lemma; isolated, cf. Remark 2)**.** Let X be a random variable supported in an interval of length ℓ a.s. under a measure μ. Then for all $\lambda \in \mathbb{R}$,

$$\log \mathbb{E}_\mu\big[e^{\lambda X}\big] \;\le\; \lambda\,\mathbb{E}_\mu[X] + \frac{\lambda^2 \ell^2}{8}.$$

In Lean this is the hypothesis named in `gain_le_of_hoeffding`; we do not prove it from first principles in our development.

Theorem 3 (Quantitative collapse of gain; `gain_le_of_hoeffding`, OSA-1d)**.** *Let* $\mathrm{spread} := \max_{z\in\mathcal{Z}} R(z) - \min_{z\in\mathcal{Z}} R(z)$ *be the reward spread (range) on $\mathcal{Z}$. Under Assumption 2,*

$$\mathcal{G} \;\le\; \frac{\mathrm{spread}^2}{8\beta}.$$

In particular $\mathrm{spread} \to 0 \Rightarrow \mathcal{G} \to 0$*: a vanishing reward spread forces a vanishing gain, uniformly in the search operator.*

Proof. Apply Assumption 2 with $\mu = q_0$, $X = R$ (supported in an interval of length spread on $\mathcal{Z}$), and $\lambda = 1/\beta$:

$$\log \mathbb{E}_{q_0}\big[e^{R/\beta}\big] \;\le\; \frac{1}{\beta}\,\mathbb{E}_{q_0}[R] + \frac{1}{\beta^2}\cdot\frac{\mathrm{spread}^2}{8}.$$

Multiply by $\beta > 0$ and rearrange to $\beta \log \mathbb{E}_{q_0}[e^{R/\beta}] - \mathbb{E}_{q_0}[R] \le \mathrm{spread}^2/(8\beta)$, which is exactly $\mathcal{G}$ by Proposition 2. The limit is immediate. □

Remark 3 (The ceiling is a loose, range-based bound; do not call it "realized")*.* Three honest qualifications attach to Theorem 3. (i) *Range, not realized spread.* Because $q_0 > 0$ everywhere on the finite $\mathcal{Z}$, the support of q_0 is all of $\mathcal{Z}$, so spread is simply the *global range* $\max R - \min R$; there is no separate "realized" spread to speak of, and we drop that adjective. The bound is driven by the two extreme reward values, regardless of how little mass q_0 places near them. (ii) *Looseness for a concentrated q_0.* A second-order (cumulant) expansion of Proposition 2 gives $\mathcal{G} = \mathrm{Var}_{q_0}(R)/(2\beta) + O(1/\beta^2)$, so for a concentrated q_0 the true gain scales with the *variance*, not the range; since $\mathrm{Var}_{q_0}(R) \le \mathrm{spread}^2/4$ (Popoviciu), the Hoeffding ceiling $\mathrm{spread}^2/(8\beta)$ over-estimates the gain by a factor that grows as q_0 concentrates. (iii) *Vacuity at small β and normalization.* As $\beta \to 0$ the ceiling $\mathrm{spread}^2/(8\beta) \to \infty$ and is vacuous; and for a normalized reward $R \in [0,1]$ it pins to $1/(8\beta)$. The non-vacuous statement is therefore the trivial range bound combined with Hoeffding: since $F(q^\star) \le \max R$ and $F(q_0) = \mathbb{E}_{q_0}[R] \ge \min R$, we always have $\mathcal{G} \le \mathrm{spread}$, so

$$0 \;\le\; \mathcal{G} \;\le\; \min\Big(\mathrm{spread},\; \tfrac{\mathrm{spread}^2}{8\beta}\Big).$$

We present the ceiling in this min form throughout and avoid calling spread "realized."

Theorem 3 and Remark 3 pair with Proposition 3 to sandwich the gain and identify the reward spread as the sole lever of a frozen model under a given reward: it is the inference-time analogue of the reward-model scaling laws that bound over-optimization [gao, 2022], and it explains why best-of-N and reranking deliver returns proportional to how much the verifiable reward actually separates candidates [bei, 2024, kha, 2025, sne, 2024]. This makes the *kind* of reward central, so we pause to reconcile the two reward families this paper uses.

Remark 4 (Two families of verifiable reward, and where Goodhart-immunity applies). Write g_θ for the frozen model's read-out map (embedding or decode). The verifiable rewards in this series split into two families. *(A) Label-derived rewards* — WER and exact-match against a ground-truth target $y^\star$ — are functions of $y^\star$ alone and are genuinely *independent* of g_θ: they cannot be satisfied by exploiting an idiosyncrasy of the frozen model, so the usual Goodhart-immunity / verifiability argument (a label-checked reward resists reward-hacking) applies to them. *(B) Model-dependent verifiable rewards* — probe accuracy and retrieval hit@k — are functions of the frozen g_θ itself. These are still *checkable*, but they are not label-independent: a probe reward reads out the same representation the policy acts in, so it *shares failure modes* with the policy and the Goodhart-immunity argument does *not* transfer to them. In particular Operator-A's read-out-selection reward (Table 3) is model-dependent: it measures how well a layer/pooling choice of g_θ supports a paralinguistic probe, and a high probe reward is evidence about g_θ's representation, not an external check on it. We therefore reserve the strong Goodhart-immunity claim for family (A) and treat family-(B) rewards (probe, retrieval) as informative-but-circular signals whose gains must be read with the model's own biases in mind.

4.3 OSA-2: Context Isolation Makes Gain Additive — but Buys No Enlargement Per Se

Theorems 2 and 3 are a diagnosis: a degenerate reward yields no gain. The natural next move is *context isolation* — factoring a monolithic decision into components decoded under separate, non-interfering contexts (distinct sub-agents, or skill/memory-conditioned roles), each carrying its own verifiable reward. Formally this replaces $\mathcal{Z}$ by a product and R by a separable sum. We prove that the gain is additive over such components, but we are careful to show, in the same breath, that isolation *per se* adds no optimization space at all.

Lemma 2 (Partition function factorizes; `Zpart_product`)**.** *Let $\mathcal{Z} = \mathcal{Z}_1 \times \mathcal{Z}_2$ with base $q_0 = q_0^{(1)} \otimes q_0^{(2)}$ and* separable *reward $R(z_1, z_2) = R_1(z_1) + R_2(z_2)$ (context isolation). Then $Z = Z^{(1)} Z^{(2)}$, where $Z^{(i)} = \sum_{z_i} q_0^{(i)}(z_i) e^{R_i(z_i)/\beta}$.*

Proof. A direct computation using independence and the multiplicativity of the exponential over the separable reward:

$$Z = \sum_{z_1, z_2} q_0^{(1)}(z_1) q_0^{(2)}(z_2)\, e^{(R_1(z_1)+R_2(z_2))/\beta} = \Big(\sum_{z_1} q_0^{(1)}(z_1) e^{R_1(z_1)/\beta}\Big)\Big(\sum_{z_2} q_0^{(2)}(z_2) e^{R_2(z_2)/\beta}\Big). \qquad \square$$

Theorem 4 (Gain is additive over isolated components; `gain_product`, OSA-2)**.** *Under the hypotheses of Lemma 2, $\mathcal{G} = \mathcal{G}(R_1) + \mathcal{G}(R_2)$, where $\mathcal{G}(R_i) = \beta \log Z^{(i)} - \mathbb{E}_{q_0^{(i)}}[R_i]$. Consequently, for k context-isolated components with product base and additively separable reward $R = \sum_{i=1}^k R_i$,*

$$\mathcal{G} = \sum_{i=1}^{k} \mathcal{G}(R_i).$$

Proof. By Proposition 2 and Lemma 2, $\beta \log Z = \beta \log Z^{(1)} + \beta \log Z^{(2)}$. Under the product base with separable reward, $\mathbb{E}_{q_0}[R] = \mathbb{E}_{q_0^{(1)}}[R_1] + \mathbb{E}_{q_0^{(2)}}[R_2]$. Subtracting term by term gives the two-factor claim; the k-fold statement follows by induction on the components. $\square$

Isolation buys no enlargement: the isolated optimum equals the monolithic optimum. The additivity of Theorem 4 must not be misread as "isolation enlarges the optimization space." It does not. The companion factorization `qstar_product` (stated and proved as Theorem 6 below) shows that the global

Gibbs optimum on the product reward factorizes exactly, $q^\star = q_1^\star \otimes q_2^\star$ — so the distribution reached by tilting each isolated, separately-rewarded block to its own local optimum is *identical* to the distribution reached by a single monolithic tilt of q_0 against the very same product reward $R = \sum_i R_i$. Both attain the same ceiling $F(q^\star) = \beta \log Z$ of Theorem 1, and Theorem 4 merely *decomposes* that one number into a sum of per-block contributions. Isolation *per se* therefore contributes *zero* additional optimization space: it is an accounting identity on a fixed objective, not an enlargement of it. The *only* source of additional gain is the introduction of *additional, non-degenerate, separable verifiable rewards* R_i — each new non-constant block adds its own strictly positive $\mathcal{G}(R_i)$, and nothing is gained by isolating a reward one already had. This is the precise sense in which OSA-2 is a statement about *adding rewards*, not about *partitioning models*.

The strict-positivity claim underlying "each new non-constant block adds gain" deserves to be a theorem in its own right, because it is what makes the k-fold sum non-vacuous.

Theorem 5 (Strictly positive gain for a non-degenerate reward; `gain_pos_of_nonconstant`)**.** *If R is non-constant on the support of q_0 (equivalently, on $\mathcal{Z}$, since $q_0 > 0$ everywhere), then $\mathcal{G} > 0$.*

Proof. By route (i) of Proposition 3, $\mathcal{G} = \beta\,\mathrm{KL}(q_0 \,\|\, q^\star)$ with $\beta > 0$. By the strict Gibbs inequality (`kl_pos_of_ne`), $\mathrm{KL}(q_0 \,\|\, q^\star) > 0$ whenever $q_0 \neq q^\star$. It remains to show $q_0 \neq q^\star$. If instead $q_0 = q^\star$, then for every z, $q_0(z) = q_0(z)e^{R(z)/\beta}/Z$, and since $q_0(z) > 0$ this forces $e^{R(z)/\beta} = Z$ for all z, i.e. $R(z) = \beta \log Z$ is constant on $\mathcal{Z}$ — contradicting non-constancy. Hence $q_0 \neq q^\star$ and $\mathcal{G} > 0$. (This is the strict converse of Theorem 2: constant $R \Leftrightarrow q^\star = q_0 \Leftrightarrow \mathcal{G} = 0$.) □

Theorem 5 is what makes the additive sum of Theorem 4 "grow in k" non-vacuously: *each* non-degenerate block contributes a strictly positive increment $\mathcal{G}(R_i) > 0$, so $k \mapsto \sum_{i \le k} \mathcal{G}(R_i)$ is strictly increasing along non-constant blocks. But the theorem also sharpens exactly what remains open. Strict positivity is a *conditional* statement — it presupposes that the block reward R_i is non-constant on the frozen model's own support. Whether a frozen speech model actually *affords* non-degenerate per-block rewards for a given factor is an *empirical* question about g_θ, not a theorem.

Conjecture 1 (Per-block non-degeneracy; empirical crux for Phase 2)**.** *Under the actual frozen-model context-isolation construction — decomposing a task across sub-agents and loading distinct skills/memories so that component i is decoded under an isolated context with its own verifiable reward R_i — there exists a constant $s_0 > 0$ such that the per-component reward spread satisfies $\mathrm{spread}_i \ge s_0$ on $\mathrm{supp}(q_0^{(i)})$ for each reward-bearing component. Equivalently, context isolation keeps each block non-degenerate (the hypothesis of Theorem 5), so that each $\mathcal{G}(R_i)$ is bounded away from zero and the additive gain of Theorem 4 is realized rather than vacuous. We state this as an* open conjecture*: it is the empirical crux on which the constructive reading of OSA-2 depends, and is precisely what the Phase 2 experiments are designed to test, not something the present theory establishes.*

Remark 5 (A standing counterexample for paralinguistic factors). Conjecture 1 is not idle: our own preliminary evidence already supplies a *standing counterexample* for the flagship's paralinguistic factors, so the conjecture is, for those factors, currently *contradicted*. (i) *Speaker.* The frozen-embedding speaker probe sits at ≈ 0.04 — near chance — at *every one* of the 37 hidden layers (Table 2, Section 6.4); the speaker block is degenerate, $\mathrm{spread}_i \approx 0$, and Theorem 5 gives no positive gain because its hypothesis fails. (ii) *Emotion.* The emotion read-out gain is fragile. The previously-reported single-seed $+0.097$ (Table 3) was the best seed under *oracle test-layer selection* — a winner's-curse / selection-leak artifact. A rigorous multi-seed re-run (dev-selected layer, paired-delta bootstrap, 5 seeds $\{42, 7, 123, 2024, 31337\}$) gives mean $\Delta = +0.037$ (range $[-0.053, +0.097]$), with the paired CI excluding 0 in only 2/5 seeds (committed artifact `projects/speech-mllm-omni-embedding-rl/_repro/emotion_pool_paired_v2.json`). Thus, for *content* and *intent*, frozen omni models plausibly do afford non-degenerate per-block rewards (Table 4), and the OSA-2 program is well-motivated; but for *speaker* and *emotion* the per-block non-degeneracy hypothesis is, on our own data, false or unsupported. The paper says this plainly: the agentic-decomposition argument is empirically backed for content/intent and is a *standing counterexample*, not a success, for paralinguistics.

With those qualifications in force, the intended — and explicitly conditional — picture is: where a flat reward affords no leverage, *adding* a non-degenerate, separable verifiable reward (e.g. a content/intent read-out, or a task-level selection reward) contributes strictly positive gain by Theorem 5, and isolating such rewards across context-isolated roles lets their gains add by Theorem 4 — *provided* Conjecture 1 holds for the factor in question, which our data supports for content/intent and contradicts for speaker/emotion. This is the conjectured mechanism by which agentic decomposition, retrieval-conditioned roles, and verifiable per-skill rewards [tul, 2024, jit, 2026] could convert a degenerate setting into a tractable one — where, and only where, the frozen model supplies the requisite non-degenerate blocks.

4.4 Reaching the Optimum: a Corollary and an Open Convergence Question

The two facts the previous subsections lean on — the static rollout deficit and the product factorization of the optimum — are corollaries of the master identity (Lemma 1) and the OSA-2 factorization (Lemma 2), respectively. We record them as such; neither is a convergence theorem, and we make no claim that they assemble into one. We deliberately avoid any "trichotomy," "stability-tax," or "third-pillar" framing: there are exactly two load-bearing results (OSA-1 on spread, OSA-2 on additivity-without-enlargement), and what follows is bookkeeping plus an honest statement of what remains open.

Corollary 1 (Rollout deficit; `rollout_deficit`)**.** *Every rollout policy q on $\mathcal{Z}$ satisfies*

$$F(q) \;=\; F(q^\star) \;-\; \beta\,\mathrm{KL}(q\,\|\,q^\star)\,, \qquad \beta\,\mathrm{KL}(q\,\|\,q^\star) \geq 0,$$

strictly positive unless $q = q^\star$. Hence a rollout that fails to land on $q^\star$ incurs a strictly positive, irreducible objective deficit.

Proof. This is Lemma 1 read as a statement about an arbitrary rollout q; strictness is the equality case of Theorem 1. ☐

Theorem 6 (Per-component tilting assembles the global optimum (static factorization); `qstar_product`)**.** *Under the isolated product structure of Lemma 2, the global Gibbs optimum factorizes:*

$$q^\star(z_1, z_2) \;=\; q_1^*(z_1)\,q_2^*(z_2), \qquad q_i^*(z_i) = \frac{q_0^{(i)}(z_i)\,e^{R_i(z_i)/\beta}}{Z^{(i)}}.$$

Therefore tilting each isolated component to its own local optimum q_i^* assembles exactly *the global optimum $q^\star$, which by Theorem 1 is $F(\cdot)$-optimal over all of $\Delta(\mathcal{Z})$. This is the factorization invoked in Section 4.3: it equates the isolated and monolithic optima on the product reward.*

Proof. Using $q_0 = q_0^{(1)} \otimes q_0^{(2)}$, the separable reward, and Lemma 2,

$$q^\star(z_1, z_2) = \frac{q_0^{(1)}(z_1)q_0^{(2)}(z_2)\,e^{R_1(z_1)/\beta}e^{R_2(z_2)/\beta}}{Z^{(1)}Z^{(2)}} = \frac{q_0^{(1)}(z_1)e^{R_1(z_1)/\beta}}{Z^{(1)}} \cdot \frac{q_0^{(2)}(z_2)e^{R_2(z_2)/\beta}}{Z^{(2)}} = q_1^*(z_1)\,q_2^*(z_2).$$

Global optimality is Theorem 1. ☐

Two caveats bound the reading of Theorem 6. First, it presupposes that credit has *already* been assigned: it requires a genuinely separable, per-component observable reward R_i on each block, and it does *not* solve the credit-assignment problem of inferring per-component advantages from a single scalar reward on the joint outcome. When only a scalar joint reward is available, the product structure is an assumption, not a consequence. Second — and this is the open question — both Corollary 1 and Theorem 6 are *static*: they describe the geometry of the fixed objective (a positive deficit off $q^\star$, a factorized optimum at $q^\star$), not the *trajectory* of any update rule. We read the $\beta\,\mathrm{KL}(q\,\|\,q^\star)$ term as a trust region penalizing drift from the tilted optimum, and we *conjecture*, by analogy with proximal/trust-region dynamics, that a large enough β keeps a retrieval-/advantage-tilting update — such as JitRL's no-gradient credit-assigned tilt [jit, 2026] — in the basin where per-component tilting decreases $\mathrm{KL}(\cdot\,\|\,q^\star)$, while a fast-drift rollout that ignores the trust region (appending its own unverified outputs as context and chasing them) is empirically observed to diverge through context collapse and error propagation [ref, 2023, ace, 2025, gao, 2022]. We stress that this link is an *analogy/conjecture*, not a theorem: a finite-time monotone convergence guarantee *remains open*, consistent with our Convergence section.

In sum: OSA-1 (Theorems 2 and 3 and Remark 3) proves that the gain is governed by the reward spread — a property of the reward, not of the model class. OSA-2 (Theorems 4 to 6) proves that context isolation makes gain *additive* while the isolated optimum *equals* the monolithic one, so isolation buys no enlargement and the only additional gain comes from introducing additional non-degenerate verifiable rewards, each strictly positive. Whether a frozen speech model affords such non-degenerate per-block rewards is the empirical Conjecture 1: supported by our data for content/intent, and — per Remark 5 — a standing counterexample for speaker and emotion. Convergence of any credit-assigned tilting dynamics is left open.

5 Convergence Analysis

A central claim of this work is that the inference-time degrees of freedom $\mathcal{Z}$ admit a sharp asymmetry: convergence is *provably* controllable at the output level, where the search distribution acts directly on a fixed generative operator, yet only *asymptotically* controllable at the agent level, where memory, skills, and context co-evolve with the policy. This section surveys and synthesizes what is established, separates proven finite-N guarantees from empirical or absent ones, and isolates the open problem that the operator-search architecture frames. Throughout we measure convergence against the Gibbs optimum $q^\star \propto q_0 \exp(R/\beta)$ of the KL-regularized objective $F(q) = \mathbb{E}_q[R] - \beta\, \mathrm{KL}(q \,\|\, q_0)$, whose maximizer is exactly $q^\star$ with optimal value $\beta \log Z$ [kor, 2022]. The question "does method M converge" is then made precise as: does the realized q approach $q^\star$ (or attain $F(q^\star)$) as the compute budget N grows, and at what rate?

5.1 Output-Level: Proven Finite-N Convergence

At the output level the action lives in $\mathcal{Z}_{\mathrm{B}}$ (and, for reranking, in $\mathcal{Z}_{\mathrm{A}}$): the generative operator is frozen and the only freedom is which candidates to draw, weight, or select. Here the literature provides genuine non-asymptotic guarantees.

Soft best-of-N tilts to $q^\star$ at rate $O(1/N)$. The cleanest result is for soft (regularized) best-of-N: drawing N i.i.d. samples from q_0 and resampling one with weights $\propto \exp(R/\beta)$ yields a distribution whose KL to the Gibbs optimum $q^\star$ decays as $O(1/N)$, so the induced q is a consistent, rate-quantified estimator of $q^\star$ [ver, 2025]. This is the tightest convergence statement available for any element of $\mathcal{Z}$ and serves as the reference behavior the rest of the architecture is measured against.

MBR and selection-style decoders. Minimum Bayes risk decoding converges to the risk-minimizing output at the Monte-Carlo rate $O(n^{-1/2})$ in the number of pseudo-references [ich, 2025], a guarantee that transfers directly to ASR, where MBR over hypothesis lists reduces word error in a way that is consistent rather than merely heuristic [mbr, 2025]. Hybrids that fuse MBR weighting with best-of-N selection inherit both the variance reduction of the former and the reward-alignment of the latter [jin, 2024], while self-consistency is the special case of majority-vote MBR under a 0/1 kernel [wan, 2022].

Hard best-of-N: KL budget and regret. Hard best-of-N does not target $q^\star$ but is nonetheless tightly characterized. Its KL to the base distribution is bounded by $\log N - (N-1)/N$ [bei, 2024], the inequality we formalize as Lean theorem T2; this caps how far selection can move the output distribution and hence bounds reward inflation. Complementarily, a smoothing-lens analysis shows the suboptimality (regret) of best-of-N against the optimal tilt grows only as $O(\sqrt{\log N})$ [ami, 2025], our Lean theorem T6. Together these bracket the operator: selection buys reward at a logarithmically growing KL price and a sub-polynomially growing regret, both finite and explicit. Related asymptotic-equivalence results connect best-of-N to KL-regularized RL in the large-N limit [yan, 2024, gui, 2024].

The over-optimization hump and the N^* controller. Convergence *toward* $q^\star$ is not the same as convergence *of measured reward toward true utility.* When R is a proxy, pushing N upward eventually over-optimizes: true performance rises, peaks, and declines, tracing a hump whose location scales predictably with the KL budget [gao, 2022]. This motivates treating N (equivalently the effective temperature) as a controlled variable. HedgeTune computes the optimal N^* — the best-of-Poisson count or tuning point that maximizes realized utility before the proxy diverges from truth — turning the budget into a closed-form set-point rather than a hyperparameter [kha, 2025]. The existence of a finite N^* is itself a convergence statement: more compute past N^* is provably counterproductive under a fixed proxy.

Compute allocation and verifier-guided variants. Several methods sharpen the constant rather than the rate. Guided speculative inference composes a small proposal with a large verifier to reach the tilted target at lower cost [geu, 2025]; CarBoN calibrates per-instance temperature to allocate samples where the reward landscape is informative [car, 2025]; certified self-consistency and confidence-guided escalation give stopping rules with coverage guarantees [cer, 2025, cge, 2025]; and test-time-compute-optimal scaling shows the budget allocation itself can be optimized [sne, 2024]. Sequential Monte-Carlo and Feynman–Kac steering extend the same tilting calculus to token-level resampling with consistency under standard SMC assumptions [zha, 2024, sin, 2025]. The common thread: every result in this subsection concerns a *fixed* operator acted on by a search distribution, which is precisely why finite-N analysis is tractable.

5.2 Agent-Level: Asymptotic Consistency Under a Trust Region

Moving from $\mathcal{Z}_\mathrm{B}$ to the full memory·skill·context space changes the operator itself between steps, and the guarantees weaken from finite-N to asymptotic.

JitRL: a closed-form KL-constrained update, consistent in probability. The strongest agent-level result is JitRL's no-gradient credit assignment [jit, 2026]. It derives the embedding update $z' = z + \beta\hat{A}$ as the closed-form solution of a KL-constrained objective, $\pi^* \propto \pi_\theta \exp(\beta A)$ [jit, 2026][2] — i.e. the same Gibbs tilt as Section 5.1, now applied in $\mathcal{Z}_\mathrm{A}$ to a frozen backbone. The advantage A is unknown and is estimated by a k-nearest-neighbor estimator $\hat{A}$; this estimator is consistent *in probability* only under a slow-policy-drift precondition: the neighborhood statistics must be drawn under a policy that changes slowly enough for the local estimate to remain valid. The resulting guarantee is therefore *asymptotic* (the realized policy approaches π^* as the sample pool grows under bounded drift), *not* a finite-time monotone-improvement statement. The trust region is implicit in the drift condition: violate it and the consistency argument lapses with no replacement bound. This is the central honest caveat — JitRL converges to the right object, but only in the limit and only inside a region the analysis assumes rather than enforces.

Tree search lacks transferable optimality. Agent planners that wrap a language model in Monte-Carlo tree search, such as LATS [lat, 2023], borrow the UCT selection rule whose regret guarantees were proven for stationary, bounded-reward bandits [koc]. Those guarantees do not transfer: the per-node reward here is a non-stationary, self-generated value estimate, the action set is unbounded, and the "environment" is the model's own context, violating the i.i.d. and stationarity premises UCT requires. Empirically LATS improves, but its convergence is not certified by the theory it invokes. Variance-reduced policy-gradient estimators such as RLOO [ahm, 2024, gre] face the analogous gap once the operator is non-stationary.

5.3 The Gap and the Failure Modes

The asymmetry is not merely a proof-technique artifact: naive rollout in the agent space is observed to *not* converge along every action axis, which is what makes the missing guarantee consequential.

Documented non-convergence across all axes. On the context/reflection axis, iterated self-refinement plateaus — Reflexion-style loops stop improving and oscillate after a few rounds [ref, 2023]. On the context-construction axis, accumulated context collapses, degrading rather than enriching the operator [ace, 2025]. On the memory axis, append-only experience logs propagate early errors forward, and added memory can actively hurt downstream accuracy [exp, 2025, rem, 2026]. These are not isolated anecdotes but a coherent

[2] Like several 2026-dated works we cite, jit [2026] is peer-reviewed (ICML 2026 Spotlight); we verified its stated closed-form KL-constrained result against the source.

Method	Convergence	Scope
Soft best-of-N [ver, 2025]	Proven, $O(1/N)$ to $q^\star$	Output ($\mathcal{Z}_\mathrm{B}$)
MBR decoding [ich, 2025, mbr, 2025]	Proven, $O(n^{-1/2})$	Output ($\mathcal{Z}_\mathrm{B}$)
Best-of-N KL bound [bei, 2024]	Proven (Lean T2)	Output ($\mathcal{Z}_\mathrm{B}$)
Best-of-N regret [ami, 2025]	Proven, $O(\sqrt{\log N})$ (Lean T6)	Output ($\mathcal{Z}_\mathrm{B}$)
N^* controller [kha, 2025]	Proven set-point	Output ($\mathcal{Z}_\mathrm{B}$)
Over-optimization hump [gao, 2022]	Proven scaling law	Output ($\mathcal{Z}_\mathrm{B}$)
GSI / CarBoN [geu, 2025, car, 2025]	Proven (target-consistent)	Output ($\mathcal{Z}_\mathrm{B}$)
Self-consistency [wan, 2022]	Proven (MBR special case)	Output ($\mathcal{Z}_\mathrm{B}$)
JitRL [jit, 2026]	Asymptotic, slow-drift precond.	Embedding ($\mathcal{Z}_\mathrm{A}$)
LATS / UCT [lat, 2023, koc]	None transferable	Agent (plan)
Reflexion loop [ref, 2023]	None (plateau)	Context
ACE context [ace, 2025]	None (collapse)	Context
Append-only memory [exp, 2025]	None (error propagation)	Memory
Memory accumulation [rem, 2026]	None (contamination)	Memory

Table 1: Convergence status of inference-time methods across the action space $\mathcal{Z}$. Finite-N guarantees exist only where the generative operator is frozen; agent-level methods are at best asymptotic under an assumed trust region, and naive rollout over memory/context is documented to diverge.

pattern: when the search distribution is allowed to rewrite the operator without a trust region, the very Gibbs-tilt logic that converges at the output level becomes a divergent feedback loop, because the reference q_0 drifts under the update. Proxy-reward exploitation compounds this [ska, 2022, gao, 2022].

What is and is not proven. Table 1 summarizes the landscape. Proven finite-N convergence exists *only* at the output level, where the operator is fixed. At the agent level the best available result is asymptotic consistency conditioned on slow drift; everything else is empirical or has no transferable guarantee. The open problem the operator-search architecture frames is therefore precise: a *finite-time, monotone-improvement* guarantee over the full $\mathcal{Z} = \mathcal{Z}_\mathrm{A} \cup \mathcal{Z}_\mathrm{B}$ space — a single trust region, analogous to the drift condition but *enforced* rather than assumed, under which each update to memory, skills, context, or decoding is certified not to decrease realized utility. The output-level results give the template (a Gibbs tilt with a KL budget and a tuned N^*); the agent-level results give the obstruction (a moving reference and a non-stationary advertised reward). Closing the gap means transporting the finite-N guarantees of Section 5.1 across an operator that is itself part of the action.

6 Feasibility Analysis

We now argue that the agent-level program of this work is not only well-posed but already *instantiable* from off-the-shelf *frozen* components, and that our own preliminary measurements both motivate and sharply *constrain* the design. The argument proceeds in four steps: we identify the two classes of frozen omni models with the two structural roles an episodic agent requires (Section 6.1); we show that a single verifiable-reward acceptance gate furnishes the control law shared by memory and skills (Section 6.2); we state an honest novelty delta against a crowded mechanism literature (Section 6.3); and we present in-house results that locate the operating regime empirically (Section 6.4).

We state at the outset the organising principle that the theory section makes precise and that our measurements bear out. The optimization gain of training-free RL—the KL-regularised tilting $q^\star \propto q_0 \exp(R/\beta)$—is governed by the reward *range* spread $= \max R - \min R$ over $\operatorname{supp}(q_0)$ (OSA-1, Lean-checked), whose leading-order discriminator is the reward variance $\operatorname{Var}_{q_0}(R)/(2\beta)$; the OSA-1 gain identity is $\mathcal{G} = \beta\,\mathrm{KL}((\,\|\,q_0)\,\|\,q^\star)$. This is a property of the *reward structure*, not of model class. A flat reward (e.g. instruction-conditioning of a frozen embedding read-out) gives vanishing gain $\mathcal{G}$; a high-range verifiable reward (e.g. task-level verifiable selection) gives large gain *even on a single frozen model*. Context-isolated composition makes gains *additive* across blocks (OSA-2, `gain_product`), and every non-degenerate block contributes *strictly positive* gain

(`gain_pos_of_nonconstant`: reward non-constant $\Rightarrow \mathcal{G} > 0$, sorry-free in Lean). But isolation *per se* adds nothing: `qstar_product` (the OSA-2 factorization) proves the context-isolated optimum *equals* the monolithic single-model optimum on the product reward $R = \sum_i R_i$. Consequently the high-range single-model gains we report below are *single-model* gains, not evidence that agentic decomposition recovers anything: the only *new* gain comes from introducing additional *non-degenerate* verifiable rewards, and whether such genuinely new per-block rewards exist for a given factor on a given frozen model is an *empirical* question (Conjecture 1, the per-block spread floor), untested at the agentic level here. Our preliminary results answer it affirmatively for content/intent on a single frozen model, and find no significant emotion gain on the one content bi-encoder we probed; we report both plainly.

6.1 The Two Omni Classes as Agent Components

The decisive enabling observation is that the contemporary omni-model ecosystem ships, off the shelf, the two computational primitives an episodic agent needs, and that both can be used *frozen*. On one side, *vector-omni bi-encoders* map an audio (or audio–text) input to a fixed embedding in a shared retrieval space [omn, 2025, 2026]; their natural role is the agent's *memory and index*, the read/write substrate over past episodes, as speech-native retrieval-augmented systems already use them [wav, 2025, spe, 2024, vox, 2025]. On the other side, *generative thinker–talker* omni models map an input to a token sequence under a decoding policy [qwe, 2025b,a, sal, 2023]; their natural role is the agent's *policy and skill executor*. Both classes are pretrained, publicly released, and, for our purposes, *frozen*: the action lives entirely in the inference-time degrees of freedom $\mathcal{Z}$, partitioned into the embedding/retrieval operator $\mathcal{Z}_{\mathrm{A}}$ realised by the bi-encoder and the generative operator $\mathcal{Z}_{\mathrm{B}}$ realised by the thinker–talker. This separation lets us treat memory construction (Operator-A) and skill execution (Operator-B) as two faces of a single search distribution q over $\mathcal{Z}$, optimised against the same reward R without touching weights or architecture.

We must, however, be honest about what one *specific* frozen omni *content bi-encoder* can and cannot index, because our own probes (Section 6.4) constrain it. On a single bi-encoder, `omni-embed-nemotron-3b`, the content axis is strong while the paralinguistic axes are not. Speaker identity is recoverable at *near chance* from this embedding at every layer (≈ 0.04 on a 91-way task, chance 0.011); since it is a *content* embedding, near-chance speaker recovery is close to definitional rather than surprising, and emotion is recoverable only weakly (with no significant gain at the across-seed level; see below). An index built solely on this omni *content* embedding would therefore retrieve on content, not on who-spoke or how-they-felt. We accordingly reframe the memory honestly: this frozen omni embedding is reserved for the *content* channel, while paralinguistic memory keys are supplied by classic, frozen, special-purpose speech encoders (an ECAPA-TDNN–style speaker embedder; a dedicated SER read-out). We do *not* generalise this negative to frozen omni models as a class, and we emphasise that it concerns a bi-encoder *probe*, not the generative Operator-B policy (which we do not test for paralinguistics). Feasibility thus reduces to whether a *control law* can compose a frozen omni policy with a heterogeneous frozen-encoder memory productively—not to whether the parts can be built, and not to a claim that one omni embedding carries every factor.

6.2 The Verifiable-Reward Acceptance Gate as the Control Law

The control law is a verifiable-reward *acceptance gate*: a candidate memory write or skill invocation is admitted only if it improves a verifiable reward on held-out probes, and is otherwise rejected. This single mechanism governs both faces of $\mathcal{Z}$. The case for making the gate load-bearing is empirical and, in our reading, strong. It rests substantially on ski [2026b], whose statistics we verified against the source (an average of +16.2 points, ranging +4.5 to +51.9, with 16/84 skills negative), and we triangulate it against the broader agent-stability literature below. SkillsBench reports that *curated* skills raise agent success by +16.2 percentage points, whereas *self-generated* skills contribute essentially zero in aggregate, with 16 of 84 skills measurably *negative* [ski, 2026b]. The lesson is not that self-evolution is futile but that unfiltered self-evolution is: without an acceptance test, an append-only memory or skill library accrues as much harm as benefit, a failure mode echoed across the agent-stability literature [exp, 2025, ace, 2025, rem, 2026, ref, 2023]. The gate is precisely the missing filter. When ground-truth labels are scarce, a surrogate verifier can stand in for the reward without collapsing the guarantee, as co-evolved skill–verifier pairs demonstrate [coe, 2026]; and for speech specifically the verifiable rewards already exist, from paralinguistic-aware speech reward models [gsr, 2026, par, 2025] to the WER/exact-match/intent rewards reused throughout this series. We therefore adopt the gate as the one control law shared by $\mathcal{Z}_{\mathrm{A}}$ (admit a memory entry) and $\mathcal{Z}_{\mathrm{B}}$ (admit a skill or decoding action), with the surrogate verifier as the fallback when labelled probes run out. This is the same KL-regularised tilting that underwrites best-of-N and reward-guided decoding [kor, 2022, mud, 2023, bei, 2024]: maximising $F(()\,q) = \mathbb{E}_q[R] - \beta\,\mathrm{KL}((\,\|\,q)\,\|\,q_0)$ with optimum $q^\star \propto q_0\,\exp(R/\beta)$, here lifted from token decisions to episode-level admission decisions. Its gain is exactly the reward-range/variance quantity of Section 6: a high-range admission reward makes the gate productive; a flat one makes it inert.

6.3 Novelty Delta

We state the contribution honestly. The *mechanism*—self-evolving episodic memory plus a growing skill library refined by feedback—is crowded: reflective self-improvement [ref, 2023, exp, 2023], library-building agents [voy, 2023, awm, 2024, mus, 2026], and gradient-free inference-time credit assignment [jit, 2026] all occupy adjacent ground. Our claim is not the mechanism in the abstract. The unfilled slot is the specific composition: a *frozen omni policy* (Operator-B) and a *frozen-encoder memory* (Operator-A) driven by *episodic self-evolution* under a *verifiable speech reward*, with no weight updates anywhere in the loop. We are explicit that this memory is heterogeneous and *not* a single-vector omni store: as established above and measured below, the one omni content bi-encoder we probed is at chance on speaker and weak on emotion, so we use it for the content channel only and delegate paralinguistic keys to classic frozen speech encoders. The closest priors fall short on the all-frozen axis. Trainable instruction embedders such as BGE-EN-ICL require gradient training to acquire their in-context behaviour [bge, 2024], violating the frozen-memory constraint; few-shot–emergent audio models such as MiMo-Audio obtain their capability through pretraining rather than through inference-time activation of an already-frozen model [mim, 2025]. Neither realises the all-frozen, reward-gated, speech-verifiable agent loop. To our knowledge, this composition has not been demonstrated, and we make the corresponding "first, to our knowledge" claim while being explicit that each ingredient in isolation has prior art—and that the load-bearing novelty is the verifiable-reward composition, not any claim that a single omni embedding affords every factor.

6.4 Preliminary Empirical Evidence

Our in-house measurements locate the operating regime along the single axis the theory identifies: the reward range spread of each (action-space, reward) pair. We organise the evidence by *channel*. A **content/intent "read" channel**—task-level verifiable selection over a single frozen model—exhibits high range and large gains. A **paralinguistic "listen" channel**—emotion and speaker read-out from one frozen content bi-encoder via a linear/kNN probe—exhibits low range; for these factors, on this bi-encoder, our probe evidence does *not* support a per-block spread floor. All numbers below are our own preliminary results on a frozen `omni-embed-nemotron-3b` with kNN probes and bootstrap confidence intervals (CREMA-D dev600/test300 and MInDS-14 dev280/test257 for the probe channel; SLURP n=500, MInDS-14 n=180, URO n=200 for the Operator-B channel); they are reported verbatim and labelled preliminary. We flag a contamination caveat: CREMA-D content is a 12-sentence closed read set, so near-ceiling content accuracy (≈ 1.00) may partly reflect memorisation of fixed transcripts, and SLURP, MInDS-14 and URO are public benchmarks that may overlap pretraining corpora; the single-model gains below should be read with that risk in mind. We stress at the outset that every Operator-B number is a *single-model* result; by `qstar_product` a context-isolated decomposition would reach the same optimum, so none of these gains is evidence of agentic recovery.

Single-model conditioning: no effect larger than the detectable minimum. Table 2 reports test probe accuracy for the frozen embedding under four instruction-conditioning regimes, crossed against three read-out factors. The same frozen embedding simultaneously supports near-perfect content read-out (≈ 1.00), weak emotion read-out (≈ 0.36), and near-chance speaker read-out (≈ 0.04; chance on this 91-way task is 0.011). Instruction conditioning does not measurably move the factor accuracies: the emotion delta from baseline to emotion-instruction is $+0.007$ with overlapping CIs, the speaker delta is -0.003, and the conditioning matrix is not diagonally dominant. We report this with the honesty its statistical power warrants.

Table 2: In-house preliminary results. CREMA-D conditioning×factor TEST probe accuracy (frozen `omni-embed-nemotron-3b`, seed 42). Instruction conditioning produces no effect larger than the MDE at $n = 300$ (overlapping CIs, `diagonal_dominant`=False)—an underpowered null, not a proven equivalence. Single-model only.

Conditioning	Content	Emotion	Speaker
Baseline	0.997	0.360	0.040
Content-instruction	1.000	0.333	0.043
Emotion-instruction	1.000	0.367	0.037
Speaker-instruction	1.000	0.367	0.033
Chance	0.08	0.17	0.011

At $n = 300$ the study is *underpowered*; we therefore report it as *no effect larger than the minimum detectable effect (MDE) at* $n = 300$, not as a positive "OSA-1 signature" and not as a proof that conditioning is exactly inert. A sweep over all 37 hidden states and two pooling read-outs leaves speaker near chance at *every* layer (full-sequence best 0.043 at L36; audio-token best 0.043 at L32), with content saturating early (full-seq 0.990 at L24) and emotion peaking only weakly (full-seq 0.400 at L0). Whatever range instruction-conditioning offers is below what this probe can resolve; we draw no stronger conclusion.

Paralinguistic "listen" channel: no significant emotion gain (null) and near-chance speaker. For the flagship's paralinguistic factors the per-block range the agent would need is small, and on this content bi-encoder it is statistically null. Choosing the read-out itself—an Operator-A ($\mathcal{Z}_{\mathrm{A}}$) action over layer and pooling—can extract an emotion gain on *some* splits, but a rigorous multi-seed re-run shows no significant gain at the across-seed level, and it forces us to retract an earlier over-claim. We previously reported a single-seed +0.097 (attentive-statistics pooling at L16 versus mean-pooling at L0). That figure was the *best seed under oracle TEST-layer selection*—a winner's-curse / selection leak, since the test layer was chosen on the test set. We have since corrected the protocol on three counts: (i) *dev*-layer selection (both the base mean-pool layer and the selected attentive-stats layer are chosen on dev and frozen before test evaluation); (ii) a *paired*-delta bootstrap 95% CI computed directly on the per-example improvement; and (iii) five seeds $\{42, 7, 123, 2024, 31337\}$. Table 3 reports the result: across the five seeds the mean improvement is $\Delta = +0.037$ with an across-seed 95% t-CI of $[-0.043, +0.116]$, which *spans zero*. The headline reading is therefore *no significant emotion gain at the across-seed level (a null)*, and the previously reported +0.097 is reproduced only as the best of five seeds under the corrected protocol.[3] Speaker read-out is worse: it stays near chance at all 37 layers (≈ 0.04, chance 0.011), which for a content embedding is near-definitional, so this embedding encodes essentially no recoverable speaker identity. For paralinguistics, then, our *probe* evidence on this one content bi-encoder does *not* support a non-degenerate per-block reward: Conjecture 1 is not supported here. We stress the scope: this is a linear/kNN probe of a frozen content bi-encoder; it does *not* bound the proposed generative Operator-B policy, which we have not tested for paralinguistics. Committed artifact: `projects/speech-mllm-omni-embedding-rl/_repro/emotion_pool_paired_v2.json`; reproduce with `PP_SEEDS=42,7,123,2024,31337 python scripts/pool_method_probe_paired.py`.

Content/intent "read" channel: high range, large single-model gains. The contrast is task-level Operator-B ($\mathcal{Z}_{\mathrm{B}}$) selection—choosing among generative candidates of a *single frozen model* under a verifiable reward, the best-of-N training-free-RL move of this series. Here the reward range is large and the gains are well separated *even on a single frozen model* (Table 4); by `qstar_product` a context-isolated decomposition would not improve on this single-model optimum, so we read these as single-model gains, not agentic recovery. We give the exact selection rule and slice for each pair. On **SLURP intent** (n=500 test slice), the baseline is a basic-label intent retriever (raw audio query against the bare intent-name text) at 0.550; admitting boundary-card candidate schemas (name + description + examples + boundary notes) under the verifiable

[3] For completeness: the per-seed paired-delta CI excludes 0 in 2 of the 5 seeds (and two seeds are nominally negative). This 2/5 vote count is not the headline; the across-seed t-CI above is the appropriate aggregate, and it is a null.

Table 3: In-house preliminary results, *corrected protocol*. CREMA-D emotion: mean-pool baseline vs attentive-statistics pooling, both layers *dev*-selected, paired-delta bootstrap 95% CI on TEST, five seeds. The earlier single-seed +0.097 was the best seed under *oracle test-layer* selection; under dev-selection the across-seed 95% t-CI is $[-0.043, +0.116]$, which spans zero: no significant emotion gain at the across-seed level (a null). Paralinguistic listen channel on one content bi-encoder; Conjecture 1 not supported here.

Seed	base (dev-sel)	sel (dev-sel)	acc_{base}	acc_{sel}	Δ	paired 95% CI
42	audio-mean L0	audio_attn L16	0.400	0.497	+0.097	$[+0.027, +0.167]$
7	audio-mean L0	audio_attn L0	0.423	0.370	−0.053	$[-0.117, +0.010]$
123	audio-mean L0	audio_attn L16	0.390	0.473	+0.083	$[+0.013, +0.150]$
2024	audio-mean L0	audio_attn L16	0.493	0.487	−0.007	$[-0.077, +0.063]$
31337	audio-mean L0	audio_attn L16	0.400	0.463	+0.063	$[-0.010, +0.130]$
mean	—	—	—	—	+0.037	across-seed 95% t-CI $[-0.043, +0.116]$ (spans 0; nu

Table 4: In-house preliminary results. Content/intent read channel: single-model Operator-B ($\mathcal{Z}_{\text{B}}$) best-of-N training-free-RL gains (seed 42, paired bootstrap). Task-level verifiable selection within one frozen model yields large, well-separated improvements, in contrast to the flat conditioning of Table 2 and the null paralinguistic gain of Table 3. One canonical baseline→after pair per task under a fixed selection rule (SLURP/MInDS: basic-label → boundary-card schema; URO: raw target text → boundary-card + raw query); the MInDS basic→tool-specific variant and the `docs/theory.md` SLURP 0.522 → 0.894 pipeline (with a later 0.894 → 0.880 regression) are noted as alternative configurations, not re-run here and not canonical. The conservative-rerank "0 regressions" is a structural property of an improve-only gate (definitional). Single-model only.

Task (n)	Baseline	After Op-B	Δ (CI)	Note
SLURP intent (500)	0.550	0.880	+0.330 [0.288,0.374]	canonical
MInDS-14 intent (180)	0.852	0.984	+0.132 [0.082,0.187]	canonical
alt. config (basic→tool-spec.)	0.883	0.972	+0.089 (no CI)	alternative config
URO spoken-QA (200)	0.380	0.715	+0.335 [0.265,0.405]	canonical
+ conservative rerank (τ=0.02)	0.715	0.845	+0.130	improve-only (0 regr. by constr.)

intent-match reward raises this to 0.880 (+0.330, paired-bootstrap CI $[0.288, 0.374]$). On **MInDS-14 intent** (n=180 slice), our canonical pair applies the same basic→boundary-card selection rule with the policy schema, improving 0.852 → 0.984 (+0.132, CI $[0.082, 0.187]$); an *alternative configuration* (basic-label → tool-specific intent on the same frozen model) gives 0.883 → 0.972 (+0.089) but was not re-run under the paired bootstrap and carries no CI, so we report it only as an alternative, not as the canonical MInDS result. On **URO spoken-QA** (n=200 slice), the baseline is raw target-text candidates at 0.380; substituting target boundary-card candidates with the raw query (policy grounding) raises this to 0.715 (+0.335, CI $[0.265, 0.405]$). A subsequent conservative low-margin rerank ($\tau = 0.02$) is an *improve-only* selector by construction—it admits a re-ranking only on a strict verified improvement—so its "0 regressions" is a structural (definitional) property of the gate, not an empirical robustness finding; it accepts 26 fixes over the 200-item slice and lifts 0.715 → 0.845 (net +0.130), which we present as such, with no separate robustness claim. We reconcile the provenance honestly. The project's own `docs/theory.md` reports a *different* boundary-card pipeline for SLURP (0.522 → 0.894, +0.372, which subsequently showed a 0.894 → 0.880 regression under a later change) together with several MInDS variants; we treat those as *alternative configurations / pipelines not re-run here* and report *one canonical baseline→after pair per task* under the fixed selection rules above. We do *not* present a single clean +0.330 as if it were the unique result. What these numbers establish—and all they establish—is that a high-range verifiable reward yields large gain on a single frozen model, exactly as OSA-1 predicts. We emphasise that this high range is demonstrated for **content/intent only**.

Reading the evidence. The picture is consistent and, in one direction, humbling. The optimization gain tracks the reward range spread of the (action-space, reward) pair, *not* the model class: a single frozen model already affords large gains under a high-range verifiable reward (content/intent: +0.330 SLURP, +0.132 MInDS, +0.335 URO), while a flat reward (instruction-conditioning) affords none we can detect at $n = 300$. For content and intent this supports composing verifiable per-block rewards across operators and episodes: by `gain_pos_of_nonconstant` each non-degenerate block adds strictly positive gain, and by `gain_product` these add. But the same theory is honest about its precondition: `qstar_product` shows that context-isolation *per se* buys nothing—the context-isolated optimum equals the monolithic single-model optimum—so the content/intent gains above are single-model gains, and whether *agentic decomposition* adds anything beyond them can only be settled by introducing genuinely new non-degenerate per-block rewards, which we have not tested. For the flagship's *paralinguistic* factors our probe evidence does not deliver a non-degenerate per-block reward on the one content bi-encoder we examined—the across-seed emotion gain is a null (mean $\Delta = +0.037$, 95% t-CI $[-0.043, +0.116]$, spans 0) and speaker is near chance at every layer—so Conjecture 1 is not supported there. This is a statement about a frozen content bi-encoder under a linear/kNN probe; it does *not* bound the untested generative Operator-B policy and does *not* generalise to frozen omni models as a class. We therefore pursue the content/intent channel as a single-model best-of-N result, reserve the agentic-decomposition question for future genuinely-new rewards, and explicitly decline a paralinguistic per-block claim, where this omni content embedding does not afford a load-bearing key and an index built on it would not perform load-bearing paralinguistic retrieval. We report all intervals, label these numbers preliminary, and state the negative result as plainly as the positive one: the design is motivated where the range is high and constrained where it is not.

7 A Speech-Agent Testbed for Falsifying the Paralinguistic Loop

We now instantiate the theory of Section 4 as a concrete, buildable agent. We are deliberate, up front, about its epistemic status: this is *not* a flagship system we claim is worth deploying now. Its novel axis—a cross-session *paralinguistic* memory loop—has, at present, *no specified mechanism* that we can show beats the strong single-enrollment ECAPA-TDNN + SER baseline, and our own Phase-2 plan (Section 8) *pre-registers a null* on exactly that axis. We therefore present the system as a **speech-agent testbed and falsification harness**: its primary deliverables are (a) the honest *negative* result on cross-session paralinguistic memory and (b) the benchmark and harness that make that result reproducible and that any future positive mechanism must clear. We build it not because we expect it to win, but because the negative is itself the contribution and the apparatus is reusable.

The design is governed by a single discipline: every component that *evolves* at inference time does so only through an acceptance gate driven by a reward signal and operating inside a β-KL trust region, never through a gradient on the base model. This is what distinguishes our agent from the prevailing self-evolving paradigm, whose append-only memories and self-generated skill libraries are documented to drift toward context collapse and error propagation [ace, 2025, exp, 2025, rem, 2026]. We design *against* that failure mode by realizing the optimization geometry of $q^\star$ (the Gibbs optimum of Section 4) as an external, mutable state whose update operator is *engineered to be contractive*. We stress at the outset that contractivity here is a design target, not a theorem: as Section 5 makes explicit, finite-time monotone-improvement over the full agent state is the open problem this work frames, and the design principles below are our attempt to approach it—not a certificate that they attain it. We are equally explicit about a second, empirical limit. The theory of Section 4 ties the realizable gain of any block to the *reward spread* of its (action-space, reward) pair—the range $\max R - \min R$ over $\mathrm{supp}(q_0)$, with a leading-order refinement $\mathcal{G} \approx \mathrm{Var}_{q_0}(R)/(2\beta)$ for a concentrated q_0 (Remark 3)—and by `gain_pos_of_nonconstant` a block contributes strictly positive gain iff its reward is non-constant. Whether the frozen omni stack affords a non-degenerate verifiable reward in each block is therefore not settled by the architecture; it is Conjecture 1 of Section 4, and our own evidence (Section 7.2) shows it *holds* for content/intent but *fails for speaker and shows no significant gain (a null) for emotion.* The system below is built to exploit the gain where it provably exists and to fail safe—degrading to an honest surrogate, never a false certificate—where it does not.

One concrete mechanism that could win—and the prior art that already achieves most of it. A testbed is only honest if it states what a *win* would look like, how it would be attributed, and—crucially—what prior art already delivers it, so that "win" is measured against the right baseline rather than a strawman. We commit to one falsifiable candidate, and we are blunt that its posterior-integration core is *not* novel. The strong baseline—single-enrollment ECAPA-TDNN with an SER head—commits to a *single* speaker template per identity and reads each turn against it. Its documented weakness is *genuine channel variation*: a speaker recorded across different devices, rooms, codecs, and health/affect states drifts away from a one-shot enrollment centroid. The obvious fix—accumulate a *posterior* over a speaker's embedding across many sessions under an explicit within-speaker channel-covariance model, and decide identity by that multi-session posterior rather than by proximity to one enrollment vector—is, up to naming, **exactly classical PLDA scoring with multi-enrollment x-vector/ECAPA speaker verification and adaptive score normalization (AS-Norm)** [xve, 2021, eca, 2020]. PLDA already factors an embedding into speaker and channel subspaces and already returns a multi-enrollment posterior; AS-Norm already integrates a cohort of enrollments to normalize the score. So "Bayesian multi-session speaker-evidence integration" is, in substance, a rediscovery of textbook speaker-verification machinery—and a one-line PLDA-over-all-enrollments baseline gets it for free. We make that baseline a first-class arm rather than pretend it does not exist.

What an agentic store would add *over* PLDA—and no more. The honest question is therefore not "does multi-session integration help" (PLDA already answers yes) but "does an evolving memory *agent* add anything *over* PLDA-over-all-enrollments." We claim the possible surplus is narrow and specific, and it is *not* the posterior integration itself: (a) *graph-structured multi-session curation*—selecting, deduplicating, and provenance-tracking *which* sessions enter the pool, rather than pooling every enrollment uniformly as vanilla PLDA does; (b) *decay / forgetting*—down-weighting or retiring stale sessions on a schedule; (c) a *trust region on writes*—the β-KL bound of Section 7.2 on how far each admission may move the store; and (d) *change-point handling*—detecting when a speaker's channel or affect has *shifted* rather than folding the shift into a stationary posterior. The posterior integration is conceded to PLDA/AS-Norm; any agentic gain must come from (a)–(d), and the ablation below is built to attribute it there and nowhere else.

Splitting a stable trait from a volatile state. There is a second, sharper reason the posterior-accumulation framing cannot be applied uniformly to the candidate: it silently conflates a *stable trait* (speaker identity) with a *volatile state* (affect). Identity is well modeled as a fixed latent to be estimated ever more sharply as sessions accumulate. Affect is not: a speaker's emotional state is *expected to change* between sessions, and an operator that accumulates toward a stationary per-speaker affect posterior would structurally *erase* precisely the change signal our benchmark's update/delete probe is built to detect ("is the caller more distressed than in session 1?"). Accumulate-toward-a-stable-posterior is therefore the *wrong* operator for affect. We split the mechanism into two operators with opposite inductive biases:

- **Identity (stable trait) $\Rightarrow$ integrate toward a stable posterior.** Estimate a fixed per-speaker trait posterior that sharpens as sessions accumulate—the PLDA half. This is where we *expect no agentic gain* beyond the curation/decay/trust-region bookkeeping of (a)–(c).
- **Affect (volatile state) $\Rightarrow$ track, don't average.** Model affect as a *time-varying* state: a per-session affect baseline *plus* an explicit change-point / drift detector, so that "affect changed since session k" is a *first-class output* of the store rather than variance to be averaged away. Here the store must report change, not converge to a point.

The affect half is the *only* place a genuinely agentic (beyond-PLDA) capability could live: PLDA has no notion of a per-speaker affect timeline or of flagging a change-point, and single-enrollment SER has no memory of prior sessions at all. If the paralinguistic loop adds anything the classic baselines cannot, that is where it must appear. Consistent with our pre-registered null and the near-flat emotion spread of Section 7.2, we do not expect it to; but change-point detection over an affect timeline is the one axis we cannot yet rule out, and it is exactly what the update/delete probe is built to test.

The ablation that isolates it. To *isolate* any agentic contribution from both encoder quality *and* from the PLDA posterior, the pre-registered ablation holds the front end fixed and varies only the memory representation across *three* arms, at matched compute and with the *same speaker/affect encoders throughout*: (i) *single-enrollment* ECAPA + SER (the original strong baseline); (ii) *PLDA-over-all-enrollments* with AS-Norm for identity and a per-session baseline for affect—the "free" multi-session posterior; and (iii) the *curated multi-session graph*—(ii) plus graph curation, decay, a β-KL write trust region, and an explicit affect change-point detector. A win is a strict improvement of (iii) over *(ii)*—not merely over (i)—on cross-session, cross-channel speaker verification/identification *and* on the affect-change probe, with matched compute. Measuring (iii) against (i) alone would credit the agent for the PLDA posterior it did not invent; the load-bearing comparison is (iii) vs. (ii). We are honest that this is a *hypothesis to be tested*, not a result: absent a positive (iii)-over-(ii) ablation, the system reduces to **PLDA-grade multi-session scoring plus memory bookkeeping over the very same encoders the baseline already reads**, and offers no paralinguistic capability those baselines lack. That reduction—and the negative result it implies—is an acceptable outcome for a testbed; what would be unacceptable is to assert the win against the wrong (single-enrollment) baseline and thereby bank the PLDA posterior as if it were the agent's own.

7.1 Architecture

The agent couples two *frozen* omni models with one external evolving state. A frozen *vector-omni* encoder [omn, 2025, 2026] serves as the memory and skill index: it maps raw spoken turns into the embedding action space $\mathcal{Z}_\mathrm{A}$ of Section 4, where retrieval is a tilting of the base index distribution q_0. A frozen *generative omni* policy [qwe, 2025b] serves as the response model, whose decoding occupies the generative action space $\mathcal{Z}_\mathrm{B}$. Neither model is fine-tuned; both are queried only. All adaptation lives in an external state S—a structured memory store and a skill library—which is mutated under a mixture of *verifiable* speech rewards and *surrogate* self-consistency signals (detailed below). Formally, the agent's inference-time degrees of freedom $\mathcal{Z} = \mathcal{Z}_\mathrm{A} \times \mathcal{Z}_\mathrm{B}$ are conditioned on S, and learning is the trajectory $S_0 \to S_1 \to \cdots$ of *state* updates rather than weight updates. Because S is the only mutable object, the convergence question reduces to the contractivity of the state-update operator—which we engineer directly, while acknowledging (Section 5) that the resulting trajectory is at best asymptotically controllable and is not yet certified to converge in finite time.

An honest positioning: omni-policy with classic-encoder memory. It is tempting to describe this system as a "two-omni" agent in which a single frozen omni embedding is simultaneously the content index and the paralinguistic memory. Our probing evidence forbids that description. The frozen vector-omni *content* embedding (a single content bi-encoder, `omni-embed`) is at or near chance on speaker identity *as a retrieval key*—a layer-wise linear/kNN speaker probe reaches accuracy ≈ 0.04 at *all* 37 encoder layers, indistinguishable from chance—and yields no significant emotion gain at the across-seed level (a null; quantified in Section 7.2). The omni embedding therefore does *not* do load-bearing paralinguistic retrieval. We scope this negative precisely: it is a statement about *this content embedding read by a linear/kNN probe as a memory key*, not about the generative Operator-B policy (never tested for paralinguistics) and not about "frozen omni models" as a class. We position the system accordingly and without euphemism: it is an **omni-policy paired with a classic-speech-encoder memory**. The omni embedding is the *content* key; speaker and emotion retrieval route through dedicated frozen encoders (ECAPA-TDNN / x-vector for speaker, a dedicated SER head for emotion). The "vector-omni memory" is load-bearing for content/intent and *not* load-bearing for the paralinguistic benchmark. Whether the omni embedding adds *any* paralinguistic signal beyond the classic encoders is an open empirical question, and we pose it as a falsifiable joint-retrieval ablation (Section 7.2) rather than assuming the affirmative.

Which rewards are verifiable. We hold to the paper's definition: a reward is *verifiable* only if it is computable from a ground-truth annotation without a learned scorer. Under this definition the agent's genuinely verifiable rewards are: **WER/CER** for ASR (against reference transcripts), **exact-match** for intent/slot tasks (against gold labels), and **speaker-ID accuracy** *only on corpora that carry ground-truth speaker labels and only on oracle (single-speaker) segments*. The speaker-ID caveat is sharper than it first appears: SID is verifiable against gold labels only when the segment boundaries are given. Under *automatic* diarization the same accuracy number is contaminated by diarization error—a missed or merged turn relabels the audio before the speaker head ever runs—so what looks like a verifiable SID reward is in fact an SID×DER product. We therefore treat automatically-diarized SID as *not* verifiable in the strict sense and, wherever we report SID under automatic diarization, we report it *jointly with the diarization error rate (DER)* so the reader can see how much of the signal is speaker accuracy and how much is segmentation luck. Paralinguistic agreement signals (speaker/emotion consistency against stored estimates) do *not* meet the verifiability bar at all and are treated separately as surrogates (Section 7.2). The architecture foregrounds the verifiable channel: wherever a ground-truth annotation exists, the acceptance gate uses it; surrogates are used only to fill gaps and are explicitly flagged as Goodhart-prone.

The deployment-time gap. The verifiable channel above exists at *training/evaluation* time, where labeled corpora are available. The testbed's distinguishing loop, however—the novel axis under test—is a *cross-session paralinguistic* loop that runs at deployment: it must decide, with no oracle present, whether a new spoken turn matches a remembered speaker/affect and whether to write it to memory. At deployment there are *no* ground-truth speaker or emotion labels. Consequently the admission reward for the cross-session paralinguistic loop cannot be verifiable; it necessarily degenerates to a *self-consistency surrogate*—agreement between two passes of the frozen extractors. We make this degeneration explicit rather than papering over it, and we name the gate honestly in Section 7.2: it is a self-consistency gate there, not a verifiable-reward gate.

Why two frozen models. Separating the index ($\mathcal{Z}_\mathrm{A}$) from the policy ($\mathcal{Z}_\mathrm{B}$) lets each operator be tilted by an independent reward channel: memory writes are gated by retrieval-utility signals, while response decoding is gated by task rewards. This factorization mirrors the Operator-A / Operator-B decomposition of Section 4 and lets us bound the two mutation rates separately (Section 7.4).

7.2 Memory

Three tiers over a bi-temporal graph. Memory is a structured store with three tiers: *episodic* (individual spoken turns), *semantic* (per-speaker persona and stable facts), and *procedural* (pointers into the skill library of Section 7.3). The tiers sit on a bi-temporal knowledge graph that records both event time and ingestion time [ari, 2024, zep, 2025, ame, 2025], so that retrieval can respect causality and the curator can reason about staleness. Episodic nodes consolidate into semantic nodes off the critical path.

Index on audio, not transcripts. A central design choice is that the retrieval key is the frozen *audio* embedding from the vector-omni encoder, not an ASR transcript [wav, 2025, vox, 2025]. Transcribing before indexing discards exactly the paralinguistic structure (prosody, speaker identity, affect) that the agent must remember, and it injects ASR error into the key. Indexing on $\mathcal{Z}_\mathrm{A}$ keeps the content key in the acoustic space. We are careful, however, not to overload it: as the next paragraph quantifies, the omni embedding earns its place as a *content* key only.

Decoupled retrieval keys, and the negative evidence that forces the decoupling. The decision to route paralinguistic retrieval away from the omni content embedding is not a stylistic preference; it is forced by our own measurements, and we report them with their failure modes intact. **Speaker.** A layer-wise linear/kNN speaker probe on the frozen vector-omni encoder reaches accuracy ≈ 0.04 at every one of the 37 layers—statistically indistinguishable from chance for the probe's label set. The content embedding carries essentially no recoverable speaker identity. **Emotion.** The emotion readout shows *no significant gain at the across-seed level*. Our first report of a $+0.097$ accuracy gain from reward-guided layer pooling was, on audit, the *best seed under oracle test-layer selection*—the pooling layer was chosen on the test split, a textbook winner's-curse / selection leak. A rigorous re-run removes the leak: the pooling layer is selected on a dev split, the effect is measured as a paired per-utterance delta, and the whole procedure is repeated over five seeds $(42, 7, 123, 2024, 31337)$. The honest headline is the *across-seed* result: mean $\Delta = +0.037$ with a 95% t-confidence interval of $[-0.043, +0.116]$, which *spans zero*—so by its own across-seed CI the emotion effect is a *null*.[4] The committed artifact

[4] A finer per-seed view (vote count) is consistent: the within-seed paired bootstrap CI excludes zero in only 2 of the 5 seeds. We report this only as a footnote; the load-bearing statement is the across-seed t-CI above, which does not establish a non-zero effect.

is `projects/speech-mllm-omni-embedding-rl/_repro/emotion_pool_paired_v2.json`. We read this plainly: for the paralinguistic factors, Conjecture 1 of Section 4 (that the frozen omni embedding affords a *non-degenerate* verifiable reward per block) *fails for speaker and is not significantly supported for emotion.* By `gain_pos_of_nonconstant`, a block with a (near-)constant reward yields (near-)zero gain. We add a theory-side caution against over-reading the content/intent successes too: by `qstar_product` (Section 4.3) the context-isolated ("agentic") optimum *equals* the monolithic single-model optimum on the same product reward, so the high-spread content/intent gains are *single-model* gains, not evidence that agentic decomposition recovers anything per se; whether decomposition adds value could only be shown by introducing genuinely new, non-degenerate rewards—untested here.

Consequences for the design. We therefore do *not* route paralinguistic retrieval through the content embedding. The composite key is built from *dedicated* frozen representations: a speaker subkey from ECAPA-TDNN / x-vector embeddings [eca, 2020, xve, 2021], an emotion subkey from a dedicated SER embedding, and a content subkey from the omni content embedding. Paralinguistic queries retrieve on the speaker and SER subspaces; content queries retrieve on the content subspace; the content embedding is reserved for what it is demonstrably good at. Because paralinguistic retrieval now routes through ECAPA/SER, we state the corollary without hedging: **the omni "vector-omni memory" is not load-bearing for the paralinguistic benchmark.** To avoid quietly assuming it adds value, we pre-register a *joint-retrieval ablation*: retrieve with (i) ECAPA/SER alone, (ii) omni-content alone, and (iii) a learned-weight fusion of all three, and admit the claim "omni adds paralinguistic signal" only if (iii) strictly beats (i) on labeled speaker/emotion retrieval. Absent that result, the honest reading stands—omni-policy plus classic-encoder memory, with the omni embedding a content key.

Composite keys and their reward status: a self-consistency gate, not a verifiable one. Each memory entry carries a composite key

$$k = (\text{speaker-ID},\ \text{emotion/SER},\ \text{turn-index}),$$

extracted by frozen diarization, an x-vector / ECAPA-TDNN speaker embedding, and an SER head [xve, 2021, eca, 2020]. These fields are *dual-use* as retrieval structure and as a feedback channel—but we are precise about what kind of feedback. When the agent later answers a speaker- or emotion-conditioned query at deployment, agreement between a freshly predicted attribute and the stored key field is **not** a verifiable reward: the stored key is itself a frozen-model estimate, so "agreement" measures the *self-consistency* of two model passes, not consistency with ground truth. We therefore name the cross-session paralinguistic admission rule what it is—a *self-consistency gate*, not a verifiable-reward gate—and we make *no* claim of Goodhart-resistance for it. On the contrary: a self-consistency objective optimized over an evolving store is Goodhart-prone in the strongest sense. A frozen extractor's systematic error is a fixed point the agent can satisfy without being correct; worse, because the store both *supplies* the stored estimate and *is updated by* the agreement signal, the loop can become *self-reinforcing*—each write that agrees with the extractor's bias makes the next agreement easier, ratcheting the store toward the extractor's error rather than the truth. We design against this directly with two decouplings. **(1) Separate the routing estimator from a *genuinely independent* reward estimator.** The model whose embedding *routes* retrieval is never the same instance/pass that *scores* admission. We do not soften this to "a sibling extractor where possible": we *commit* to a genuinely independent reward extractor—a different-architecture, separately-trained speaker/affect model (e.g. an x-vector TDNN scoring against an ECAPA-routed key, and vice versa)—and we report *its* empirical agreement-with-truth on labeled corpora as a precondition for trusting the decoupling. If that independent extractor's measured agreement-with-truth is not established on the evaluation corpora, we *drop* the decoupling claim rather than assert it; the gate then carries only the weaker within-extractor augmentation/window decoupling, and we say so. **(2) Periodic calibration against held-out human labels—and its honest cost.** One option re-anchors the self-consistency signal, on a schedule, to a small held-out set of *human-labeled* speaker/emotion probes: the surrogate's measured agreement-with-truth there sets its down-weight, and a drop quarantines recent paralinguistic writes. We are blunt that this *reintroduces deployment-time human labels* and so partially contradicts the label-free premise, and we cost it explicitly: it requires fresh human labels *per deployment population, per detected drift event, and per newly-enrolled* *speaker*—an ongoing annotation tax that scales with population churn and channel drift, not a one-time setup. We therefore compare it head-to-head with the trivial alternative of *simply re-reading the frozen encoder* (no calibration labels, accept the surrogate's bias) and pre-register a **deployment-without-calibration ablation**: run the loop with the calibration anchor disabled and measure how fast the store drifts toward the extractor's fixed point. Calibration is justified only if it beats label-free re-reading by enough to repay its per-population/per-drift/per-speaker label cost. Either way it converts an unverifiable, drift-prone signal into a *periodically audited* one—it does not make it verifiable, and we do not pretend otherwise. Where corpus speaker labels exist, we substitute true speaker-ID accuracy for the surrogate (reported with DER under automatic diarization, per Section 7.1). This is a deliberate departure from approaches that close the loop with a *trained* memory manager [mem, 2025d]: we obtain a weak, honestly-labeled self-consistency signal from the keys, not a verifiable reward, and we lean on the genuinely verifiable channels (WER, exact-match, labeled oracle-segment speaker-ID) for the load-bearing credit assignment.

Retrieve–rerank. Retrieval is two-stage. A fast first stage does late-interaction matching in $\mathcal{Z}_\mathrm{A}$ [col, 2020]; a precise second stage reranks the shortlist by reward-weighted importance, weighting each candidate by its historical contribution to verifiable reward (with surrogate self-consistency signals entering only as a discounted, calibration-gated secondary term) [gen, 2023]. The reranker is the discrete analogue of the importance weight $\exp(R/\beta)$ that defines $q^\star$ in Section 4.

Non-destructive curation with provenance. Writes are mediated by a curator that emits one of $\{\textsc{Add}, \textsc{Update}, \textsc{Delete}, \textsc{Noop}\}$ [mem, 2025a]; Noop is first-class and dominant, which is what keeps the store from bloating. Salience decays over time on an Ebbinghaus schedule [mem, 2023b], so unreinforced episodic traces fade unless consolidated. Every node carries provenance and a trust score; low-trust or anomalous writes are quarantined to resist memory poisoning [age, 2024a], the dominant attack surface for evolving stores. The calibration check above feeds this quarantine: paralinguistic writes admitted only on the self-consistency surrogate carry a lower trust score than writes admitted on a verifiable channel, and are the first to be revoked when calibration degrades.

Gradient-free credit assignment. The curator does *not* learn its policy by gradient. Given a candidate set of memory-write operations, it admits the write only if a best-of-N comparison shows a positive reward delta against the no-write baseline—computed on the verifiable channel wherever ground truth is available, and otherwise on the discounted, calibration-gated self-consistency surrogate. This replaces the trained manager of mem [2025d] with a training-free selection rule whose acceptance weight is the tilting weight of Section 4, inheriting the best-of-N KL bounds of bei [2024], ver [2025]—noting, as Section 5 records, that those bounds are finite-N guarantees for a *fixed* operator and transfer only conditionally once the operator (the store) is itself mutated. We also flag the obvious corollary of the spread analysis: on the paralinguistic channel, where the surrogate's reward spread is small (the across-seed emotion delta above is $\approx +0.037$ with a CI spanning zero, and the speaker channel is flat), the best-of-N gain is correspondingly small—so the curator's paralinguistic writes are expected to be *conservative*, and we treat large paralinguistic gains as a warning sign of the self-reinforcing pathology rather than as success.

β-KL trust region on mutation rate. The first instantiation of the control law bounds the *rate* of memory mutation: the curator may not move the store's distribution over keys by more than a β-controlled KL budget per episode,

$$\mathrm{KL}\big(q_{S_{t+1}} \,\|\, q_{S_t}\big) \le \epsilon_{\mathrm{mem}}(\beta).$$

This trust region is the mechanism by which we *design against* the context collapse documented for append-only agents [ace, 2025, exp, 2025]: a bounded-KL update operator is intended to be contractive in the sense Section 4 requires, so that the state trajectory is steered away from the divergent feedback loop rather than left to drift. We state this as a design principle, not a theorem: bounding per-step KL is necessary for stability but we do not claim it is sufficient for finite-time convergence, which remains open (Section 5.3). It is also *not* a defense against the Goodhart fixed point of the self-consistency gate—a slow drift toward the extractor's bias can stay inside the KL budget at every step—which is precisely why the routing/reward decoupling and periodic calibration above are separate, non-KL safeguards. Consolidation (episodic→semantic compaction, decay sweeps, re-embedding) runs asynchronously off the critical path, in the spirit of sleep-time and asynchronous-retrieval memory services [let, mos, 2026], so the trust region never blocks live inference.

7.3 Skills

The speech-skill object. A skill is a structured SKILL.MD-style object (C, π, T, R)—a usage context C, a callable procedure π, a set of test cases T, and a verifiable reward R [sok, 2026, ant]. The procedure π is drawn from a small set of speech-native bodies: generative error correction for ASR rare-word repair [ger, 2025], voice tool-use orchestration [aur, 2025], and paralinguistic-preserving speech-to-speech rewriting [par, 2025]. Each body is a callable over the frozen policy, never a fine-tune of it.

Verification gate. A candidate skill is admitted to the library only if it produces a *positive verifiable-reward delta* relative to the no-skill baseline on its test set T; a resident skill whose running delta turns negative is deprecated [ski, 2026b]. Here we require the delta on a genuinely verifiable channel (WER, exact-match, or labeled oracle-segment speaker-ID), precisely because the skill gate is load-bearing and must not be satisfiable by a surrogate. This is a deliberate asymmetry with the memory subsystem: skills are admitted *only* on the high-spread, genuinely verifiable channels where Conjecture 1 holds, so a paralinguistic-only "skill" with no verifiable reward simply cannot be admitted—there is no surrogate back-door into the library. This operationalizes the curation-quality result of the skills literature: curated, verified skills dominate self-generated ones, and the gate is what enforces curation. At selection time the agent retrieves skills body-first—matching the situation to skill *bodies* rather than to their natural-language descriptions [ski, 2026a]—which avoids the description-mismatch failure of name-based routing.

Gradient-free improvement under a Pareto trust region. Skills improve without touching base weights through reflective, evolutionary search: GEPA-style Pareto editing of skill prompts and procedures [gep, 2025] and AFlow-style workflow search over skill compositions [afl, 2024]. The governing *design principle* is the second instantiation of our control law: an edited skill is accepted only if it is *Pareto non-regressive*—it must not lose verifiable reward on any task in its competency set. We intend Pareto non-regression to act like a β-KL trust region on the skill library's behavior, keeping the post-edit policy within a bounded-KL neighborhood of the pre-edit policy on each verified task, so as to *discourage* the reflection-plateau and skill-thrash pathologies [ref, 2023, lat, 2023]. We do *not* claim this guarantees monotone non-degradation: non-regression is checked only on the finite, observed competency set T, so it bounds measured regression on T but cannot certify global non-degradation, and the convergence of the resulting edit sequence is exactly the open problem of Section 5.3. The construction is engineered to be contractive; certifying that it is remains future work.

Bloat control and composition. Library growth is contained by polymorphic abstraction—collapsing near-duplicate skills into parameterized families [pol, 2025]—and by CRAFT-style validate-and-deduplicate at admission [cra, 2023]. Genuinely new competence accretes by AWM-style snowballing, where verified skills compose into higher-order workflows that are themselves gated [awm, 2024]. Thus the library grows in capability while its effective description length stays bounded, the skill-side analogue of the memory decay schedule.

7.4 The Unified Control Law

The two subsystems are governed by one law, stated once:

> *Admit an inference-time state mutation iff it yields a positive reward delta against the no-mutation baseline—measured on a verifiable channel wherever ground truth exists, and otherwise on a down-weighted, calibration-gated self-consistency surrogate (the acceptance gate)—and only within a β-KL trust region that bounds how far the induced action distribution may move (the stability constraint).*

The acceptance gate is the discrete realization of the reward tilting that defines the Gibbs optimum $q^\star \propto q_0 \exp(R/\beta)$ of the objective $F(q) = \mathbb{E}_q[R] - \beta\,\mathrm{KL}(q \,\|\, q_0)$; at the output level it inherits the $O(1/N)$ tilting rate and the KL–reward trade-off of Section 5. The strength of any admitted step is governed by the *spread* of its reward channel (Section 4)—to leading order the discriminator is $\mathrm{Var}_{q_0}(R)/(2\beta)$: on the high-spread verifiable channels (WER, exact-match) the gate moves the state materially, while on the low-spread paralinguistic surrogate it is—by construction and by our own measurements—near-inert. The trust region is the regularizer $\beta\,\mathrm{KL}(\cdot \,\|\, q_0)$ that keeps the search distribution q from collapsing. The law has exactly two instantiations: the memory-mutation-rate bound $\epsilon_{\mathrm{mem}}(\beta)$ of Section 7.2, and the skill-library Pareto non-regression bound of Section 7.3. Both are bounded-KL, and both are *engineered* to yield contractive state-update operators. We are deliberate about the strength of this claim on two fronts. First, the finite-N convergence results of Section 5 hold for the *fixed*-operator (output-level) case, whereas here the operator—the external state S—is itself part of the action, which is exactly the regime where only asymptotic, trust-region-conditioned guarantees are currently available and finite-time monotone convergence is unproven (Section 5.3). Second, the law's *otherwise*-branch (the self-consistency surrogate) carries no verifiability and no Goodhart-resistance guarantee; it is held in check not by the KL trust region but by the explicit routing/reward decoupling and periodic human-labeled calibration of Section 7.2. We therefore present the control law as the architecture's *convergence design principle*: each admitted step is constructed to strictly improve a (preferably verifiable) reward while staying inside a β-controlled neighborhood, with no base-model gradient anywhere in the loop. Whether the resulting trajectory $S_0 \to S_1 \to \cdots$ provably converges—rather than merely being steered toward convergence—and whether the frozen omni stack ever affords a non-degenerate *paralinguistic* verifiable reward to drive it—rather than the near-flat surrogate our evidence currently shows—are the central open questions this proposal sets out to answer, not results we assert here.

8 Research Plan

We organize the empirical program as a partially-observed sequential decision problem over experiments: each phase is selected to maximize value-of-information per unit of compute, so that cheap, high-diagnostic measurements gate the expensive ones. The ordering is deliberate, and it is now disciplined by a result we obtained while writing this plan. The thesis of the paper is that the optimization gain $\mathcal{G}$ of training-free RL—Gibbs tilting $q^\star \propto q_0 \exp(R/\beta)$ of a frozen base q_0—is governed by the *reward range* spread $= \max_{\mathrm{supp}\, q_0} R - \min_{\mathrm{supp}\, q_0} R$ of the (action-space, reward) pair (OSA-1, Theorem 3); to leading order in $1/\beta$ the operative discriminator is the base-measure variance $\mathrm{Var}_{q_0}(R)/(2\beta)$ (a remark refines the certified range-based ceiling to this variance form). This is a property of the *reward structure*, not of model class: a high-range verifiable reward yields large gain *even on a single frozen model* (our own single-model Operator-B results, e.g. SLURP intent +0.330, Table 4), while a flat reward—instruction-conditioning of a frozen embedding read-out—yields vanishing gain regardless of how the model is wrapped. Context isolation makes gain *additive* across blocks (OSA-2, Theorem 4) and each non-degenerate block contributes *strictly positive* gain (`gain_pos_of_nonconstant`: reward non-constant $\Rightarrow \mathcal{G} > 0$, sorry-free in Lean), but Theorem 6 (`qstar_product`) proves the context-isolated optimum equals the monolithic single-model optimum on the product reward $R = \sum_i R_i$—so *isolation per se adds nothing*; the only new gain comes from introducing additional *non-degenerate verifiable rewards*. In particular, our high-range content/intent gains are *single-model* optima, *not* evidence that agentic decomposition recovers anything a single model could not. Whether a frozen omni embedding read-out affords such non-degenerate per-block rewards for our flagship paralinguistic factors is therefore an empirical question (Conjecture 1), and the answer we have so far is humbling.

The result that re-scoped this plan. We ran the paired-delta bootstrap that Section 6 flagged as outstanding Phase-0 work, now committed at `projects/speech-mllm-omni-embedding-rl/_repro/emotion_pool_paired_v2.json`. With the read-out layer selected on a dev split (no oracle test-layer peeking) and a paired-delta bootstrap over 5 seeds $(42, 7, 123, 2024, 31337)$, the emotion-pooling Operator-A effect is a *null* at the across-seed level: the mean is $\Delta = +0.037$, but the 95% across-seed t-confidence interval is $[-0.043, +0.116]$ and spans 0, so the effect is not significant.[5] The previously reported single-seed +0.097 was the best seed under *oracle test-layer selection*—a winner's-curse / selection-leak artifact, not a stable effect. In parallel, a frozen linear/kNN probe on the omni *content* bi-encoder (Omni-Embed) sits near chance (≈ 0.04) for speaker identity at *every* one of its 37 layers. We are careful about the scope of this negative: it is established on *one content bi-encoder* via a linear/kNN read-out probe, and it bounds only that Operator-A read-out path; it does *not* bound the proposed *generative* Operator-B policy, which we have not yet tested on paralinguistics. The honest conclusion is that the spread-floor conjecture holds for content/intent, while for the flagship paralinguistic factors the one frozen embedding we have probed exposes no non-degenerate per-block reward (emotion null, speaker at chance). We therefore re-scope: high-range content/intent verifiable rewards yield large *single-model* gains, whereas whether *agentic decomposition* adds anything for speaker/affect is unresolved and—by Theorem 6—can be settled only by introducing genuinely new non-degenerate verifiable rewards, which remain untested. This re-scoping is load-bearing for every baseline and admission band below; the plan is written so that a null paralinguistic result is a reportable outcome, not a failure of the instrument.

We first close the loop on a single GPU with frozen weights (Phase 0), because no downstream claim can be trusted until the verifiable-reward gate is demonstrably faithful end-to-end and the single-model spread is measured honestly. We then build the missing measurement instrument (Phase 1), since the central question—whether reward-guided inference-time optimization over $\mathcal{Z}$ can *compound* competence across sessions—cannot be answered without a benchmark that does not yet exist, and that benchmark must be *falsifiable against a real, teethed baseline*. Only then do we run the minimal self-evolving loop (Phase 2) and ablate its controls (Phase 3). Throughout, the only quantities that drive selection are verifiable rewards R, and the locked test set is touched exactly once. We stress at the outset that the benchmark, and the baseline the agent must beat, are the load-bearing risks in this plan: a poorly constructed cross-session benchmark, or a strawman baseline, can manufacture an apparent compounding effect out of artifacts, so a substantial part of Phase 1 is devoted to ruling out such artifacts and to pre-registering a classic-pipeline baseline that the agent must *exceed* before any agentic claim is made.

8.1 Phase 0: Closed-Loop Bring-Up

The objective of Phase 0 is a verified, reproducible inference-time loop on the in-place RTX 5090 (Blackwell, sm_120; torch `cu128`) with *no weight updates and no structural change.* The frozen stack couples an omni-embedding model (Omni-Embed-Nemotron, omn, 2025, 2026) for the embedding operator $\mathcal{Z}_{\mathrm{A}}$ with Qwen3-Omni [qwe, 2025b] served via GGUF/`llama.cpp` for the generative operator $\mathcal{Z}_{\mathrm{B}}$. Concretely, this realizes the two-operator action space $\mathcal{Z} = \mathcal{Z}_{\mathrm{A}} \cup \mathcal{Z}_{\mathrm{B}}$ used throughout: $\mathcal{Z}_{\mathrm{A}}$ selects pooling/probe read-outs of a frozen representation; $\mathcal{Z}_{\mathrm{B}}$ is reward-guided decoding (best-of-N, reranking, soft-BoN) over text. In light of the result above, we make a structural commitment up front: the omni embedding is reserved for *content* read-out and retrieval, where its spread is real, while *speaker* and *affect* memory keys are carried by classic speech encoders (ECAPA-TDNN speaker embeddings eca, 2020 and a dedicated SER classifier), not by the omni embedding. The "two-omni" system is thus an omni *policy* (Operator-B) plus an omni *content* memory, paired with classic paralinguistic encoders—we do *not* claim the omni embedding performs load-bearing paralinguistic retrieval, because on our own content-bi-encoder probe it is at chance on speaker.

We reproduce two known frozen baselines as correctness anchors: SER accuracy ≈ 0.49 and intent accuracy ≈ 0.25 under zero-shot prompting of the backbone. Matching these within bootstrap tolerance certifies that data loading, prompt templating (`speechrl_common.models`), and decoding are wired correctly. Because these anchors come from public corpora that may overlap a backbone's pretraining data, we treat them as wiring checks only, not as evidence of generalization, and we subject every public-dataset preliminary number to the contamination analysis described below before quoting it. We then verify the verifiable-reward gate: for ASR we instrument WER-based exact rewards, and for SER/SID we instrument label-exact rewards (`speechrl_common.rl`), confirming that (i) best-of-N selection under R strictly dominates greedy on a held-in probe, consistent with the regret/winrate scaling of soft and hard best-of-N [ver, 2025, bei, 2024, ami, 2025], and (ii) the implied search distribution q tilts the base q_0 toward the tilted optimum $q^\star \propto q_0 \exp(R/\beta)$ of the objective $F(q) = \mathbb{E}_q[R] - \beta\,\mathrm{KL}(q \,\|\, q_0)$ as β is lowered, with the partition function Z and objective $F(\cdot)$

[5] As a diagnostic only, the per-seed paired CI excludes 0 in 2/5 seeds; this seed-vote count is *not* the inferential statistic—the across-seed CI is, and it does not exclude 0.

tracked as in the KL-as-Bayes view [kor, 2022]. Critically, Phase 0 also *re-confirms the spread map* that drives the rest of the plan: we log per-utterance $\mathcal{G}$ and spread for each candidate reward and report which rewards are high-range (content/intent best-of-N) and which are flat (paralinguistic instruction-conditioning), using the dev-selected, multi-seed paired-delta protocol that produced the null +0.037 emotion result above—never oracle test-layer selection. The exit criterion is a green end-to-end run logging $\mathcal{G}$ and spread per utterance to local MLflow, a passing `pytest` smoke suite, and a committed, dev-selected paired-delta artifact for every paralinguistic readout claim. Phase 0 buys the highest information per dollar: it eliminates instrumentation confounds, and it forces every paralinguistic gain through the same selection-leak-proof, across-seed-CI protocol before any benchmark or loop claim is made.

Contamination analysis of preliminary public-dataset numbers. Before any of the Phase 0 anchor numbers (or other figures computed on existing public datasets) enter the paper, each is passed through a contamination screen, not only the newly constructed benchmark. The screen combines (i) train-cutoff reasoning, comparing each corpus's release date against the backbone's reported pretraining cutoff and flagging any corpus plausibly in-distribution; (ii) n-gram / transcript overlap between corpus references and any recoverable pretraining-adjacent text, reported as an overlap rate per split; and (iii) a canary / memorization probe that checks whether the backbone can complete held-out references above chance without the audio [con, 2024, tas]. Numbers that fail any screen are reported as wiring checks with an explicit contamination caveat rather than as generalization evidence. We make no claim that this screen is exhaustive; it is a documented best effort whose results we report alongside the affected numbers.

8.2 Phase 1: A Speech Cross-Session Paralinguistic Memory Benchmark

The gap. Existing long-horizon memory benchmarks are text-only (lon, 2024, loc, 2024; the agentic memory line amb, 2026, ana, 2026), and audio memory work targets within-document or RAG retrieval [wav, 2025, vox, 2025, aud, 2025]. No raw-audio, *cross-session*, paralinguistically-keyed memory benchmark exists. This is the intended contribution: a benchmark whose queries can only be answered by retaining *who spoke* and *how they sounded* across sessions, not by transcribing content. Whether a cheaply constructed benchmark can faithfully isolate this competence is itself an open question, and—given the negative paralinguistic evidence above—we treat the construction below as a hypothesis to be validated rather than a settled instrument, and we equip it with a real baseline that the agent must beat before any agentic claim is asserted.

Primary baseline: the strongest simple pipeline the agent must exceed. The single most important addition to this plan is a *teethed* primary baseline with a *pre-registered, locked* accumulation rule, and it must be the genuinely strongest *simple* pipeline—not a strawman—or the comparison cannot isolate agentic gain. We split the baseline *by factor*, because the identity (trait) and affect-change (state) sub-problems have different strongest-simple solutions.

Identity (trait) baseline: ECAPA + PLDA with full multi-enrollment. For speaker identity we pre-register, in the locked release artifact, a frozen **ECAPA-TDNN** front-end [eca, 2020, xve, 2021] scored by **PLDA** (equivalently, cosine with adaptive symmetric normalization, AS-Norm) under *full multi-enrollment*: all of a speaker's enrolled utterances across sessions are combined under the PLDA model, not merely averaged into a recent-k centroid. This is a deliberate *upgrade* from the cosine running-centroid dictionary we first considered, and the reason is structural: the candidate agentic mechanism of Phase 2—Bayesian multi-session evidence integration, i.e. accumulating a per-speaker posterior over identity from per-session ECAPA read-outs treated as noisy observations—*is essentially PLDA*. PLDA is exactly the generative model that treats each embedding as a latent speaker factor plus channel/session noise and integrates evidence across enrollments. A cosine-centroid baseline would therefore let the agent book a "gain" that is nothing more than PLDA-over-cosine (proper channel-aware multi-enrollment scoring)—a classical, non-agentic improvement—so it *cannot isolate agentic gain from PLDA-over-cosine*. Only by pre-registering ECAPA+PLDA-with-multi-enrollment as the baseline do we force the agentic store to demonstrate that it adds something *beyond* correct Bayesian evidence integration—concretely: graph-structured curation of which enrollments to keep, UPDATE/DELETE decay of stale keys, a trust region on the per-episode store change, and explicit change handling (speaker re-enrollment on detected drift). The agentic store must *strictly exceed* ECAPA+PLDA multi-enrollment, or its identity claim is unsupported.

Affect-change (state) baseline: per-session-independent SER. For affect we pre-register a **per-session-independent SER** baseline—each session's affect is predicted from that session alone by the frozen SER labeler, with *no* cross-session state, optionally smoothed by **majority vote with exponential decay** for a running-state variant—together with an explicit **change-detection threshold** on affect drift. The agentic change-point store must beat this at *detecting affect changes* across sessions: the test is whether a curated cross-session store detects a genuine affect change (or its absence) better than independently labeling each session and differencing. As with identity, the baseline's exact UPDATE/DELETE and change-detection-threshold rule is locked in the release artifact.

We require the self-evolving agent to *strictly exceed* the relevant baseline on each factor, with every contrast tested by a paired bootstrap on per-item deltas [bis] over ≥ 5 seeds with across-seed CIs and FWER correction across the sweep. We pre-register two *separate* falsifiers so that a null on one does *not* over-generalize to "the agentic claim for paralinguistics" as a whole: **(trait/identity)** if multi-session curation does not exceed ECAPA+PLDA multi-enrollment with a paired-bootstrap CI excluding 0, the agentic claim *on identity* is recorded as unsupported; **(state/affect-change)** if the change-point store does not beat the per-session-independent SER baseline at detecting affect changes with a paired-bootstrap CI excluding 0, the agentic claim *on affect change* is recorded as unsupported. A null on identity leaves the affect-change question open, and vice versa. These baselines are deliberately strong precisely because the one frozen content bi-encoder we probed is at chance on speaker: the classic pipeline encodes exactly the paralinguistic competence the omni *content* embedding lacks, so beating it—*beyond* mere PLDA evidence integration—is the real test of whether the *agentic* composition (memory $\times$ policy $\times$ verifiable gate) adds anything. Both baselines, with their locked rules, ship in the release artifact so any reader can re-run the comparison.

Minimal spec. (i) *Stimuli*: multi-session spoken dialogues delivered as raw audio, with ≥ 3 sessions per persona separated by simulated time gaps. (ii) *Queries*: speaker-ID-keyed and SER-keyed (e.g., "which speaker was angry in the previous call," "has this caller's affect changed since session 1"), so that the intended answer is a function of paralinguistic state, not lexical content—guarding against the read-not-listen shortcut [vox, 2026, lis, 2025]. (iii) *Ground truth*: SID labels are treated as verifiable; SER labels are *not* treated as ground truth from a frozen labeler (see the SER gating paragraph), and only the human-validated, high-confidence SER subset is admitted for exact rewards or reporting. (iv) *Probes*: a knowledge-update probe (a speaker's emotion/role changes across sessions, testing UPDATE/DELETE semantics mem, 2025a), a contamination probe [con, 2024, tas], and a memory-poisoning probe [age, 2024a, mem, 2026].

Query-generalization split (recall is not memory). A positive cross-session slope is meaningless if it is answer-cache recall. We therefore split queries into a *repeated* set (the same question re-asked across sessions) and a *novel-query* set (new questions about the *same* speaker's stored paralinguistic state, never asked before). Stored state must *transfer* to the novel queries; we report the compounding slope *separately* on repeated vs. novel queries, and treat the novel-query slope as the primary compounding evidence. A slope that appears only on repeated queries is reported as caching, not memory.

Leakage control: raw embeddings only, no gold labels in store. To make a positive slope unattributable to key-value recall, the memory store is forbidden from writing gold labels: only *raw embeddings* (ECAPA speaker vectors, SER logits/embeddings, omni content vectors) may be persisted, never the benchmark's gold SID/SER answer. Thus a positive slope cannot be produced by reading back a stored label; it must arise from re-reading the acoustic state. The store schema is validated automatically to contain no gold-label field.

Construction and its realism risks. The cheap construction we initially considered—re-keying an existing diarized corpus (SLURP/SUPERB-style intent+speaker material, slu, sup, 2021, dyn, 2023) into "multi-session dialogues"—has a serious validity flaw we make explicit: stitching single-speaker clips of the same speaker into separate "sessions" yields *channel-matched, exact-audio* continuity, so cross-session speaker linking can be solved by low-level acoustic/channel fingerprinting rather than by any genuine paralinguistic memory, trivializing the competence the benchmark is meant to measure. We therefore require *genuine* cross-session variation as an admission condition: either (a) the same persona is re-recorded under different recording conditions (device, room, codec, background), or (b) we source a real repeated-caller corpus in which a speaker recurs naturally across genuinely distinct sessions. Channel-matched exact-audio re-keying is used, if at all, only as an explicitly labeled *upper-bound/ceiling* control, never as the headline cross-session set.

Numeric admission bands (DER / SER / SV-EER), fixed in advance. To prevent "disqualify if near-trivial" from being post-hoc tunable, we pre-register numeric bands and freeze them before construction. The speaker-linking sub-problem is admitted only if the frozen ECAPA speaker-verification *equal-error rate* (SV-EER) on the constructed cross-session pairs lies in $[8\%, 30\%]$: an EER below 5% signals channel fingerprinting (trivial; disqualified), above 40% signals no recoverable speaker signal (degenerate; disqualified). Diarization difficulty is admitted only if the frozen-diarizer *DER* lies in $[10\%, 30\%]$ (below 5% trivial, above 40% unusable). The SER sub-problem is admitted only if frozen-labeler-vs-human balanced accuracy on a held-out calibration slice lies in $[0.35, 0.75]$—above the k-way chance floor but well below ceiling, so the task is neither trivial nor saturated. These three bands are reported on every constructed split; any split outside a band is rejected or relabeled a ceiling control. To quantify residual leakage we additionally report DER and SER error on the constructed set so the difficulty of the speaker-linking and affect sub-problems is measured rather than assumed.

Listen-vs-read control as a primary metric, run on the agent's own transcripts. Every paralinguistic query must pass a transcript-only / listen-only control before admission: a strong text LLM given only the gold transcripts (no audio) must perform at chance on the paralinguistic queries; queries solvable from transcript alone are removed. We harden this control in three ways. First, we run the transcript condition on the agent's *own ASR transcripts* in two variants—a *surface-marked* version preserving the cues a careless transcriber might leak (capitalization, "!!!", `[laughter]`, disfluency markers) *and* a *normalized* version that strips them—so a gap that vanishes only under normalization is exposed as surface leakage, not listening. Second, we *balance topic across speakers*, so that lexical topic cannot proxy for speaker or affect identity. Third, we add a *positive control*: a model independently known to attend to acoustics is run on the same queries, and we require it to show a substantial audio−transcript gap; if even the positive control cannot, the gap reflects benchmark difficulty rather than a single model's deafness, and we say so. We report the *audio-vs-transcript gap*—audio-conditioned minus transcript-only performance—as a *primary* metric: a benchmark with a small or zero gap is, by construction, not measuring listening. Construction uses *frozen* diarization (pyannote, pya, per, 2019, xve, 2021) and a frozen SER labeler only as *proposal* tools whose outputs are audited and, where used for ground truth, human-validated; their raw outputs are never the final labels.

SER-keyed ground truth, spontaneous and acted affect reported separately. We do not treat a frozen SER labeler as ground truth. Any SER-keyed write into memory, and any SER-keyed reward, is gated behind an SER-confidence threshold: low-confidence affect predictions are abstained on rather than committed. For the *evaluated* subset, SER labels are obtained by human annotation, with multiple annotators per item; we report inter-annotator agreement (Krippendorff's α/Fleiss' κ) and exclude items below a pre-registered agreement floor. Acted affect is known to overstate SER reliability, so we source at least one *spontaneous* (non-acted) affect corpus—in-the-wild, naturally elicited speech—and report it *separately* from acted CREMA-D, never pooled. For the knowledge-update probe specifically we report the *fraction of affect-change items that survive the IAA floor*: if few affect-change items reach inter-annotator agreement, the update probe is under-powered and we report it as such rather than quoting a slope on a handful of items. Exact SER rewards are computed only against human-agreed, high-confidence labels; the frozen labeler serves solely to pre-filter and scale annotation, never to define correctness.

Evaluation design. Scoring uses verifiable rewards only, reported with paired bootstrap confidence intervals over the per-utterance deltas [bis], standard frozen splits [gor], and an audited memory-eval rubric [ana, 2026]. We report, as primary and *per factor*, the agent-minus-baseline paired-bootstrap contrast (agent vs. ECAPA+PLDA for identity, agent vs. per-session-independent SER for affect change); as secondary, DER/SV-EER/SER admission bands, the audio-vs-transcript gap, the repeated-vs-novel-query slopes, the spontaneous-vs-acted affect split, *maintenance latency* (wall-clock cost of the memory ADD/UPDATE cycle), and we run every metric across ≥ 2 backbones (Qwen3-Omni and Qwen2.5-Omni, qwe, 2025a) to separate benchmark signal from backbone idiosyncrasy. Phase 1 is sequenced second because it is the precondition for any compounding claim, yet is independent of the loop's internals, so it can be built and frozen—and its realism controls, numeric bands, and baselines passed—before Phase 2 risks overfitting to it.

8.3 Phase 2: Minimal Self-Evolving Loop

Phase 2 instantiates the smallest loop that could plausibly compound: a paralinguistic *memory* store (raw-embedding keys with ADD/UPDATE/DELETE mem, 2025a,c, ECAPA/SER keys for paralinguistics and omni keys for content, never gold labels), a *skill gate* that admits a candidate skill only when a held-out verifiable reward improves [ski, 2026b, coe, 2026], and a *trust region* that bounds the per-episode change in the search distribution via $\mathrm{KL}(q \,\|\, q_0) \leq \epsilon$, capping the inference-time tilt to limit reward over-optimization [gao, 2022, ska, 2022]. The loop is gradient-free: credit is assigned by reward comparison across rollouts [jit, 2026, ahm, 2024], and selection over $\mathcal{Z}_{\mathrm{B}}$ uses reranking/MBR among N samples [ich, 2025, mbr, 2025]. SER-keyed writes inherit the confidence gating and abstention discipline of Phase 1, so unreliable affect predictions do not silently become persistent state. The loop's design is consistent with Theorem 6: because the context-isolated optimum equals the monolithic single-model optimum on the product reward, the loop adds value only insofar as it introduces *additional non-degenerate verifiable rewards*—which, by our Phase-0 evidence, are abundant for content/intent and scarce for the paralinguistic read-out we probed. Concretely, Phase 2 tests candidate paralinguistic mechanisms *by factor*: for identity, **Bayesian multi-session evidence integration** (treating each session's ECAPA read-out as a noisy observation and accumulating a per-speaker posterior over identity) against the pre-registered ECAPA+PLDA multi-enrollment baseline; for affect, a **change-point store** over per-session SER read-outs against the per-session-independent SER baseline. Because the identity mechanism is essentially PLDA (Section 8.2), only the ECAPA+PLDA baseline can isolate genuine agentic gain from PLDA-over-cosine—the agentic store must add graph curation / decay / trust region / change handling *on top of* correct evidence integration. A null on the identity test disqualifies the *agentic claim for identity*, and a null on the affect-change test disqualifies the *agentic claim for affect change*, but *neither* disqualifies the other factor nor the underlying theory (Theorem 6 already predicts no gain absent a genuinely new non-degenerate reward). We therefore expect and pre-register an *asymmetric*, *per-factor* outcome.

We *pre-register* success and kill criteria before running, separately by factor. **Success (content/intent):** a positive, bootstrap-significant cross-episode slope on intent/spoken-QA reward, on the *novel-query* split, that *exceeds the classic-pipeline baseline* under matched per-speaker evidence with a paired-bootstrap CI excluding 0, with the gate's admit-rate stable and $\mathrm{KL}(q \,\|\, q_0)$ inside the trust region. **Trait/identity hypothesis (pre-registered as a null):** we do *not* predict that multi-session curation exceeds ECAPA+PLDA multi-enrollment on SID; a null or baseline-not-exceeded result on the novel-query SID split is an *expected and reportable* outcome, given the negative read-out evidence and Conjecture 1. **State/affect-change hypothesis (pre-registered as a null):** we likewise do *not* predict that the change-point store beats the per-session-independent SER baseline at detecting affect changes on the human-validated SER subset; a null there is *separately* expected and reportable, and does *not* generalize to the identity claim. **Kill:** no slope after a fixed episode budget, admit-rate collapse, trust-region saturation (drift), or—decisively—failure of a factor's agentic mechanism to exceed *its* pre-registered baseline (ECAPA+PLDA for identity, per-session-independent SER for affect change), which we record as *not supporting the agentic claim* on *that factor only.* The headline measurement is *compounding*: reward as a function of episode index. The *primary* contrast is agent-minus-baseline where *both receive the identical accumulated per-speaker evidence*—the same enrolled utterances, in the same order—so the comparison isolates the agentic update rule from the sheer amount of evidence. We further impose an **evidence-matched / fixed-audio-budget control**: the total per-speaker enrolled audio (equivalently, the number of enrolled utterances) is held *constant* across episode index, so a positive compounding slope cannot be driven by mere acoustic-data accumulation rather than by self-evolution of the memory. The vs-empty-memory single-shot contrast (best-of-N with non-persistent memory) is *demoted to a secondary diagnostic*: it cannot separate self-evolution from accumulation and so cannot, on its own, establish compounding. The compounding measurement isolates cross-session state from raw test-time compute [sne, 2024, zuo, 2025] and from a trivial accumulating dictionary. Because a spurious slope could arise from benchmark artifacts or answer caching, this measurement is interpretable only on a Phase 1 set that has passed the DER/SV-EER/SER admission bands, the audio-vs-transcript-gap control, the raw-embeddings-only leakage control, and the novel-query split.

8.4 Phase 3: Scale and Ablate

Phase 3 stress-tests each control by removal, with pre-registered directional predictions. (i) *Remove the skill gate*: we predict SkillsBench-style degradation as unvetted skills accumulate [ski, 2026b, ace, 2025]. (ii) *Remove the trust region*: we predict drift, i.e., $\mathrm{KL}(q \,\|\, q_0)$ growth correlated with falling locked-test reward, the inference-time analogue of reward hacking [gao, 2022, rem, 2026]. (iii) *Append-only vs. curated memory*: we predict append-only error propagation and context collapse [exp, 2025, ace, 2025], with curated UPDATE/DELETE recovering it [mem, 2025a]. (iv) *Omni vs. classic paralinguistic keys*: we predict that swapping the omni embedding in for the ECAPA/SER keys *degrades* speaker/affect compounding toward chance, directly testing the negative read-out finding that motivated this plan. Each ablation is a single-variable contrast against the Phase 2 configuration, scaled across $N \in \{4, 8, 16, 32\}$ and across backbones to chart the reward-vs-compute and reward-vs-tilt frontiers. These predictions are hypotheses; a null or reversed result is reported as such rather than reinterpreted post hoc.

8.5 Evaluation Protocol

The protocol is fixed in advance and applies uniformly. (1) *Rewards*: verifiable only (WER/exact-match/label-exact), with SER rewards restricted to the human-validated, confidence-gated subset; no LLM-judge in any selection or reporting path, avoiding self-preference and position bias [sel, 2024, pos, 2024]. (2) *Test discipline*: model selection and all tuning—including read-out layer/pooling selection—occur on a disjoint dev split only [dwo, roe]; *oracle test-layer selection is prohibited*, having already produced the spurious single-seed $+0.097$ emotion gain that the dev-selected re-run corrected to a null $+0.037$ (across-seed CI $[-0.043, +0.116]$, spanning 0). After dev-only selection, the locked test set is evaluated in a single, untouched pass at the end of each phase. (3) *Demonstration hygiene*: in-context demonstrations are drawn disjoint from both dev and test, since demonstrations alter format more than accuracy [min, 2022, ali, 2026], and we report true few-shot results [per, 2021]. (4) *Uncertainty*: every comparison is run over ≥ 5 seeds with across-seed confidence intervals [hen, 2017], and deltas between conditions are tested with a *paired* bootstrap on the per-item differences [bis, dro]—reporting, as the emotion result does, the across-seed mean and CI as the inferential statistic (with the fraction of seeds whose paired CI excludes 0 given only as a diagnostic, never as the headline)—with a power analysis fixing the minimum detectable effect [car, 2020]. (5) *Multiplicity*: family-wise error-rate (FWER) correction is applied across the config/ablation sweep *and across the two pre-registered per-factor baseline contrasts*, so that selecting among many configurations (or many read-out layers) does not inflate the apparent significance of the single test-pass deltas. (6) *Contamination*: the Phase 0 contamination screen (train-cutoff reasoning, n-gram/transcript overlap, canary) is applied to *every* public-dataset number, not only the constructed benchmark, and its results are reported alongside the affected figures. (7) *Negative results are reported as first-class outcomes*—the null emotion result and the at-chance speaker probe (on one content bi-encoder) are reported in the paper, not suppressed, and per-factor nulls are reported per factor without over-generalization—and the full configuration follows the reproducibility checklist [pin, 2020, dod, 2019].

8.6 Milestones, Compute, and Reproducibility

Compute is already provisioned: an in-place RTX 5090 with torch `cu128`, all five models (including the Qwen3-Omni GGUF and Omni-Embed-Nemotron omn) and all 28 datasets resident on WSL ext4 disk, frozen to a pinned manifest. Every run is reproducible by construction: pinned data revisions + fixed seeds + a single `reproduce:` command logged to MLflow, so any reported number maps to one invocation—including the committed paired-delta artifact `emotion_pool_paired_v2.json` that re-scoped this plan.

The release artifact bundles the benchmark builder (including the realism controls: cross-session variation requirement, numeric DER/SV-EER/SER admission bands, listen-vs-read filtering on own ASR transcripts

Milestone	Phase	Deliverable	Exit gate
M0	0	Closed loop on frozen $\mathcal{Z}_{\mathrm{A}}+\mathcal{Z}_{\mathrm{B}}$	SER ≈ 0.49, intent ≈ 0.25 reproduced (wiring
M1	0	Reward-gate + contamination screen + spread map	best-of-N > greedy, CI-sig.; dev-selected pai
M2	1	Cross-session benchmark + ECAPA+PLDA / SER baselines	genuine variation; DER/SV-EER/SER in-ba
M3	1	Listen-vs-read + human-SER + novel-query split	audio−transcript gap > 0 (incl. positive cont
M4	2	Minimal self-evolving loop	pre-registered success/kill adjudicated per fa
M5	2	Compounding vs. baselines (evidence-matched)	slope exceeds per-factor baseline (ECAPA+F
M6	3	Control ablations (incl. omni-vs-classic keys)	gate/TR/append-only/key-type contrasts co
M7	3	Frontier sweep (N, backbone)	reward-vs-compute/tilt curves + release

Table 5: Value-of-information–ordered milestones. Each gate is a verifiable, bootstrap-tested criterion; the locked test set is touched once per phase, after dev-only selection (no oracle test-layer selection), with FWER correction across the sweep and the two per-factor contrasts. The compounding gate (M5) requires *exceeding the per-factor baselines—ECAPA+PLDA multi-enrollment for identity, per-session-independent SER for affect change—under matched per-speaker evidence*, not merely a positive slope, and a null is adjudicated per factor.

with surface-marked and normalized variants, the spontaneous-affect split, and the SER human-annotation pipeline), the two per-factor baselines with their pre-registered locked rules—the ECAPA+PLDA identity baseline (frozen ECAPA-TDNN front-end, PLDA/AS-Norm scoring under full multi-enrollment) and the per-session-independent SER affect baseline (SER majority-with-decay, explicit change-detection threshold)—the raw-embeddings-only memory schema, the loop configuration, pinned manifests, the contamination-screen scripts, the committed paired-delta artifacts, and per-milestone `reproduce:` commands, so that all of Table 5 can be re-derived end-to-end on equivalent hardware. This ordering guarantees that the expensive compounding study (M4–M7) is only attempted after the instrument and its baselines (M2–M3) and the gate and spread map (M0–M1) are independently certified—including their artifact, contamination, and selection-leak controls—maximizing the information returned for the fixed single-GPU budget, and ensuring that an honest negative on either the identity or the affect-change agentic claim is a publishable, per-factor result of this plan rather than a hidden failure.

9 Threats to Validity and Limitations

We hold this work to the standard of self-criticism it advocates, and enumerate the threats in descending order of how much they could change the paper's claims. We distinguish *threats to the thesis* (is the gain of training-free RL really governed by the reward range, and does context isolation add anything beyond what a single model already affords?) from *threats to the moat* (is the contribution durable and novel?) and *threats to the empirics* (do the measurements support the conclusions?). The most consequential item is not an external risk but an internal, honest negative result of our own, so we state it first.

Our strongest finding is a negative one: on our embedding probe, the paralinguistic per-block rewards the flagship would need are absent. The constructive reading of OSA-2 depends entirely on Conjecture 1: that the frozen-model context-isolation construction actually yields *non-degenerate* per-block verifiable rewards, i.e. a per-component spread spread_i bounded away from zero so that each $\mathcal{G}(R_i) > 0$. This is an empirical conjecture, not a theorem, and our own measurements indicate it fails for the flagship's paralinguistic factors — but we are deliberate about the *scope* of that evidence. It is established on a *single content bi-encoder* (the omni-embed read-out) via a linear/kNN probe, and it therefore bounds only *that read-out*: it does *not* bound the proposed generative Operator-B policy, which we have never tested for paralinguistics, and it is not a verdict on frozen omni models as a class. For speaker, the kNN/linear probe on the omni-embed embedding sits at ≈ 0.04 — near chance — at *all* 37 hidden layers and under both pooling readouts, so that one content read-out affords essentially no speaker spread to optimize over. For emotion, a rigorous multi-seed re-run replaces the earlier single-seed headline: with the readout layer selected on dev, a paired-delta bootstrap, and five seeds $(42, 7, 123, 2024, 31337)$, the *across-seed* mean improvement is $\Delta = +0.037$ with a 95% t-CI of $[-0.043, +0.116]$ that spans zero — i.e. *no significant emotion gain at* *the across-seed level (a null).*[6] The previously reported single-seed +0.097 was the *best* seed under *oracle test-layer* selection — a winner's-curse / selection-leak artifact, not a stable effect. The committed artifact is `projects/speech-mllm-omni-embedding-rl/_repro/emotion_pool_paired_v2.json`. We present this as a central finding of the paper, not as a failure to be hidden: it sharply scopes where the reward-range thesis has empirical teeth — *content/intent*, where single-model task-level selection already exhibits large, well-separated spread (e.g. SLURP +0.330, Section 6) — and where, on this read-out, it does not: paralinguistics, where on our data Conjecture 1 is, for now, contradicted.

Context isolation buys nothing structurally: the isolated optimum equals the monolithic one. A reader might suppose that decomposing a task into context-isolated blocks is itself the source of gain. Theorem 6 (`qstar_product`) forbids that reading: under a product base and a separable reward $R = \sum_i R_i$, the global Gibbs optimum factorizes as $q^\star = \prod_i q_i^*$, so tilting each isolated block to its own optimum assembles *exactly* the same distribution — and the same objective value $F(q^\star) = \beta \log Z$ — as a single monolithic tilt against R. The high-spread content/intent gains are therefore *single-model* gains; isolation *per se* adds zero. The only way to increase $\mathcal{G}$ is to introduce additional *non-degenerate* verifiable rewards R_i (each strictly positive by `gain_pos_of_nonconstant`); the architectural act of separating contexts is a convenience for credit assignment, not a source of optimization headroom. Whether agentic decomposition adds anything a single model could not is therefore an *empirical* claim about the availability of genuinely new non-degenerate rewards — a claim we have not tested, and one the previous paragraph shows is, for paralinguistics, currently unsupported on the omni-embed read-out.

The moat is an absence of evidence (fast-follower risk). Our central feasibility argument rests on the observation that no prior work couples training-free, reward-guided inference-time optimization with a frozen omni model's disentangled embeddings as the search variable. An absence of evidence is structurally weaker than evidence of absence: it is falsified the moment a competing group publishes the same coupling, and the individual ingredients — best-of-N and reranking [bei, 2024, ver, 2025], controlled decoding [mud, 2023], no-gradient credit assignment [jit, 2026], test-time RL [zuo, 2025, li2, 2025] — are all public. We therefore do not claim a defensible technical monopoly; we claim a *first-mover synthesis* whose value is the formalization (the OSA theorems) and the machine-checked qualitative core, not the mechanism itself. A reader should weight this contribution as integrative rather than as a primitive discovery.

The mechanism is not novel; it is a domain transfer with a now-doubtful precondition. The Gibbs-optimal tilting $q^\star \propto q_0 \exp(R/\beta)$ and its KL-regularized variational characterization are textbook results in the alignment-as-Bayesian-inference literature [kor, 2022, yan, 2024], and the best-of-N/KL frontier is now sharply understood [bei, 2024, gui, 2024]. Our contribution is to instantiate this machinery over the *embedding* action space $\mathcal{Z}_{\mathrm{A}}$ of a frozen speech model rather than the token space. The transfer assumes the relevant degrees of freedom $\mathcal{Z}$ already carry the pretrained structure the reward asks for [fro, 2024, sal, 2023]. Our results split this assumption: on the omni-embed content read-out the *semantic/content* structure is clearly present and high-spread, but the *paralinguistic* structure (speaker, emotion) is, on that same probe, at or near chance — so for those factors the hidden precondition of the transfer is violated on the read-out we measured, which is exactly why the negative result above is not incidental but predicted by the theory.

OSA-2 idealizes context isolation as exact separability. OSA-2 models a multi-key agent reward as a sum of per-context-isolated components and proves additivity under *exact* separability of R across the content, speaker, emotion, and language/intent factors. Real agent rewards are at best *approximately* separable: an ASR-conditioned downstream reward leaks into the emotion channel through prosody, and intent classification is not independent of transcription quality. Under a residual coupling of magnitude ϵ in the cross-factor Hessian, the Gibbs optimum of the separable surrogate is displaced from the true optimum, and the additive $\mathcal{G}$ we certify is an idealization that degrades as R departs from additivity. We have not characterized this degradation quantitatively; the gap between idealized and approximate separability is a

[6] The per-seed paired CI happens to exclude zero in 2 of the 5 seeds; that vote count is *not* the headline — the across-seed interval, which spans zero, is.

large theory-to-practice risk, and the empirical separability of the embedding factors [xia, 2020, wan, 2020, lea, 2023] is an assumption our probes support only locally for content, and not at all for speaker.

Reaching the optimum is static geometry, not convergence; finite-time convergence is open. We carry exactly *two* load-bearing results — OSA-1 (gain governed by the reward range) and OSA-2 (additivity with the isolated optimum equal to the monolithic one). What we record beyond them is bookkeeping, not a third pillar. The rollout deficit (`rollout_deficit`, Corollary 1) is a corollary of the master identity: every rollout q off the optimum incurs a *nonnegative* objective deficit $\beta\,\mathrm{KL}(q \,\|\, q^\star) \geq 0$ — which at the base rollout is exactly the gain $\beta\,\mathrm{KL}(q_0 \,\|\, q^\star)$ — a static deficit, not a bounded "stability tax." The product factorization $q^\star = \prod_i q_i^*$ (`qstar_product`, Theorem 6) belongs to OSA-2, not to any separate dynamical result. Neither statement is a convergence theorem, and they do not compose into one. The operational update we actually run is JitRL's [jit, 2026] no-gradient credit assignment, whose guarantee is *asymptotic under slow drift* of the rollout distribution; the two regimes do not assemble into a finite-time statement, so we do *not* have a monotone-improvement guarantee for the gated update over the full, continuous, only-approximately separable space at a finite horizon. We read the β-KL term as a trust region by analogy with proximal/trust-region dynamics, but this analogy is an open conjecture, not a theorem, and it bears on the stability claims that agentic deployments care about most [ace, 2025, exp, 2025, rem, 2026]. We accordingly treat convergence of the gated update as an open problem.

What "machine-checked" covers: one isolated lemma, one Pinsker hypothesis, one documented `sorry`. Our machine-checked development is sorry-free in its qualitative core — including the gain identity, its nonnegativity, the flat-reward collapse, the partition-function and Gibbs-optimum product factorizations, the strict per-non-degenerate-block positivity `gain_pos_of_nonconstant`, and `rollout_deficit` — but the *quantitative* results carry named, undischarged hypotheses, and we separate three statuses carefully so no reader mistakes "machine-checked" for "unconditionally proved." (i) Hoeffding's lemma, which underwrites the spread$^2/(8\beta)$ ceiling, is a standard result *discharged on paper* but isolated in Lean as a named hypothesis (`gain_le_of_hoeffding`) rather than reproved from first principles; if the per-context reward is heavier-tailed than the assumed sub-Gaussian envelope, its constants are optimistic. (ii) Pinsker's inequality, behind the $O(\sqrt{\log N})$ best-of-N regret reading, is *consumed as a hypothesis, not discharged.* (iii) The order-statistics step in the best-of-N KL bound carries the *single documented* ***`sorry`*** in the development (`klBoN_le_klBoundBoN_TODO`): the asymptotics of the maximum order statistic [yan, 2024, bei, 2024] are invoked as a hypothesis, not proved in our formalization. The checked artifact is sound *relative to* these three.

Goodhart on verifiable rewards, and the sharper risk of *model-dependent* rewards. Inference-time optimization against a fixed reward is, by construction, an over-optimization engine: pushing N or lowering β moves probability mass toward whatever the reward measures, including its defects [gao, 2022, ska, 2022]. Verifiable speech rewards (WER, exact match, intent accuracy) are more robust than learned reward models, but they are not immune, and "spurious rewards" can induce gains that do not reflect the intended capability [spu, 2025, rlv, 2025]. A sharper, easily overlooked variant arises whenever the reward is *computed from the same frozen model* — e.g. a probe read off the model's own embedding, or a surrogate verifier instantiated from the model itself. Such a model-dependent reward can be high-spread for reasons internal to the model rather than because it tracks the target capability, so the certified $\mathcal{G}$ may measure the model's self-consistency, not the skill. Because we change no weights the failure is recoverable (lower N, raise β, decouple the verifier), but the gate must be tuned against held-out, contamination-controlled splits [con, 2024, gor] with an *externally* grounded reward, or the gain is an artifact of the read-out.

Paralinguistic keys are dual-use: a mis-estimated label both mis-routes and mis-rewards. The same emotion/speaker estimate that *routes* a query (selecting the context and skill) also *rewards* the rollout that produced it. A mis-estimated SER label therefore enters two coupled loops with the same sign of error: it sends the query to the wrong context *and* reinforces the embedding that yielded the wrong label, a positive-feedback pathway that idealized separability hides. This is sharpened by evidence that omni models tend to "read, not listen" on paralinguistic cues [vox, 2026, lis, 2025] and that speaker identity resists in-context correction [spk, 2026] — exactly the regime where our speaker probe sits at chance on the omni-embed read-out. The mitigation — decoupling the routing estimator from the reward estimator — is future work, not a property of the current gate.

The two-omni "vector-omni memory" collapses for paralinguistics on this read-out; we reposition the design. A natural reading of the system is two frozen omni models — one as policy (Operator-B), one as a vector memory whose embeddings index past episodes (Operator-A) — with the omni memory carrying paralinguistic keys. Our speaker probe refutes the load-bearing version of that design: at ≈ 0.04 accuracy (near chance) across all layers, the omni-embed *content* read-out does *not* support paralinguistic retrieval, so a memory keyed on it would route on noise. We are careful that this is a statement about the omni-embed read-out under a probe, not about a generative omni policy, which we have not tested for paralinguistics. We nonetheless reposition the contribution honestly: the omni embedding is reserved for *content* retrieval, where its spread is real, and paralinguistic memory is delegated to classic, purpose-built speech encoders (ECAPA-style speaker embeddings, a dedicated SER head) rather than to the omni read-out. The "two-omni" framing is downgraded to "omni policy + omni content memory + classical paralinguistic encoders."

Memory poisoning via acoustic spoofing. When the optimized embedding is written to agent memory, the acoustic channel becomes an injection surface: a spoofed speaker or emotion can plant a poisoned key that later retrieval amplifies [age, 2024a, mem, 2026]. Unlike text poisoning, the attack needs no lexical trigger — it rides paralinguistic features the model already over-trusts. We provide no audit defense here; memory hygiene and provenance [ana, 2026] are required before the embedding-as-memory-key design is safe to deploy.

Small-sample intervals and provenance discrepancies under reconciliation. Several preliminary results are reported on modest evaluation sets, where bootstrap intervals [bis] are wide and statistical power is low [car, 2020, dro]; effects near the resolution of a few hundred utterances should be read as directional, not confirmed, and adequately powered, seed-averaged [hen, 2017] replication on the frozen, contamination-controlled split is a precondition for any quantitative headline. Relatedly, the emotion re-run above already surfaced a provenance discrepancy between an earlier single-seed, oracle-test-layer number and the multi-seed, dev-selected protocol whose across-seed interval spans zero; we are reconciling all reported figures to the committed artifacts and the dev-selected, paired-delta protocol, and any residual mismatch between an in-text number and its artifact should be read as the artifact governing.

10 Conclusion

We have argued, and for its qualitative core machine-checked, a single reframed thesis: the optimization gain available to training-free RL — reward-guided, inference-time search with no gradient and no weight or structure change — is governed by the *reward range* (the spread spread $= \max R - \min R$ over $\mathrm{supp}(q_0)$) of the (action-space, reward) pair, a property of the *reward structure* and not of model class. This is the proven core, OSA-1: the gain is the divergence $\mathcal{G} = \beta\,\mathrm{KL}(q_0 \,\|\, q^\star)$ between what the model already does and what the reward would have it do, sandwiched as $0 \le \mathcal{G} \le \min\bigl(\text{spread},\ \text{spread}^2/(8\beta)\bigr)$; to leading order, for a concentrated q_0, the true discriminator is $\mathrm{Var}_{q_0}(R)/(2\beta)$, with the range-based Hoeffding term as a certified ceiling. A flat reward — e.g. instruction-conditioning of a frozen embedding read-out — yields vanishing gain; a high-spread reward — e.g. task-level verifiable selection — yields large gain *even on a single frozen model*, as our own single-model Operator-B results show (SLURP +0.330, URO +0.335). The lever is the range, not the agent.

OSA-2 sharpens this for composed rewards. We prove that gain is *additive* across context-isolated blocks (`gain_product`) and that each *non-degenerate* block contributes *strictly* positive gain (`gain_pos_of_nonconstant`, sorry-free: a non-constant reward forces $\mathcal{G} > 0$). But we also prove (`qstar_product`) that the isolated optimum *equals* the monolithic optimum on the product reward $R = \sum_i R_i$ — so context isolation *per se* adds nothing structurally, and the only new gain comes from introducing additional non-degenerate verifiable rewards. These are the two proven, load-bearing results. Whether a *frozen speech model* actually affords non-degenerate per-block rewards is an empirical question (Conjecture 1), not a theorem; and whether agentic decomposition adds anything a single model could not is, equally, untested — it can be settled only by introducing genuinely new non-degenerate rewards and measuring their spread, which we have not done.

On that empirical question our own data return a central, honest negative for the flagship's paralinguistic factors — established, we stress, on the omni-embed *content* read-out via a linear/kNN probe, and so silent about the untested generative Operator-B policy. A rigorous multi-seed re-run of the emotion-pooling gain (dev-selected layer, paired-delta bootstrap, seeds $42, 7, 123, 2024, 31337$) gives an across-seed mean $\Delta = +0.037$ with a 95% t-CI of $[-0.043, +0.116]$ that spans zero: *no significant emotion gain at the across-seed level (a null)* — the earlier $+0.097$ was the best seed under oracle test-layer selection, a selection leak — and the speaker probe is at chance (≈ 0.04) across all 37 layers. So the reward-range thesis has teeth for *content/intent*, where the gains are single-model and well-separated, but on this read-out it does not for paralinguistics. We report this not as a setback but as a result: it scopes the thesis to where it has teeth and refutes the load-bearing version of a two-omni paralinguistic memory, which we accordingly reposition as an omni policy plus omni *content* memory plus classical speaker/SER encoders.

The static geometry of reaching the optimum we keep as bookkeeping, not a third result: a nonnegative rollout deficit $\beta\,\mathrm{KL}(q \,\|\, q^\star) \geq 0$ (`rollout_deficit`) off the optimum, and the OSA-2 factorization $q^\star = \prod_i q_i^*$ (`qstar_product`). A finite-time monotone-convergence guarantee for the gated, credit-assigned update over the full, only-approximately-separable space remains *open*, and the trust-region reading of the β-KL term is an analogy we conjecture, not a theorem we prove.

We are therefore honest about what this system is today: a *testbed*, not a flagship to build out now. Its value is threefold — a falsifiable negative result (paralinguistic per-block rewards are absent on the omni-embed read-out), a frozen, contamination-controlled benchmark on which that negative and any future positive can be re-measured, and exactly one falsifiable candidate mechanism (the reward-range thesis and its per-block non-degeneracy conjecture, Conjecture 1) — rather than a deployable agentic stack. We close by reaffirming the discipline that makes the claim falsifiable: a survey-first posture that treats our own moat as an absence of evidence to be defended, not assumed; a commitment to *externally verifiable* rewards over learned or model-internal proxies, precisely to blunt the Goodhart pressure that inference-time optimization invites; and a no-gradient, no-weight-change constraint that keeps every failure recoverable at inference rather than frozen into parameters. The promise of *activating*, rather than retraining, the latent competence of frozen speech models is real and near-term for content and intent; for paralinguistics it is, on present evidence, an open and likely negative hypothesis — and we state the whole only to the precision that the proofs and the measurements can currently bear.

A Lean 4 Formalization

The *qualitative core* of our training-free RL theory is machine-checked in **Lean 4 + mathlib** (toolchain and mathlib both pinned to `v4.31.0`; the exact mathlib commit is recorded in the `lake-manifest.json` committed alongside the proof project, see Section A), in the project `proofs/tfrl` (Lean namespace `TfrlProofs`). We are careful about what "machine-checked" means here, and Section A below states it precisely. In brief: the structural backbone — the master identity and T1 optimality, the flat-reward collapse, the product factorizations, the gain identity together with its nonnegativity *and its strict positivity under any non-constant reward*, and the rollout deficit (`gain_eq`, `gain_nonneg`, `gain_pos_of_nonconstant`, `kl_pos_of_ne`, `flat_no_gain`, `Zpart_product`, `gain_product`, `qstar_product`, `rollout_deficit`) — is **fully `sorry`-free**. This qualitative core is exactly the formal content of our reframed thesis: the optimization gain of training-free RL is governed by the *reward spread* (the range $R_{\max} - R_{\min}$ over $\mathrm{supp}(q_0)$) of the (action-space, reward) pair, a property of the reward *structure* and not of model class. Two facts make the honesty of this framing precise and are both `sorry`-free below. (i) `gain_pos_of_nonconstant` shows every non-degenerate verifiable reward contributes *strictly* positive gain, even on a single frozen model. (ii) `qstar_product` shows the context-isolated optimum *equals* the monolithic optimum on the product reward $R = \sum_i R_i$, so context isolation *per se* adds no gain; the only new gain comes from introducing additional *non-degenerate* verifiable rewards. Whether a frozen speech model affords such non-degenerate per-block rewards for a given factor is an *empirical* question (Conjecture 1 in the main text), not a theorem — and for the flagship's paralinguistic factors our own evidence (§6.4 of the main text; committed artifact `_repro/emotion_pool_paired_v2.json`) shows no significant gain: a *null* for the emotion factor (the across-seed t-CI spans 0) and a near-degenerate reward for the speaker factor. We stress the scope of this negative: it is measured on *one* frozen content bi-encoder (omni-embed) via a linear/kNN probe over the embedding action space $\mathcal{Z}_\mathrm{A}$; it does *not* bound the proposed generative Operator-B policy, which is untested for paralinguistics.

The *quantitative* results, by contrast, are Lean theorems **conditional on named, undischarged analytic hypotheses** — each is a *Lean-checked consumption of an external bound*, not an end-to-end formalization: the spread$^2/(8\beta)$ gain ceiling consumes Hoeffding's lemma as a hypothesis (the analytic bound itself is discharged by the *paper* proof of Lemma 3 in this appendix, not inside Lean), the best-of-N KL bound (`kl_best_of_n_le`) is itself `sorry`-free and consumes the Beirami order-statistics bound as a hypothesis — that hypothesis being the lone documented `sorry`, isolated in a *separate* lemma `klBoN_le_klBoundBoN_TODO` — and the $O(\sqrt{\log N})$ regret consumes Pinsker's inequality as a hypothesis. **We state this plainly: no quantitative bound is discharged end-to-end inside Lean.** What is machine-checked `sorry`-free is the qualitative core *together with the new strict lemmas* `kl_pos_of_ne` and `gain_pos_of_nonconstant`; the quantitative corollaries are sound Lean derivations from explicit analytic inputs that are standard and are proved on paper here. Throughout, $\mathcal{Z}$ is a `Fintype` (the inference-time action space), $q_0 : \mathcal{Z} \to \mathbb{R}$ is a strictly positive reference distribution with $\sum_z q_0(z) = 1$, $R : \mathcal{Z} \to \mathbb{R}$ is a reward, and $\beta > 0$ a temperature. Here OSA abbreviates *Optimization-Space Adequacy* (and only that) throughout.

Verification status: `sorry`-free vs. conditional. To prevent any over-reading of "machine-checked", we tabulate the exact status of each Lean result.

Lean lemma	Status
`kl_nonneg`	`sorry`-free
`kl_pos_of_ne` (strict Gibbs)	`sorry`-free
`F_sub_eq_beta_mul_kl`	`sorry`-free
`tilting_optimal` (T1)	`sorry`-free
`qstar_eq_q0_of_const` (T3)	`sorry`-free
`gain_eq` (OSA-1a)	`sorry`-free
`gain_nonneg` (OSA-1b)	`sorry`-free
`gain_pos_of_nonconstant` (OSA-1b′)	`sorry`-free
`flat_no_gain` (OSA-1c)	`sorry`-free
`Zpart_product` (OSA-2)	`sorry`-free
`gain_product` (OSA-2)	`sorry`-free
`qstar_product` (OSA-2 factorization)	`sorry`-free
`rollout_deficit` (rollout-deficit corollary)	`sorry`-free
`klBoundBoN_nonneg`	`sorry`-free
`gain_le_of_hoeffding` (OSA-1d)	*conditional (consumes bound)*
`kl_best_of_n_le` (T2)	*conditional (consumes the Beirami order-statistics bound as a hypothe*
`klBoN_le_klBoundBoN_TODO` (T2 order-stat step)	*contains the single documented* `sorry` *(Beirami order-statistics bound)*
`regret_O_sqrt_log` (T6)	*conditional (consumes bound)*

Every entry in the upper block compiles with no axioms beyond mathlib's and no `sorry`; the exact mathlib commit against which all of these compile is pinned in the `lake-manifest.json` committed with the `proofs/tfrl` project, so the build is reproducible bit-for-bit. Each entry in the lower block is a genuine Lean theorem, but it is a *Lean-checked consumption of an external bound*: it *takes its analytic input as an explicit hypothesis* (the Hoeffding ceiling for OSA-1d, the Beirami order-statistics bound for T2, Pinsker for T6) — and for T2 the consuming wrapper `kl_best_of_n_le` is itself `sorry`-free, while the order-statistics step it consumes is isolated in a *separate* lemma, `klBoN_le_klBoundBoN_TODO`, which carries the lone documented `sorry`; the conclusion is then derived rigorously in Lean from that input. In particular, the spread$^2/(8\beta)$ ceiling is **not** discharged inside Lean — Lean only performs the substitution $S \mapsto \mathcal{G}$ via `gain_le_of_hoeffding`; the analytic content lives in the paper proof of Lemma 3. To restate the boundary precisely: *no quantitative bound is discharged end-to-end in Lean*; what is machine-checked `sorry`-free is the qualitative core plus the two new strict lemmas `kl_pos_of_ne` and `gain_pos_of_nonconstant`.

Tilting primitives and the Gibbs inequality. The exponentially tilted optimum $q^\star$, its partition function Z, and the KL-regularized objective $F(\cdot)$ are defined exactly as in the main text. We reproduce the Lean signatures verbatim.

```
noncomputable def Zpart (q0 R : Z → ℝ) (β : ℝ) : ℝ := ∑ w, q0 w * Real.exp (R w / β)
noncomputable def qstar (q0 R : Z → ℝ) (β : ℝ) (z : Z) : ℝ :=
  q0 z * Real.exp (R z / β) / Zpart q0 R β
noncomputable def F (q0 R : Z → ℝ) (β : ℝ) (q : Z → ℝ) : ℝ :=
  (∑ z, q z * R z) - β * ∑ z, q z * Real.log (q z / q0 z)
```

```
theorem kl_nonneg {p r : Z → ℝ} (hp : ∀ z, 0 ≤ p z) (hr : ∀ z, 0 < r z)
    (hpsum : ∑ z, p z = 1) (hrsum : ∑ z, r z = 1) :
    0 ≤ ∑ z, p z * Real.log (p z / r z)
```

`kl_nonneg` is Gibbs' inequality: the pointwise estimate $p(z) - r(z) \le p(z)\log(p(z)/r(z))$ (trivial when $p(z) = 0$; otherwise $\log x \le x - 1$ with $x = r(z)/p(z)$ via `Real.log_le_sub_one_of_pos`) is summed only over $\mathrm{supp}(p)$. On that index set $\sum_{z\in\mathrm{supp}(p)} p(z) = 1$, while $\sum_{z\in\mathrm{supp}(p)} r(z) \le \sum_z r(z) = 1$ (the partial sum over the support is $\le$ the full sum, since $r \ge 0$); the latter inequality is exactly what we need, because it gives $\sum_{z\in\mathrm{supp}(p)} \big(p(z) - r(z)\big) \ge 1 - 1 = 0$, hence $\mathrm{KL}(p \,\|\, r) \ge 0$. Writing the support sum as an equality with the full sum is never required — the one-sided bound "$\le$ the full sum" is sufficient for $\mathrm{KL}(\ge \,\|\, 0)$.

We additionally prove the *strict* form, new in this revision and `sorry`-free:

```
theorem kl_pos_of_ne {p r : Z → ℝ} (hp : ∀ z, 0 ≤ p z) (hr : ∀ z, 0 < r z)
    (hpsum : ∑ z, p z = 1) (hrsum : ∑ z, r z = 1) (hne : p ≠ r) :
    0 < ∑ z, p z * Real.log (p z / r z)
```

`kl_pos_of_ne` upgrades Gibbs to its equality case: $\mathrm{KL}(p \,\|\, r) = 0 \iff p = r$, so $p \ne r \Rightarrow \mathrm{KL}(p \,\|\, r) > 0$. The pointwise slack $p(z)\log(p(z)/r(z)) - \big(p(z) - r(z)\big) \ge 0$ is *strictly* positive at any z with $p(z) > 0$ and $p(z) \ne r(z)$ (strict form of $\log x < x - 1$ for $x \ne 1$, via `Real.log_lt_sub_one_of_ne`); since $p \ne r$ and both sum to 1 there must exist such a z, so summing the nonnegative slacks — at least one of which is strictly positive — gives a strictly positive total. This strict inequality is the engine of the new strict gain lemma below: it is exactly why a non-constant reward yields *strictly* positive optimization gain rather than merely nonnegative gain.

Master identity and T1. The decomposition $F(q^\star) - F(q) = \beta\,\mathrm{KL}(q \,\|\, q^\star)$ is the engine of optimality.

```
theorem F_sub_eq_beta_mul_kl [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
    {q : Z → ℝ} (hq : ∀ z, 0 ≤ q z) (hqsum : ∑ z, q z = 1) :
    F q0 R β (qstar q0 R β) - F q0 R β q = β * ∑ z, q z * Real.log (q z / qstar q0 R β z)
```

```
theorem tilting_optimal [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
    {q : Z → ℝ} (hq : ∀ z, 0 ≤ q z) (hqsum : ∑ z, q z = 1) :
    F q0 R β q ≤ F q0 R β (qstar q0 R β)
```

T1 (`tilting_optimal`) is then immediate: $F(q^\star) - F(q) = \beta\,\mathrm{KL}(q \,\|\, q^\star) \ge 0$ by `kl_nonneg` and $\beta > 0$.

T3 — flat reward, no tilt.

```
theorem qstar_eq_q0_of_const [Nonempty Z] {q0 : Z → ℝ} {β c : ℝ}
    (hq0sum : ∑ z, q0 z = 1) {R : Z → ℝ} (hR : ∀ z, R z = c) :
    ∀ z, qstar q0 R β z = q0 z
```

When $R(z) \equiv c$, $Z = \exp(c/\beta)$ collapses and $q^\star(z) = q_0(z)\exp(c/\beta)/\exp(c/\beta) = q_0(z)$: a degenerate action space with zero reward spread admits no steering.

T2 — best-of-N KL bound.

```
noncomputable def klBoundBoN (N : ℕ) : ℝ := Real.log N - ((N : ℝ) - 1) / N
theorem klBoundBoN_nonneg {N : ℕ} (hN : 1 ≤ N) : 0 ≤ klBoundBoN N
theorem kl_best_of_n_le (klBoN : ℕ → ℝ) {N : ℕ}
    (hBeirami : klBoN N ≤ klBoundBoN N) :
    klBoN N ≤ Real.log N - ((N : ℝ) - 1) / N := hBeirami
```

We prove rigorously that $\text{spread}_{\text{BoN}}(N) := \log N - (N-1)/N \geq 0$ (it equals $\log N - 1 + 1/N$, nonnegative since $\log N \geq 1 - 1/N$). The order-statistics derivation $\text{KL}(\pi_{\text{BoN}} \,\|\, q_0) \leq \texttt{klBoundBoN}\, N$ [bei, 2024] is the lone documented `sorry` (`klBoN_le_klBoundBoN_TODO`); `kl_best_of_n_le` takes that estimate as a hypothesis and concludes. Thus the best-of-N KL bound is a *conditional* Lean theorem — a Lean-checked consumption of the Beirami bound: the analytic order-statistics input is not discharged in Lean, only consumed.

T6 — $O(\sqrt{\log N})$ regret.

```
theorem regret_O_sqrt_log {gain R_max kl : ℝ} {N : ℕ} (hN : 1 ≤ N)
    (hRmax : 0 ≤ R_max) (hPinsker : gain ≤ R_max * Real.sqrt (kl / 2))
    (hKL : kl ≤ klBoundBoN N) :
    gain ≤ R_max * Real.sqrt (Real.log N / 2)
```

Chaining the Pinsker estimate $\mathcal{G} \leq R_{\max}\sqrt{kl/2}$ (taken as the hypothesis `hPinsker`, not proved in Lean) with $kl \leq \texttt{klBoundBoN}\, N \leq \log N$ and monotonicity of $\sqrt{\cdot}$ yields $\mathcal{G} = O(\sqrt{\log N})$. Like T2, this is therefore a Lean theorem *conditional* on its named analytic input (here Pinsker's inequality) — another Lean-checked consumption of an external bound.

The OSA suite (`TfrlProofs.OptSpace`). We define the optimization gain $\mathcal{G} := F(q^\star) - F(q_0)$ and prove its structural properties. Every lemma in this module compiles `sorry`-free; the only one whose *quantitative* conclusion is conditional is `gain_le_of_hoeffding`, which is a sound Lean proof that simply consumes the Hoeffding bound as an explicit hypothesis (discharged on paper in Lemma 3, not in Lean).

```
noncomputable def gain (q0 R : Z → ℝ) (β : ℝ) : ℝ :=
  F q0 R β (qstar q0 R β) - F q0 R β q0

theorem gain_eq [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
    (hq0sum : ∑ z, q0 z = 1) :
    gain q0 R β = β * Real.log (Zpart q0 R β) - ∑ z, q0 z * R z
theorem gain_nonneg [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
    (hq0sum : ∑ z, q0 z = 1) : 0 ≤ gain q0 R β
theorem gain_pos_of_nonconstant [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
    (hq0sum : ∑ z, q0 z = 1) {z₁ z₂ : Z} (hR : R z₁ ≠ R z₂) :
    0 < gain q0 R β
theorem flat_no_gain [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
    (hq0sum : ∑ z, q0 z = 1) {c : ℝ} (hR : ∀ z, R z = c) : gain q0 R β = 0
theorem gain_le_of_hoeffding [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
    (hq0sum : ∑ z, q0 z = 1) {S : ℝ}
    (hHoeff : β * Real.log (Zpart q0 R β) - ∑ z, q0 z * R z ≤ S) :
    gain q0 R β ≤ S

theorem Zpart_product :
    Zpart (fun p : Z1 × Z2 => q01 p.1 * q02 p.2) (fun p => R1 p.1 + R2 p.2) β
      = Zpart q01 R1 β * Zpart q02 R2 β
theorem gain_product [Nonempty Z1] [Nonempty Z2] (hq01 : ∀ z, 0 < q01 z)
    (hq02 : ∀ z, 0 < q02 z) (hβ : 0 < β)
```

```
    (hq01sum : ∑ z, q01 z = 1) (hq02sum : ∑ z, q02 z = 1) :
    gain (fun p : Z1 × Z2 => q01 p.1 * q02 p.2) (fun p => R1 p.1 + R2 p.2) β
      = gain q01 R1 β + gain q02 R2 β
theorem qstar_product (p : Z1 × Z2) :
    qstar (fun p : Z1 × Z2 => q01 p.1 * q02 p.2) (fun p => R1 p.1 + R2 p.2) β p
      = qstar q01 R1 β p.1 * qstar q02 R2 β p.2

theorem rollout_deficit [Nonempty Z] (hq0 : ∀ z, 0 < q0 z) (hβ : 0 < β)
    {q : Z → ℝ} (hq : ∀ z, 0 ≤ q z) (hqsum : ∑ z, q z = 1) :
    F q0 R β q = F q0 R β (qstar q0 R β)
        - β * ∑ z, q z * Real.log (q z / qstar q0 R β z)
      ∧ 0 ≤ β * ∑ z, q z * Real.log (q z / qstar q0 R β z)
```

`gain_eq` rewrites the gain as $\beta \log \mathbb{E}_{q_0}[e^{R/\beta}] - \mathbb{E}_{q_0}[R]$ (**OSA-1a**); `gain_nonneg` (**OSA-1b**) is Jensen / Gibbs. `gain_pos_of_nonconstant` (**OSA-1b′**) is the new strict companion: if the reward takes two distinct values, $R(z_1) \neq R(z_2)$, then $q^\star \neq q_0$ (the tilt is non-trivial), so by the strict Gibbs lemma `kl_pos_of_ne` (applied with the strictly-positive argument $r = q^\star > 0$) the master identity $\mathcal{G} = \beta\, \mathrm{KL}(q_0 \,\|\, q^\star)$ (read off `gain_eq`/`F_sub_eq_beta_mul_kl` at $q = q_0$) is *strictly* positive. This is the formal statement of the reframed thesis at the single-block level: **any** non-degenerate verifiable reward yields strictly positive gain — crucially, *even on a single frozen model*, since nothing in the hypotheses references a model class. Conversely `flat_no_gain` (**OSA-1c**) recovers T3: a flat reward (e.g. instruction-conditioning of a frozen embedding read-out that does not separate the labelled classes, hence a degenerate $\mathcal{Z}$ under that reward) affords *zero* RL improvement. Together OSA-1b′/OSA-1c make the dichotomy sharp and model-agnostic: gain is governed by whether the reward is non-constant, not by single-model-vs-agent. `gain_le_of_hoeffding` (**OSA-1d**) isolates the analytic bound S as a hypothesis and substitutes it into $\mathcal{G}$, so that *given* the paper-proved ceiling $S = \mathrm{spread}^2/(8\beta)$ one obtains $\mathcal{G} \le \mathrm{spread}^2/(8\beta)$ and hence spread $\to 0 \Rightarrow \mathcal{G} \to 0$ — the Lean step is the substitution, the analytic content is Lemma 3. The product lemmas (**OSA-2**) show that over a context-isolated product $\mathcal{Z}_1 \times \mathcal{Z}_2$ with $q_0 = q_0^{(1)} \otimes q_0^{(2)}$ and separable $R(z_1, z_2) = R_1(z_1) + R_2(z_2)$, the gain is *additive* and $q^\star$ factorizes; iterating over k isolated blocks gives $\mathcal{G} = \sum_i \mathcal{G}_i$, growing with k *only insofar as each added block carries a non-degenerate reward* (each summand is then strictly positive by OSA-1b′, and zero by OSA-1c otherwise). We stress the honest reading of `qstar_product` (the **OSA-2** factorization): the context-isolated optimum *equals* the monolithic single-model optimum on the product reward $R = \sum_i R_i$, so context isolation *per se* contributes no gain — the additivity $\mathcal{G} = \sum_i \mathcal{G}_i$ is a decomposition of the *same* optimum, and the only source of new gain is the introduction of additional non-degenerate verifiable rewards. In particular the high-spread content/intent gains are *single-model* gains, not evidence that agentic decomposition recovers anything; whether decomposition adds gain can only be settled by introducing genuinely new non-degenerate rewards (untested). Whether a frozen omni speech model in fact supplies such non-degenerate per-block rewards for a target factor is therefore *not* settled by these lemmas; it is the empirical Conjecture 1, which (per the main-text cautionary study) holds for content/intent but is *not* established for paralinguistic factors — a null emotion result (across-seed t-CI spanning 0) and a near-degenerate speaker reward, both measured on the frozen content bi-encoder's embedding probe and not on the generative Operator-B policy. Finally `rollout_deficit` (the rollout-deficit corollary) shows any rollout policy q sits below $q^\star$ by exactly $\beta\, \mathrm{KL}(q \,\|\, q^\star) \ge 0$: a naive rollout that fails to reach $q^\star$ incurs this nonnegative deficit (at $q = q_0$ it equals the gain itself, $\beta\, \mathrm{KL}(q_0 \,\|\, q^\star)$), while the β-KL trust region enforces the slow-drift precondition of credit-assigned tilting [jit, 2026].

B Extended Proofs

We give in full the two analytic results that the Lean development isolates as hypotheses or pure algebra. The first — the $\mathrm{spread}^2/(8\beta)$ ceiling — is precisely the input that `gain_le_of_hoeffding` leaves undischarged: it is established here, on paper, not inside Lean.

B.1 Hoeffding instance: the gain ceiling $\mathrm{spread}^2/(8\beta)$

Lemma 3 (Bounded-reward gain ceiling)**.** *Let $R : \mathcal{Z} \to [R_{\min}, R_{\max}]$ with reward spread* $\mathrm{spread} := R_{\max} - R_{\min}$*, and let q_0 be any distribution on $\mathcal{Z}$ with $\mathbb{E}_{q_0}$ the corresponding expectation. Then for every $\beta > 0$,*

$$\beta \log \mathbb{E}_{q_0}\big[e^{R/\beta}\big] - \mathbb{E}_{q_0}[R] \;\le\; \frac{\mathrm{spread}^2}{8\beta},$$

and hence, by `gain_le_of_hoeffding` *with $S = \mathrm{spread}^2/(8\beta)$, $\mathcal{G}(q_0, R, \beta) \le \mathrm{spread}^2/(8\beta)$.*

Proof. Let $X := R(z)/\beta$ for $z \sim q_0$, a random variable supported in $[a, b]$ with $a = R_{\min}/\beta$, $b = R_{\max}/\beta$, so $b - a = \mathrm{spread}/\beta$. Define the cumulant generating function $\psi(\lambda) := \log \mathbb{E}[e^{\lambda(X - \mathbb{E}X)}]$. We use Hoeffding's lemma: for a random variable X supported in $[a, b]$ and any $\lambda \in \mathbb{R}$,

$$\psi(\lambda) \;=\; \log \mathbb{E}\big[e^{\lambda(X - \mathbb{E}X)}\big] \;\le\; \frac{\lambda^2(b-a)^2}{8}.$$

To prove this standard bound, set $\varphi(\lambda) := \log \mathbb{E}[e^{\lambda X}]$. Then $\varphi(0) = 0$, $\varphi'(0) = \mathbb{E}X$, and $\varphi''(\lambda) = \mathrm{Var}_\lambda(X)$, where Var_λ is the variance under the tilted law $d\mu_\lambda \propto e^{\lambda x} d\mu$. Since $X \in [a, b]$ almost surely, the same holds μ_λ-a.s., and a random variable bounded in an interval of length $b - a$ has variance at most $((b-a)/2)^2 = (b-a)^2/4$ (the extremal case is the two-point distribution at the endpoints). Hence $\varphi''(\lambda) \le (b-a)^2/4$ for all λ. By Taylor's theorem with the Lagrange (mean-value) form of the remainder, for some ξ between 0 and λ,

$$\varphi(\lambda) = \varphi(0) + \lambda\varphi'(0) + \tfrac{1}{2}\lambda^2\varphi''(\xi) \le \lambda\,\mathbb{E}X + \frac{\lambda^2(b-a)^2}{8},$$

so $\psi(\lambda) = \varphi(\lambda) - \lambda\mathbb{E}X \le \lambda^2(b-a)^2/8$, which is Hoeffding's lemma.

Now take $\lambda = 1$. Then

$$\log \mathbb{E}[e^X] - \mathbb{E}[X] = \psi(1) \le \frac{(b-a)^2}{8} = \frac{1}{8}\Big(\frac{\mathrm{spread}}{\beta}\Big)^2 = \frac{\mathrm{spread}^2}{8\beta^2}.$$

Multiplying by $\beta > 0$ and substituting $X = R/\beta$ (so $\mathbb{E}[X] = \mathbb{E}_{q_0}[R]/\beta$ and $\mathbb{E}[e^X] = \mathbb{E}_{q_0}[e^{R/\beta}]$) gives

$$\beta \log \mathbb{E}_{q_0}[e^{R/\beta}] - \mathbb{E}_{q_0}[R] \le \frac{\mathrm{spread}^2}{8\beta}.$$

By `gain_eq`, the left side is exactly $\mathcal{G}$, completing the proof. This is the analytic step that supplies the hypothesis $S = \mathrm{spread}^2/(8\beta)$ to `gain_le_of_hoeffding`; Lean itself only carries out the final substitution. □

Lemma 3 makes the scaling explicit: the inference-time RL gain is quadratically throttled by the reward spread, $\mathcal{G} = O(\mathrm{spread}^2/\beta)$, so a flat or near-flat reward landscape (T3, **OSA-1c**) yields vanishing headroom regardless of search budget — the analytic counterpart of reward over-optimization saturation [gao, 2022]. This is precisely the lever behind our reframed thesis: it is the reward spread, not the model class, that sets the ceiling, and Lemma 3 (upper bound) together with `gain_pos_of_nonconstant` (strict lower bound for any non-constant reward) bracket the gain from both sides.

B.2 Product factorization (OSA-2) in full

We verify the algebra underlying `Zpart_product`, `meanR_product`, and `gain_product` (all of which are `sorry`-free in Lean; this subsection is the paper transcription of that pure-algebra content). Let $\mathcal{Z} = \mathcal{Z}_1 \times \mathcal{Z}_2$, $q_0(z_1, z_2) = q_0^{(1)}(z_1)\, q_0^{(2)}(z_2)$ with $\sum q_0^{(1)} = \sum q_0^{(2)} = 1$, and $R(z_1, z_2) = R_1(z_1) + R_2(z_2)$ (context isolation).

Proposition 4 (Partition factorization)**.** *$Z(q_0, R, \beta) = Z(q_0^{(1)}, R_1, \beta) \cdot Z(q_0^{(2)}, R_2, \beta)$.*

Proof. Expanding over the product index set and using $e^{(R_1(z_1)+R_2(z_2))/\beta} = e^{R_1(z_1)/\beta} e^{R_2(z_2)/\beta}$,

$$Z = \sum_{z_1,z_2} q_0^{(1)}(z_1) q_0^{(2)}(z_2)\, e^{R_1(z_1)/\beta} e^{R_2(z_2)/\beta} = \Big(\sum_{z_1} q_0^{(1)}(z_1) e^{R_1(z_1)/\beta} \Big) \Big(\sum_{z_2} q_0^{(2)}(z_2) e^{R_2(z_2)/\beta} \Big),$$

the last step by distributing the double sum (`Finset.sum_mul_sum`), which is the product of the two partition functions. □

Proposition 5 (Mean reward additivity)**.** $\mathbb{E}_{q_0}[R] = \mathbb{E}_{q_0^{(1)}}[R_1] + \mathbb{E}_{q_0^{(2)}}[R_2]$.

Proof. For fixed z_1, $\sum_{z_2} q_0^{(1)}(z_1) q_0^{(2)}(z_2)(R_1(z_1)+R_2(z_2)) = q_0^{(1)}(z_1) R_1(z_1) \sum_{z_2} q_0^{(2)}(z_2) + q_0^{(1)}(z_1) \sum_{z_2} q_0^{(2)}(z_2) R_2(z_2) = q_0^{(1)}(z_1) R_1(z_1) + q_0^{(1)}(z_1) \mathbb{E}_{q_0^{(2)}}[R_2]$, using $\sum q_0^{(2)} = 1$. Summing over z_1 and again using $\sum q_0^{(1)} = 1$ gives $\mathbb{E}_{q_0^{(1)}}[R_1] + \mathbb{E}_{q_0^{(2)}}[R_2]$. □

Corollary 2 (Additive gain, OSA-2)**.** $\mathcal{G}(q_0, R, \beta) = \mathcal{G}(q_0^{(1)}, R_1, \beta) + \mathcal{G}(q_0^{(2)}, R_2, \beta)$.

Proof. By `gain_eq`, $\mathcal{G} = \beta \log Z - \mathbb{E}_{q_0}[R]$. Substituting the two propositions and $\log(xy) = \log x + \log y$ (legitimate since each partition function is strictly positive, `Zpart_pos`),

$$\mathcal{G} = \beta\big(\log Z_1 + \log Z_2\big) - \big(\mathbb{E}_{q_0^{(1)}}[R_1] + \mathbb{E}_{q_0^{(2)}}[R_2]\big) = \mathcal{G}_1 + \mathcal{G}_2.$$

Iterating over k context-isolated components yields $\mathcal{G} = \sum_{i=1}^{k} \mathcal{G}_i$, strictly increasing in k *only* as each added component has nonzero reward spread — in which case its summand is strictly positive by `gain_pos_of_nonconstant`, and otherwise zero by `flat_no_gain`. We re-emphasize the honest reading via `qstar_product` (the **OSA-2** factorization): the companion identity $q^\star(z_1, z_2) = q^{\star(1)}(z_1)\, q^{\star(2)}(z_2)$ follows by the same exponential split and shows the isolated optimum *coincides* with the monolithic single-model optimum on $R = \sum_i R_i$. Thus the additive decomposition redistributes the *same* total gain across blocks; isolation per se manufactures none, and the additional gain that an agentic decomposition can realize is exactly the gain of whatever *new* non-degenerate verifiable rewards it introduces — an empirical, factor-dependent matter (Conjecture 1), not a consequence of the algebra. In particular the algebra gives no warrant for crediting the high-spread content/intent gains to agentic decomposition: they are single-model gains. □

C Notation

Symbol	Meaning
$\mathcal{Z}$	inference-time action space (degrees of freedom changing no weights)
$\mathcal{Z}_\mathrm{A}, \mathcal{Z}_\mathrm{B}$	embedding/Operator-A space; generative/Operator-B space
q_0	base reference policy q_0 (strictly positive, $\sum_z q_0(z) = 1$)
q	a search distribution q over $\mathcal{Z}$
$q^\star$	Gibbs / tilted optimum $q^\star(z) = q_0(z) e^{R(z)/\beta}/Z$
R	reward function $\mathcal{Z} \to \mathbb{R}$
β	temperature (> 0); KL-regularization strength
Z	partition function $\sum_w q_0(w) e^{R(w)/\beta}$
$F(q)$	KL-regularized objective $\mathbb{E}_q[R] - \beta\,\mathrm{KL}(q \,\Vert\, q_0)$
$\mathrm{KL}(p \,\Vert\, q)$	Kullback–Leibler divergence
$\mathbb{E}$	expectation $\mathbb{E}$
$\mathcal{G}$	optimization gain $F(q^\star) - F(q_0) \geq 0$
spread	reward spread $R_{\max} - R_{\min}$
N	number of candidates in best-of-N / search budget

D Verifiable Reward Functions

The reward callables used as verifiable signals R are exposed by the shared library `speechrl_common.rl`, and are deliberately deterministic and contamination-resistant [tul, 2024]. Heavy dependencies (`jiwer`, `scikit-learn`) are imported lazily inside each function so the package and its smoke tests import without the full stack.

Content / ASR (`rl.reward`). `wer(reference, hypothesis)` returns the word error rate after normalization; `asr_reward` returns $\max(0, 1 - \mathrm{WER}) \in [0, 1]$, a bounded reward whose spread directly governs the gain ceiling of Lemma 3; `exact_match_reward` returns $\mathbf{1}[\text{normalized strings equal}] \in \{0, 1\}$ for closed-form tasks (intent, keyword), matching the binary RLVR signal of T4-style plurality selection. These content/intent rewards are the ones for which our experiments realize a large, non-degenerate spread (e.g. the single-model Operator-B selection gains reported in the main text), consistent with the reframed thesis.

Retrieval / embedding (`rl.embedding_metrics`). For the flagship embedding action space $\mathcal{Z}_{\mathrm{A}}$, `recall_at_k(query, gallery, labels_q, labels_g, k)` and `mean_reciprocal_rank(...)` score a frozen model's pooled embeddings against a labelled gallery under cosine similarity (`cosine_sim_matrix` with ℓ_2 normalization); `retrieval_reward(..., k)` packages recall@k as a $[0, 1]$ reward for speaker-ID / language retrieval. We flag honestly that for the *speaker* factor this reward is near-degenerate on the frozen omni embedding (probe ≈ 0.04, near-chance across all 37 layers): a near-flat landscape under which Lemma 3 predicts vanishing gain — which is why the deployed memory reserves the omni embedding for content and delegates speaker retrieval to a classic encoder (ECAPA), rather than treating the omni vector as a load-bearing paralinguistic store.

Probe accuracy (`rl.probe`). `linear_probe_accuracy` / `knn_probe_accuracy` fit a frozen-feature classifier (held-out split) to score disentanglement of emotion (SER) and language/intent factors; `probe_reward(..., kind)` returns the held-out accuracy as the reward, providing the verifiable signal for the content/speaker/emotion/language axes the flagship disentangles. For the *emotion* factor this signal shows *no significant gain at the across-seed level (a null)*: a multi-seed re-run with dev-selected pooling layer (5 seeds: $42, 7, 123, 2024, 31337$) gives mean $\Delta = +0.037$ with a 95% across-seed t-CI of $[-0.043, +0.116]$, which spans 0 — not significant.[7] The previously-reported single-seed $+0.097$ was the best seed under oracle test-layer selection (a selection-leak / winner's-curse artifact). The committed artifact is `_repro/emotion_pool_paired_v2.json`. We scope this negative precisely: it is established on *one* frozen content bi-encoder (omni-embed) via a linear/kNN probe over the embedding action space $\mathcal{Z}_{\mathrm{A}}$, and it does *not* bound the proposed generative Operator-B policy, which is untested for paralinguistics. It is the empirical non-confirmation of Conjecture 1 for the emotion factor on this encoder that the qualitative theory deliberately does not assert. Auxiliary metrics (`macro_f1`, `bleu`, `chrf`, `eer`) in `rl.metrics` are used for evaluation only, not as RL rewards, to avoid coupling the reward and the test statistic [ska, 2022].

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